How Hidden Local Government Debt Shapes Shadow Banking Competition

Evidence from Wealth Management in China

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Abstract

Using granular holdings data on China's RMB 33 trillion market for wealth-management products operated by banks or their subsidiaries, we study how government support for borrowers shapes competition in shadow banking. Smaller banks sought an edge over larger rivals by reaching for yield in home-province bonds issued by local government financing vehicles (LGFVs). The 2023 debt resolution initiative strengthened expected support for LGFVs and compressed their bond spreads. The estimated product-return premium associated with home LGFV bonds fell by 69%. Larger local banks expanded into higher-yielding, privately negotiated LGFV-linked credit, while smaller banks substituted little and lost ground in relative product returns and wealth-management balances. Our model explains this unequal adjustment through fixed contracting costs that make investment scale important. Government support for borrowers can thus erode smaller banks' competitive edge and shift shadow banking toward larger intermediaries and more opaque credit.

JEL classification: G11, G21, G23, H63, H74

Keywords: local credit, bank competition, shadow banking, wealth-management products, debt resolution, local government financing vehicles, negotiated credit

1 Introduction

Borrowing by local government financing vehicles (LGFVs) and China's shadow banking system grew together. After the 2009 fiscal stimulus, these state-owned companies, which finance local public projects, refinanced maturing bank loans through bonds and shadow-banking channels (Chen et al. 2020). Wealth-management products (WMPs) pool savers' money in investment portfolios operated by banks or their wealth-management subsidiaries, separately from banks' own balance sheets. These products became a central part of China's shadow banking system (Acharya et al. 2025). At the end of 2025, WMPs held RMB 33 trillion, nearly a quarter of China's GDP, for about 143 million investors, almost all of them individuals. In 2025 alone, they earned RMB 730 billion for these investors.[1] At the end of 2024, WMPs held an estimated 28.1% of outstanding LGFV bonds (Figure A5). The exposure was especially important for local banks: LGFV bonds accounted for 38.5% of identified standard-bond holdings in their WMPs, compared with 13.2% at national banks.[2] LGFV bond yields therefore matter both for borrowers' financing costs and for the returns that local banks can offer WMP savers.

In this paper, we study how LGFV debt shapes competition in shadow banking. Because WMPs cannot guarantee principal or returns, the performance their products deliver is an important way for banks to compete for savers. Smaller banks, with less extensive brands and branch networks, reached for yield in LGFV bonds from their home provinces. China's 2023 local-government debt resolution undercut this strategy. By allowing provinces to issue special refinancing bonds to repay selected LGFV obligations, the program strengthened expected support for LGFVs and compressed their bond spreads. The estimated product-return premium associated with home LGFV bonds fell by 69%. Larger local banks expanded

[1] Industry statistics are from the Banking Wealth Management Registration and Depository Center's annual market reports (2025; 2026), referred to below as the annual WMP market reports. China's 2025 GDP of RMB 140.2 trillion is the National Bureau of Statistics' preliminary estimate.

[2] These are shares by holding value over 2022Q1–2023Q3. The denominator includes all positions identified as standard bonds in the holdings data, including government and policy-bank bonds, but excludes negotiated credit. Internet Appendix D.8 gives the calculation and distinguishes these bond-only shares from the debt-allocation shares used below.

into higher-yielding, privately negotiated LGFV-linked credit. Smaller banks substituted little, and their product returns and WMP balances declined relative to those of larger rivals.

In the WMP market, large state-owned and nationwide joint-stock banks, which we call national banks, compete with local banks serving provincial, city, and rural markets. Higher-yielding investments accessible to WMPs are scarce. Products can hold government and policy-bank bonds, deposits, negotiable certificates of deposit, and corporate debt, but safer instruments offer limited spreads and China's public high-yield bond market is narrow (Schipke et al. 2019). LGFV bonds help fill this gap. All eligible investors can buy them, and local banks can draw on familiarity with the borrowers. Privately negotiated credit, known as non-standard debt (NS), can finance the same borrowers through less transparent, nontraded claims such as trust loans and financing through asset-management plans. Unlike buying a marketable bond, arranging these investments entails sourcing, evaluation, and contracting costs. This distinction motivates the scale mechanism we develop below.

We link quarterly holdings of 60,396 WMPs from 171 banks to product returns, investor flows, issuer credit events, and bank-level WMP balances. Local banks concentrated their LGFV bond holdings in their headquarters provinces; we refer to these as home LGFV bonds. Across all banks, smaller institutions held larger home-LGFV shares. These holdings mattered commercially: before debt resolution, home LGFV exposure carried the largest estimated product-return premium among the asset categories we examine, including LGFV bonds from other provinces. Within risk and withdrawal categories, top-return-quintile products attracted subsequent net inflows exceeding those of bottom-quintile products by 12.2% of beginning assets. The portfolio and demand evidence thus connects home bonds to local banks' competition for savings.

Why do local banks seek yield at home rather than elsewhere? We examine whether local information and support for local borrowers help explain this choice. Among otherwise similar LGFV bonds, greater primary market participation by banks elsewhere in the issuer's province is associated with lower issue prices relative to public secondary-market valuations. These

banks appear to obtain better investment terms, consistent with informed purchasing.[3] For banks in the issuer's own city, the pricing pattern reverses: greater participation accompanies higher issue prices and lower yields, reducing the borrower's financing cost. Subsequent distress estimates point toward better selection of same-city issuers, with bonds held by same-city issuers less likely to have future distress events. Together, these patterns suggest that local information and borrower support can coexist with reaching for yield.

Debt resolution eroded this source of yield. Average LGFV issuance spreads over Treasury yields fell from about 200 to 97 basis points, with greater compression in more intensively refinanced provinces. For a 20-percentage-point allocation, the annual product-return premium from home LGFV bonds fell from about 21 to 7 basis points, the largest decline among the asset categories. Banks with greater pre-program exposure to these bonds also experienced larger product-return declines. As refinancing progressed, institutions initially more tilted toward home investments reduced their home LGFV bond shares more.

Negotiated credit retained considerably more yield than LGFV bonds: from 2023Q2 to 2024Q4, its average contractual yield fell from 6.6% to 5.9%, compared with 4.6% to 2.7% for bonds, widening the gap from 2.0 to 3.2 percentage points.[4] Home negotiated credit also retained a positive product-return premium. Before resolution, national banks held 82.7% of the identified negotiated credit in our sample, mostly outside their headquarters provinces. Afterward, larger local banks expanded their use at home. The largest third increased home negotiated credit from 7.4% to 18.3% of disclosed bonds and negotiated credit, more than offsetting their reduction in home bonds, while the smallest third kept this share below 1%. The difference persists after accounting for initial portfolios and common provincial conditions. Most expansion in these negotiated credit are still contracted with the LGFV sector. Smaller banks continued launching products but rarely included this substitute.

[3]Participation is measured by ownership at issue-quarter end, not observed primary-market orders.

[4]The financing-amount-weighted series are in Figure 2. Comparing the same legal borrower and financing month, with maturity controls, also shows a widening gap: from 1.9 percentage points in 2023Q2 to 2.9 in 2024Q4 (Figure 3).

Smaller local banks lost ground in both product returns and WMP balances. After implementation, annual product returns at a local bank fell by about 10 basis points relative to those at a bank with twice its initial assets; the larger bank's WMP balances also grew about 11% more. Our model explains why smaller banks rely more on high yields yet have less scope to replace them. Larger banks attract cautious savers through reputation; smaller banks compete for savers willing to bear risk by investing more in higher-yielding LGFV bonds. When debt resolution compresses bond yields, negotiated credit offers a substitute but requires a fixed contracting cost. A larger investor base makes the additional savings attracted by the substitute sufficient to cover that cost. Smaller banks cannot justify the same cost, so they substitute less and lose ground in both product returns and savings business.

These findings reveal the other side a two-way relationship between LGFV borrowing and shadow banking less explored by literature. Shadow banks finance local public investment, but the yield available on LGFV debt also shapes how they compete for savings. Stabilizing those borrowers can weaken smaller intermediaries' return advantage while encouraging larger institutions to finance the same sector through more opaque claims.

Our paper contributes to three strands of the literature. First, we connect shadow banks' role in financing local public investment to their competition for savings. Existing work traces off-budget fiscal expansion through LGFVs (Bai et al. 2016), the refinancing of stimulus loans through bonds and shadow banking (Chen et al. 2020), and the crowding out of private investment (Huang et al. 2020). Related studies show how deposit competition (Acharya et al. 2025), liquidity rules (Hachem and Song 2021), and monetary tightening (Chen et al. 2018) encouraged banks' shadow-banking activities. Implicit guarantees also shaped the pricing and funding of trust products and WMPs (Allen et al. 2023; Huang et al. 2023).

More broadly, research explains shadow banking through demand for claims perceived as safe (Gennaioli et al. 2013), regulation and technology (Buchak et al. 2018), and competition with insured deposits (Xiao 2020); low interest rates can also change funds' product choices and industry structure (Di Maggio and Kacperczyk 2017). We identify an investment channel

of competition: LGFV bond yield helped smaller banks offset weaker franchises, while the scale required to replace that yield reshaped their position relative to larger intermediaries.

Second, we connect local specialization to competition on returns. Distance and organizational form shape lenders' use of borrower information (Petersen and Rajan 2002; Degryse and Ongena 2005; Hauswald and Marquez 2006; Agarwal and Hauswald 2010; Stein 2002; Berger et al. 2005). Geographic and industry concentration are associated with investment performance (Coval and Moskowitz 2001; Kacperczyk et al. 2005; Loutskina and Strahan 2011), while lending specialization makes credit relationships difficult to substitute (Paravisini et al. 2023). Performance-sensitive flows and redemption risk shape asset managers' incentives (Sirri and Tufano 1998; Chen et al. 2010; Goldstein et al. 2017); competition (La Spada 2018; Choi and Kronlund 2018) and regulatory constraints (Becker and Ivashina 2015) can encourage reaching for yield. We show how familiar local bonds supplied a return-based competitive strategy for smaller banks, and why replacing that yield required greater scale.

Third, we study how borrower support affects financial intermediaries. Governments influence credit allocation (Sapienza 2004; Khwaja and Mian 2005) and, under fiscal stress, banks' holdings of government debt (Becker and Ivashina 2018; Ongena et al. 2019). In China, creditor political power affects selective defaults (Gao et al. 2021), and expected government support shapes the pricing gap between state-owned and private borrowers (Geng and Pan 2024). Geng et al. (2025) trace the government-led LGFV credit cycle and show that debt resolution narrows local banks' subordinated-bond spreads while reducing profitability and market shares at banks in riskier cities. Using granular holding data, we uncover an investment channel through which LGFV support changes shadow banking competition for household savings. Linking individual WMP holdings to product returns, we reveals how home LGFV bonds supported smaller banks' return advantage. The portfolios also identify the replacement investment: higher-yielding negotiated credit linked to the same borrower sector, increasingly held by larger local banks. This evidence, together with the fixed-cost

model, shows why borrower stabilization can change shadow bank's competation for savings and the opacity of the credit they provide.

Section 2 introduces the market, policy, and data. Section 3 establishes the local investment strategy, its contribution to returns, investor demand, and evidence on information and borrower support. Section 4 studies yield compression and bond retreat. Section 5 examines unequal substitution, product returns, and WMP balances. Section 6 explains these findings with a model in which savers differ in risk aversion and negotiated credit requires a fixed cost. Section 7 concludes.

2 Setting and Data

This section describes the WMP market and its place in household savings and shadow banking, the investments available to WMPs, the debt-resolution program, and the data we use.

2.1 Banks, products, and investment opportunities

A wealth-management product pools savers' money in a separately accounted investment portfolio run by a bank or its wealth-management subsidiary. For Chinese households, WMPs are a major alternative to bank deposits, and they form a central part of the country's shadow banking system (Acharya et al. 2025; Chen et al. 2020). Since the 2018 asset-management reform, products must report net asset values (NAVs) and cannot guarantee principal or returns.[5] Banks sell WMPs through branches and online platforms. National banks comprise large state-owned and nationwide joint-stock commercial banks, with extensive branch networks and nationwide name recognition. Local banks include provincial, city, and rural institutions. They compete for savings with both national banks braches and among themselves of different bank size.

[5]See the asset-management reform guidelines (2018).

WMPs invest primarily in fixed-income and similar assets. Permitted investments primarily include government and policy-bank bonds, deposits, negotiable certificates of deposit, and corporate debt.[6] Most readily accessible investments offer limited credit spreads, while the public high-yield corporate bond market is narrow (Schipke et al. 2019). Figure A1 shows the industry's asset mix, and Figure A2 traces its growth.

LGFV bonds provide one source of higher yield. LGFVs are state-owned companies that finance local infrastructure and public services. They borrow in their own names, separately from formal government debt. Repayment depends on project assets, local fiscal resources, and expected government support (Hendy et al. 2024; Yang and Jiang 2021). Favorable issuance review helped qualifying vehicles obtain bond financing.[7] In our pre-program sample, LGFV issuers were smaller and their bonds offered higher issuance yields than those of other issuers. Median FY2021 assets were RMB 28.4 billion, compared with RMB 35.3 billion for other nonfinancial issuers and RMB 52.9 billion for financial issuers (Table A3, Panel E). Average fixed-rate issuance yields were 4.26%, 3.03%, and 2.77%, respectively. The LGFV yield differences persist after controlling for issue characteristics and ratings (Table A15). All eligible investors can buy these bonds, but local banks can draw on lending, deposit, and payment relationships to assess familiar but opaque borrowers.

WMPs can also finance the same borrowers through privately negotiated claims, classified by regulators as non-standard debt (NS). We call these investments negotiated credit throughout. They include identified trust loans, leasing receivables, and claims held through trust or asset-management plans. Unlike buying marketable bonds, investing in negotiated credit requires arranging, documenting, and monitoring a contract without a continuously observable trading price. These activities entail costs even for a small investment. The same borrower can issue both bonds and negotiated credit.

[6] Article 35 of the commercial-bank WMP rules lists permitted investments (2018).

[7] The enterprise-bond review notice (2013) provided accelerated and simplified review for a qualifying urban investment company's first application when its debt-to-assets ratio was below 50%. Exchange-market corporate bonds and interbank corporate debt instruments follow separate review and registration frameworks (2021; 2015).

The WMP–LGFV bond link is economically important for both markets. At year-end 2024, we estimate that WMPs held about RMB 3.1 trillion in LGFV bonds, representing 10.2% of total WMP net assets and 28.1% of outstanding LGFV bonds (Figure A5). These industry totals are extrapolated from disclosed holdings; Internet Appendix D.8 explains the calculation and its denominators.

2.2 Debt resolution and the linked data

The policy announcement of July 24, 2023 called for comprehensive local-debt resolution. Provinces began issuing special refinancing bonds in October, raising about RMB 1.4 trillion in 2023Q4 (Xinhua News Agency 2023; Wang 2023; Wong 2024). The proceeds repaid selected obligations as part of a broader program that also included loan restructuring and restrictions on new borrowing. These refinancing bonds are provincial-government obligations, distinct from the LGFV bonds held by WMPs. An expansion announced in November 2024 added RMB 6 trillion of debt-swap capacity for 2024–2026 (Ministry of Finance 2024). The program strengthened expected government support without formally guaranteeing every LGFV claim. The resulting spread compression can raise the prices of existing bonds while reducing the yields available when banks reinvest. We date implementation to 2023Q4, or October in monthly tests.

We start from Wind's product-level WMP holdings and link them to NAVs, bond issuance, and borrower-side negotiated-credit disclosures from a commercial data vendor. The holdings data identify each product's largest disclosed positions at the individual-asset level.[8] Quarterly disclosures over 2022Q1–2025Q3 contain 2,279,252 holding-level records for 60,396 products, covering 253,484 product-quarters. The sample includes 171 banks headquartered in 30 provinces: 18 national banks (six state-owned and twelve joint-stock) and 153 local banks (100 provincial or city commercial banks and 53 rural banks). Tables A1–A3 define the variables and report sample coverage and descriptive statistics.

[8]The headline allocation and return-premium findings survive disclosure controls and alternative exposure denominators (Appendix E.3).

3 Local Credit and Competition for Savings

Local banks compete for savings against national banks with stronger reputations and broader distribution networks. A return advantage can offset that disadvantage. We begin by asking which investments provide this margin, whether it attracts savers, and why local banks seek it through familiar borrowers.

Our investment and performance outcomes describe banks' WMP businesses, not their own balance-sheet portfolios or profitability. References to a bank's holdings or product returns therefore mean those of its WMPs, whether operated directly or through a wealth-management subsidiary. We assign each WMP its underlying bank's national or local category. An investment is home when its borrower is in that bank's headquarters province; either national or local banks' WMPs can hold home and non-home assets.

Throughout the product regressions, i denotes a product, a its issuing bank or subsidiary, b the underlying bank, q a quarter, m a month, and p_b the bank's headquarters province. Monthly returns distinguish separately priced NAV series. Variable names describe what is measured; units and timing are specified with each regression. Log bank size_b always refers to fixed FY2021 bank size, whereas log bank assets in the control rows varies over time. Controls collects the covariates listed for that specification, so its contents can differ across tests. Fixed-effect subscripts identify the units held fixed; $\mathbb{1}\{\cdot\}$ indicates that a condition holds.

3.1 Overall asset allocation

We first compare LGFV bonds with negotiated credit, the two higher-yielding WMP investments central to the analysis. To place them on a common basis, we express each holding as a share of the combined value of identified bonds and negotiated credit with known borrower

locations, excluding central-government, central-bank, and policy-bank obligations. These shares describe this part of the WMP portfolio, not total WMP assets.[9]

The initial WMP portfolios differ sharply by bank type. Within these bond and negotiated-credit holdings, LGFV bonds accounted for 43.1% at local banks, compared with 12.6% at national banks; negotiated credit accounted for 9.3% and 24.1%, respectively (Table D1). Local banks' WMPs thus relied more on marketable LGFV bonds, whereas national banks' WMPs already made substantial use of privately negotiated claims.

The bank-type comparison leaves open how these investments vary with size. We therefore estimate the initial size gradient across national and local banks:

$$\text{Asset share}_{iq} = \beta_S \,\text{Log bank size}_b + \gamma' \,\text{Controls}_{iq} + \alpha_q + u_{iq}, \tag{1}$$

Here the outcome is the total LGFV bond share or total negotiated-credit share, in percentage points of the WMP bond and negotiated-credit holdings defined above. Log bank size$_b$ is log FY2021 underlying-bank interest-bearing assets in RMB trillion; bank and subsidiary products share that measure. A doubling increases Log bank size$_b$ by $\ln 2$. The ten variables in Controls$_{iq}$ cover product size, risk, and redemption terms; bank nonperforming loans, cost–income ratio, and net interest margin; and home-city GDP, growth, LGFV debt/GDP, and urban income.[10] The coefficient β_S compares allocations across bank sizes, holding these characteristics and quarter conditions fixed.

Across national and local banks, smaller banks' WMPs allocate more to LGFV bonds and less to negotiated credit (Table 1). For a bank twice as large, the estimated LGFV bond

[9]The tables refer to this denominator as included debt holdings. It also excludes deposits, funds, and other assets. The introduction's 38.5% and 13.2% instead divide LGFV bond value by identified standard-bond value in the broader disclosed sample; those bond-only shares exclude negotiated credit and retain government and policy-bank bonds. Neither comparison uses total WMP net assets, the denominator of the 10.2% industry estimate in Section 2. Appendix D.8 reconciles the samples and denominators; Table A11 reports the broader disclosed portfolio.

[10]The sample is 2022Q1–2023Q3. Category zeros remain; shares are not benchmark-adjusted. Regressions weight by included debt value and cluster by underlying bank. Fixed size replaces contemporaneous bank assets; quarter effects enter without bank effects or a local-bank indicator. Appendix D.9 reports sensitivity checks.

share is about 4.9 percentage points lower and the negotiated-credit share 4.7 points higher. The underlying coefficients are −7.13 and 6.83 points per log point of bank assets. Negotiated credit is also scarce at the smallest local banks: it accounts for only 1.1% of their WMP bond and negotiated-credit holdings, compared with 11.4% at the largest third (Table D1, Panel B). Smaller banks thus enter the program with portfolios more dependent on marketable LGFV bonds and little use of the privately arranged alternative.

3.2 Home-province allocation

The asset mix is only part of the specialization: local banks also concentrate investment in their home provinces. Home LGFV bonds account for more than one-fifth of the WMP bond and negotiated-credit holdings in every local-bank size group (Table D1, Panel B). Using the same denominator and size specification as in equation (1), we next restrict the numerator to home-province holdings. Across all banks, a bank twice as large has a home LGFV bond share about 3.8 percentage points lower; the coefficient is −5.46 points per log point of bank assets (Table 1). The greater LGFV allocation at smaller banks therefore has a substantial home-province component.

Large home holdings could simply reflect a large local bond market. We therefore compare each product's home share with its province's share of the nationwide market for the same asset class. The difference measures home investment beyond market weights. Over 2022Q1–2025Q3, we estimate

$$\begin{aligned} \text{Home bias}_{iq}^{n} = \beta_n \, \mathbb{1}\{\text{Local bank}\}_b \\ + \gamma_n' \, \text{Controls}_{iq} + \alpha_q + \varepsilon_{iq}. \end{aligned} \tag{2}$$

For asset class n, $\text{Home bias}_{iq}^{n}$ is the product's home-province portfolio share minus the home province's weight in the nationwide market portfolio for that class. Zero means the product matches the market portfolio; a positive value means it overweights home borrowers. The

local-bank indicator identifies products issued by local banks or their subsidiaries. Thus β_n measures the local–national difference in home bias.[11]

Local banks overweight home borrowers even after accounting for provincial market size (Table 2). Their home bias exceeds that of national banks by 32.0 percentage points for LGFV bonds, 16.1 points for other bonds, and 48.9 points for negotiated credit. The LGFV coefficient of 0.3205 is a 32.0-point difference because the dependent variable is a fraction. The specialization is therefore geographic as well as a choice of asset class.

Product design provides another way to see how local banks compete. Periodic-open and closed-end products account for 80.2% of their product assets, compared with 47.3% at national banks (Table A6). These products restrict withdrawals, giving banks more room to hold investments that are difficult to sell quickly. Daily-open products instead compete more directly on access to cash.

To connect withdrawal terms to the local investment strategy, we estimate

$$\begin{aligned}\text{Home allocation bias}_{iq}^{n} = {} & \beta_1 \mathbb{1}\{\text{Local bank}\}_b + \beta_2 \mathbb{1}\{\text{Restricted redemption}\}_i \\ & + \beta_3 \mathbb{1}\{\text{Local bank}\}_b \times \mathbb{1}\{\text{Restricted redemption}\}_i \\ & + \gamma' \text{Controls}_{iq} + \alpha_q + \varepsilon_{iq}.\end{aligned} \tag{3}$$

$\mathbb{1}\{\text{Restricted redemption}\}_i$ identifies products without daily withdrawals. The outcome Home allocation bias$_{iq}^{n}$ is home investment in class n divided by included debt holdings, less the corresponding provincial benchmark; category zeros remain. The coefficient β_3 asks whether the local–national allocation difference is greater in restricted-redemption products.[12]

The local–national home-allocation difference is 20.0 percentage points greater in restricted-redemption products, including 7.7 points in LGFV bonds and 7.8 in negotiated credit

[11] Only positive class holdings enter. Controls are product size, risk, withdrawal terms, disclosure count and coverage; the bank's average product term and risk; bank nonperforming loans, cost–income ratio, net interest margin; and home-city GDP growth. Errors cluster by headquarters city–quarter. Appendix D.9 defines both denominators.

[12] The controls are those in equation (1), excluding redemption, which enters separately, and adding time-varying log bank assets. The sample is 2022Q1–2025Q3. Quarter effects enter; errors cluster by bank city–quarter. Appendix D.3 gives the construction.

(Table A7). Local banks' greater use of withdrawal restrictions thus accompanies greater investment in home borrowers: their product terms and portfolios support the same return-based strategy.

The geographic and product-design evidence shows how local banks use these investments to differentiate their WMPs: they concentrate on home borrowers and offer withdrawal terms that accommodate less liquid holdings. For the smallest banks, substantial home LGFV positions coexist with little negotiated credit and few accessible high-yield alternatives. We next ask whether their reliance on home bonds delivers a return margin to savers.

3.3 Local credit and product returns

Did these portfolios help local banks offer higher returns before debt resolution? We compare product returns across otherwise similar portfolios with different asset allocations, distinguishing home LGFV bonds from bonds elsewhere and from negotiated credit. To compare initial return premiums with their subsequent changes on the same basis, we estimate one model that allows the allocation–return relationship to differ before and after implementation. This subsection focuses on the initial premiums. We relate monthly returns to holdings at the preceding quarter end:

$$\begin{aligned}\text{Monthly return}_{im} = {} & \beta' \,\text{Asset shares}_{i,h(m)} \\ & + \delta' \,\text{Asset shares}_{i,h(m)} \times \mathbb{1}\{\text{Post}\}_m \\ & + \gamma' \,\text{Controls}_{i,h(m)} + \lambda\, \mathbb{1}\{\text{Local bank}\}_b + \alpha_m + \varepsilon_{im}.\end{aligned} \tag{4}$$

The outcome Monthly return$_{im}$ is the change in log distribution-adjusted NAV, in basis points. In this specification i is a separately priced NAV series, assigned its product's holdings. The five fractions Asset shares$_{i,h(m)}$ are home LGFV bonds, home negotiated credit, home other bonds, non-home LGFV bonds, and non-home negotiated credit, relative to included debt holdings. Non-home other bonds are omitted. Holdings and controls are measured at the

preceding quarter end $h(m)$; June holdings explain July–September returns. Post starts in October 2023.[13]

Before implementation, Post equals zero, so β measures each asset's return premium relative to non-home other bonds. These are the coefficients we interpret here. The interactions δ allow the premiums to change afterward; Section 4.3 examines those changes and the remaining premiums $\beta + \delta$. For a 20-percentage-point allocation replacing non-home other bonds, the annualized comparison is 12×0.20 times the coefficient (Table 3).

Home LGFV bonds provide the largest estimated pre-program return premium (Table 3). Reallocating 20 percentage points from non-home other bonds to home LGFV bonds is associated with about 21 annual basis points more product return, compared with about 7 points for LGFV bonds from other provinces. The raw monthly coefficients are 8.852 and 3.041 per unit share. The home premium is almost three times the non-home premium, and their difference is statistically significant.[14]

Negotiated credit also has a larger return premium at home: the same 20-point allocation corresponds to about 16 annual basis points for home claims and 7 for claims elsewhere, from raw coefficients of 6.765 and 2.776. Home other bonds contribute little, with a coefficient of 0.506. This pattern ties the return margin to LGFV bonds and negotiated credit rather than to home investment generally. Home LGFV bonds also have enough weight to matter: their initial mean allocation is 26% and the 90th percentile 72% (Table A3).

We also ask whether higher reported returns came with more frequent visible losses. Using the full-period specification in equation (D.8), we relate reported volatility and downside outcomes to benchmark-adjusted home LGFV exposure, retaining the eleven controls, local-bank indicator, and month effects (Table A8). A 20-point exposure difference is associated

[13]The eleven controls add time-varying log bank assets to the ten controls in equation (1). The regression includes a local-bank indicator and month effects, with underlying-bank-clustered errors. It uses April 2022–December 2025 returns, excluding observations outside the 1st–99th percentile ranges of returns or trailing volatility. September 2025 holdings support October–December returns. These are conditional reported-return premiums, not risk-adjusted alphas. Internet Appendix E examines measurement sensitivity.

[14]The pre-program home-minus-non-home LGFV contrast is 5.811 monthly basis points per unit share (bank-clustered SE 1.756; $p = 0.0012$), calculated from the same regression and its covariance matrix. At a 20-point allocation difference, the contrast is about 14 annual basis points.

with a 0.8-percentage-point lower probability of a negative month, against a 9.9% mean. Savers therefore observed relatively smooth performance alongside the return margin.[15]

3.4 Performance and investor demand

A return margin gives banks a commercial advantage when savers direct more money to better-performing products. We test that link by ranking products' prior-quarter returns within the same risk rating and withdrawal category, including products from both local and national banks. Following the performance–flow approach in Sirri and Tufano (1998) and La Spada (2018), we estimate

$$\begin{aligned}\text{Net flow}_{iq} = &\sum_{k=2}^{5} \beta_k \mathbb{1}\{\text{Return quintile}_{i,q-1} = k\} \\ &+ \gamma' \,\text{Controls}_{i,q-1} + \alpha_{g(i,q-1),q} + \varepsilon_{iq}.\end{aligned} \tag{5}$$

The outcome Net flow$_{iq}$ is asset growth net of investment gains, as a percentage of beginning assets. Return quintile$_{i,q-1}$ is prior return quintile, with the bottom fifth omitted; $g(i, q-1)$ identifies prior risk and withdrawal categories. Category–quarter effects hold common demand conditions fixed, and Controls$_{i,q-1}$ contains the lagged covariates listed below. The coefficient β_5 is the top–bottom flow difference.[16]

Flows increase across the performance ranks (Table A10). Top-quintile products receive subsequent net inflows 12.2% of beginning assets greater than bottom-quintile products—RMB 12.2 million for a product initially holding RMB 100 million—and are 8.1 percentage points more likely to receive positive flows. Comparing products of banks in the same home province and quarter gives a similar 12.8% flow difference. This evidence explains the

[15] Exposures lag one quarter; errors cluster by bank city–month. Reported NAV fluctuations need not capture all credit risk. Appendix D.6 defines the outcomes and specification.

[16] The eleven controls and local-bank indicator enter; risk and withdrawal levels are absorbed. Product assets are dated at the flow quarter's start and other controls one quarter earlier. The sample is 2022Q3–2025Q3; errors cluster by underlying bank. Top and bottom groups differ by about one point in prior quarterly return. Appendix D.5 explains inferred assets and peer groups.

incentive behind the investment strategy: higher reported performance helps attract savings, and home LGFV bonds supplied local banks with a return margin.

3.5 Local information and borrower support

The return and flow results establish why banks value higher yield, but not why they obtain it at home. Local knowledge may improve investment choices, while a desire to support local borrowers may lead banks to accept less favorable terms. We distinguish banks in the issuer's own city from banks elsewhere in its province to examine whether both motives appear in pricing and subsequent issuer distress.

We compare the yield at issuance with CSI's public secondary-market valuation of the same bond. A lower issue price and higher yield relative to this benchmark mean better terms for the buyer; a higher price and lower yield reduce the borrower's financing cost. Our outcome, the yield concession, is the valuation spread minus the issue spread in basis points. A positive value therefore means that the issuer borrows more cheaply than the public valuation implies.

For otherwise similar bonds j issued in the same city and quarter q, we estimate

$$\begin{aligned} \text{Yield concession}_{jq} = {} & \beta_C \, \text{Same-city ownership}_{jq} \\ & + \beta_P \, \text{Other-city provincial ownership}_{jq} \\ & + \gamma' \, \text{Controls}_j + \alpha_{c(j),q} + \varepsilon_{jq}. \end{aligned} \tag{6}$$

The two ownership shares are percentage points of issue size held by same-city local banks and by local banks elsewhere in the province. Both enter together. Issuer-city–quarter effects $\alpha_{c(j),q}$ hold the local issuing environment fixed. Controls_j contains log issue amount, maturity, guarantee and call indicators, coupon, county- and prefecture-level issuer indicators, WMP holder count, and total WMP ownership.[17]

[17] Issue-quarter-end ownership proxies participation, not observed primary orders; CSI provides a valuation rather than necessarily a transaction price. The sample is 2022Q1–2025Q3. Table errors are heteroskedasticity-robust; issuer-clustered inference is also reported for the ownership contrast.

The contrast $\beta_C - \beta_P$ measures the pricing difference from shifting one point of ownership from other-city provincial banks to same-city banks, holding their combined ownership fixed. Positive values mean the closest banks accept a larger yield concession.

The two groups display opposite pricing patterns (Table A12). Bonds with greater participation by banks elsewhere in the issuer's province are issued at lower prices and higher yields relative to the benchmark, consistent with informed investors obtaining better terms. Greater same-city participation is associated with higher prices and lower yields. Shifting one percentage point of ownership from other-city provincial banks to same-city banks increases the yield concession by 1.01 basis points. Where proximity should make information particularly useful, the closest banks instead accept less favorable investment terms. This pattern points to support that lowers local borrowers' financing costs alongside the use of local information.

A favorable price for the issuer could reflect support, but pricing alone does not reveal the quality of the selected borrowers. We therefore replace the pricing outcome with Issuer distress$_{jq}$, equal to 100 when the issuer records its first credit event within eight quarters and zero otherwise. Previously undistressed issuers of 2022Q1–2023Q4 bonds have complete outcomes through 2025Q4. We retain both ownership shares, city–quarter effects, and pricing controls, adding the public valuation spread to compare issuers with similar observed ex ante risk. Coefficients are distress-probability differences in percentage points per ownership point.

Greater same-city ownership is associated with fewer subsequent issuer distress events (Table A13), and the same-city-minus-other-city contrast is negative.[18] Favorable pricing for same-city issuers therefore accompanies a lower incidence of subsequent distress, conditional on observed initial risk. The pricing and distress evidence together are consistent with local information and borrower support coexisting.

[18]The contrast is about -0.10 percentage points per ownership point, against a 2.7% mean; its issuer-clustered p-value with issuer-city-by-quarter effects is 0.132. Alternative event windows are in Table A4. Ownership is measured at issue-quarter end, not from primary orders, and distress concerns the issuer rather than default on that bond.

Selective lending also appears when borrowers face repayment pressure. Greater scheduled city-quarter LGFV maturities predict lower home net purchases and more protected new claims (Table A9; equations (D.5) and (D.6) in Appendix D.4). Home investment is therefore compatible with both borrower support and selection. Accepting a lower yield on some same-city bonds need not eliminate the portfolio-return margin from investing more in a higher-yielding asset class. We next examine what happens when government support compresses that asset class's yield.

4 Debt Resolution and the Loss of Bond Yield

The preceding evidence identifies a return-based strategy built on familiar local bonds. Debt resolution changed the income those bonds could provide. We first establish how bond yields changed relative to Treasury yields and negotiated-credit yields, then trace the change into product-return premiums and portfolio allocations.

4.1 LGFV bond-spread compression

LGFV bonds lost much of the extra yield they offered over government bonds. Among newly issued fixed-rate bonds, the average spread over the Treasury yield fell from 200 basis points in 2022Q1–2023Q3 to 97 basis points in 2023Q4–2025Q4, roughly halving. Average LGFV issuance yields fell from 4.18% to 2.70%, while the corresponding Treasury yields fell from 2.18% to 1.72%. Thus, the yield available from LGFV bonds declined by substantially more than the general fall in government borrowing rates.[19]

[19]These are descriptive, equal-issue-weighted averages for 7,128 pre-implementation and 9,763 post-implementation LGFV issues in the bond-pricing database, without conditioning on WMP ownership. This sample differs from the issuer-type comparison in Table A15. We retain fixed-rate, at-par, nonstructured, nonconvertible, nonperpetual, nonsubordinated bonds, so the reported issue rate measures yield at issuance. Each issue's spread subtracts the Treasury-curve yield on its issue date at the nearest available maturity node. The before period includes the announcement quarter, consistent with implementation beginning in 2023Q4; these averages are not a composition-adjusted policy estimate.

The regional gap also narrowed. Provinces in the highest and lowest quintiles of initial spreads differed by 564 basis points in 2023Q1 but only 64 by 2025Q4 (Figure A4). Keeping group membership fixed, almost nine-tenths of the gap disappeared: formerly high-spread regions supplied much less additional yield for reinvestment.

Was compression greater where the program refinanced more debt, and when did that difference emerge? We estimate a quarterly event study that compares provinces with different refinancing intensity:

$$\begin{aligned}\text{Bond spread}_{pq} = \alpha_p + \alpha_q \\ + \sum_{k \neq q_0} \beta_k \, \text{Final refinancing}_p^{GDP} \mathbb{1}\{q = k\} + \varepsilon_{pq},\end{aligned} \qquad q_0 = 2023\text{Q2}. \tag{7}$$

Bond spread_{pq} is province p's secondary-market spread in basis points. Final intensity $\text{Final refinancing}_p^{GDP}$ is cumulative special-refinancing bond issuance at the last observed date, as a percentage of 2020 provincial GDP. It is fixed across quarters. Province and quarter effects remove persistent provincial differences and common movements. The plotted β_k measures additional spread change since 2023Q2 per percentage point of final intensity (Figure 1).[20]

The event study links the timing of compression to refinancing intensity. Pre-announcement estimates are imprecise around zero; a province with one percentage point higher final intensity experiences 6.5 basis points more compression by 2023Q4 and 18.6 by 2025Q4. The difference deepens through 2024, indicating a persistent reduction in the yield available from formerly higher-spread regions.

The cumulative-intensity regression asks the same economic question using refinancing completed by each date rather than a fixed final exposure. With province and quarter effects, its coefficient is -17.67 basis points per percentage point of refinancing/GDP (Table A5). Allowing initially high spreads to converge separately, or each province its own trend, reduces

[20]The sample is 2022Q1–2026Q1; errors cluster by province. Spreads and intensity are winsorized at the 1st and 99th percentiles. Intensity levels are absorbed by province effects. Equation (D.1) uses cumulative refinancing instead of fixed final intensity.

the magnitude but preserves the negative association. Both comparisons connect greater refinancing to greater loss of bond yield.

4.2 Contractual yields on alternative claims

Replacing lost bond income is possible only if another accessible instrument retains yield. We therefore compare new LGFV bonds with new negotiated contracts (Figure 2). From 2023Q2 to 2024Q4, average negotiated-credit yields fall from 6.6% to 5.9%, while bond yields fall from 4.6% to 2.7%, widening the gap from about 2.0 to 3.2 percentage points. To see whether this difference reflects the financing instrument rather than different borrowers or maturities, we next hold the legal borrower and financing date fixed and control for tenor.

We use an instrument-by-quarter event study to compare the yield gap before and after the announcement. For contract or bond issue j of borrower f in month m, we estimate

$$\begin{aligned}\text{Yield}_{jfm} = \alpha_{fm} + \sum_q \beta_q \mathbb{1}\{\text{Negotiated credit}\}_j \mathbb{1}\{q(m) = q\} \\ + \sum_q \gamma_q \,\text{Maturity}_j \mathbb{1}\{q(m) = q\} + e_{jfm}.\end{aligned} \tag{8}$$

The annual outcome Yield_{jfm} is in percentage points, $\mathbb{1}\{\text{Negotiated credit}\}_j$ identifies negotiated credit, and Maturity_j is maturity in years. Borrower–month effects compare the same legal borrower's two financing channels, excluding affiliates; quarter-specific maturity slopes adjust tenor. Thus β_q measures the quarterly NS-minus-bond yield gap, not a change in borrower composition.[21]

To present the event study relative to 2023Q2, Figure 3 plots

$$\delta_q = \beta_q - \beta_{q_0}, \qquad q_0 = 2023\text{Q2}, \qquad \delta_{q_0} = 0. \tag{9}$$

[21]There are 17,303 observations for 589 borrowers over 2021Q1–2025Q3. Errors cluster by borrower province. Mixed-channel borrower-months identify the direct comparison; other months inform maturity controls. Appendix D.2 gives the matching rules.

The same-borrower comparison confirms that negotiated credit retains more yield. Zero on the plot means an unchanged gap from the 1.90 percentage points in 2023Q2, not equal yields. The gap widens from the second half of 2023, reaching 2.86 points in 2024Q4—0.96 points above the reference—and remaining 1.06 points above it in 2025Q3. Confidence intervals use the covariance of both gap estimates. The substitute becomes more attractive relative to bonds because its yield falls less, not because its own yield must rise.

Alternative maturity controls, financing-cell weights, a fixed borrower cohort, and inference methods retain the later widening (Tables D4–D5). Pre-announcement movements, especially the 2023Q1 trough, remain: a fixed-borrower rate-level comparison attributes that trough to a temporary rise in bond yields, not a collapse in negotiated-credit yields (Figure D1; Appendix D.2).

A broader reporting-issuer comparison provides a second check on the financing-channel difference. Within issuer and origination quarter, the negotiated-credit-minus-bond gap widens by 1.30 percentage points between 2022 and 2024, controlling for maturity and persistent instrument differences. On common issuer-years, it widens by an additional 1.04 points in the twelve priority provinces relative to elsewhere (Table 4; equations (D.2) and (D.4)). This sample can include affiliates, unlike the exact-borrower comparison. The common finding is that an alternative instrument retained yield within the local borrower sector; whether banks used it is a separate question.

4.3 Compression of the bond-return premium

Contractual yields measure the income available on new financing. We now ask whether the bond-yield decline also reduced the return margin that WMPs offered savers, and how that loss compares across investment categories. Section 3.3 established the initial premiums using β in equation (4); we now turn to the Post interactions δ in that same regression. Each interaction measures the change in an asset's premium, and $\beta + \delta$ gives its post-program

premium. We retain the same annualized comparison for a 20-point allocation replacing non-home other bonds (Table 3).

Home LGFV bonds lose the most of the initial return margin. Their annual premium falls from 21.2 to 6.5 basis points for the common 20-point allocation: a loss of 14.7 basis points, or 69%. The corresponding non-home LGFV premium falls from 7.3 to 1.9 basis points, a loss of 5.4. Home bonds therefore lose about 9.3 basis points more than LGFV bonds elsewhere. The asset supplying the strongest initial home return advantage also experiences the largest erosion.[22]

Negotiated credit retained more of its contribution to product returns. Its home premium fell from 16.2 to 10.5 annual basis points, a loss of 5.7; its non-home premium rose from 6.7 to 10.0. Home other bonds had little premium in either period, at 1.2 and 1.4 basis points. The decline in the home LGFV premium is statistically larger than the change in each of these categories.[23] In particular, home LGFV bonds lost about 9.0 annual basis points more than home negotiated credit. This difference matters for substitution: negotiated credit still offered a positive return premium from local borrowers after much of the bond premium had disappeared.

Is the disappearing premium related to the income on the bonds, rather than merely a change in the products holding them? We keep the rate-observed sample fixed and add the portfolio's holdings-weighted original bond issue yield and its Post interaction to the return regression. The annual home-LGFV premium loss falls from about 15 to 6 basis points for a 20-point allocation difference (Table E1). Accounting for contractual bond income thus absorbs about three-fifths of the estimated decline on the same observations.

[22]The raw monthly coefficients for home LGFV exposure are 8.852 before implementation and -6.133 for its Post interaction. Their sum gives the remaining premium; multiplying each comparison by 0.20×12 gives the annualized 20-point allocation comparison.

[23]Bank-clustered tests using the regression's full covariance matrix give $p = 0.014$ for the difference in changes between home and non-home LGFV bonds, $p = 0.015$ between home LGFV bonds and home negotiated credit, $p = 0.002$ against home other bonds, and $p < 0.001$ against non-home negotiated credit. The 20-point comparisons multiply the raw monthly coefficients by 0.20×12; they do not standardize the regressors.

We next ask whether the premium loss appears only because monthly NAVs recognize gains and losses at different times. Holding exposures fixed before complete six- and twelve-month performance windows, the annualized home-LGFV premium loss remains about 16 and 13 basis points, while home negotiated credit retains premiums of about 13 and 12 (Table E2). Internet Appendix E gives the starting-exposure specification, window-date effects, bank clustering, and measurement checks. Measuring performance over longer windows preserves the distinction between lost bond income and the return still associated with the substitute.

Together, the financing and product-return comparisons establish the change in investment opportunities: home LGFV bonds lost much of their yield contribution, while negotiated credit retained a return premium. We next ask how banks adjusted the bond positions that had supported their strategy.

4.4 Initial reliance and bond retreat

If the lost yield mattered for portfolio choice, institutions initially more oriented toward home investments should reduce home bonds more as refinancing proceeds. We compare local banks within the same province and quarter:

$$\begin{aligned}\text{Home LGFV bond share}_{iq} = \beta\, \text{Initial home bias}_a \times \text{Refinancing}^{debt}_{p_b,q} \\ + \gamma' \,\text{Controls}_{iq} + \alpha_a + \alpha_{p_b,q} + \varepsilon_{iq}.\end{aligned} \tag{10}$$

The outcome is the home LGFV bond share in percentage points. Initial reliance Initial home bias$_a$ is the issuing institution's product-size-weighted 2022Q1–2023Q3 home tilt across bonds and negotiated credit, net of provincial market shares, not simply its LGFV share. Refinancing$^{debt}_{p_b,q}$ is cumulative special-refinancing bond issuance through quarter q divided by 2020 LGFV debt in the underlying bank's home province. Institution

effects absorb fixed reliance; province–quarter effects absorb intensity and common regional conditions. A negative β indicates greater bond retreat by initially more reliant institutions.[24]

Initially more reliant banks cut their home-bond allocations more (Table 5). For two banks in the same province differing by ten percentage points in prior reliance, a ten-point increase in refinancing relative to pre-program LGFV debt is associated with an additional 0.46-percentage-point reduction in home LGFV bond allocation at the more reliant bank (province wild-bootstrap $p = 0.010$). Measuring reliance strictly before the July announcement or allowing it separate quarterly paths gives similar estimates. Banks whose portfolios depended more on local yield therefore adjusted more strongly as that yield declined.

4.5 The geography of bond retreat

A fall in home holdings could reflect less local issuance as well as a change in banks' demand. We distinguish these channels by comparing home and outside buyers of the same issuer city's bonds at the same date:

$$\begin{aligned}\text{LGFV bond share}_{acq} = \beta\, \mathbb{1}\{\text{Home buyer}\}_{ac} \times \mathbb{1}\{\text{Post}\}_q \\ + \mu_{ac} + \mu_{aq} + \mu_{cq} + \varepsilon_{acq}.\end{aligned} \tag{11}$$

Here a buys LGFV bonds from issuer city c in quarter q. The outcome is that city's bonds as a percentage of the buyer's total bonds. Home identifies a local bank investing in its headquarters province. Buyer–city effects remove persistent relationships, buyer–quarter effects bank-wide changes, and city–quarter effects local bond-supply conditions. Thus β measures home buyers' additional allocation change relative to outside buyers of the same city's bonds.[25]

[24] Both Initial home bias$_a$ and Refinancing$^{debt}_{p_b,q}$ are fractions. The eleven time-varying controls enter. Errors and wild-bootstrap draws cluster by home province. Unlike the spread tests, intensity uses an LGFV-debt denominator, not GDP.

[25] Home and Post levels and bank/city controls are absorbed. Other outcomes are net holdings changes relative to current bond holdings and log one plus new purchases in RMB million. Errors cluster by bank city; Appendix D.4 gives construction details.

Home banks retreat more than outside investors in the same city market (Table 6). Their allocation declines by 0.45 percentage points of the bank's bond portfolio per issuer city, about 12% of the initial mean for home bank-city pairs. Replacing Home with separate own-city and rest-of-home-province indicators shows that the decline is greatest in the bank's own city: 1.94 percentage points, compared with 0.24 elsewhere in its province. Net holdings changes and new purchases also fall. In the new-purchase regression, interacting Home buyer × Post with the city's pre-program LGFV spread, in percentage points, shows a larger purchase reduction in formerly higher-spread cities. The retreat is therefore concentrated in the local positions that had supplied higher yield and does not simply reflect less local issuance.

We then estimate an event study to establish when home-bond retreat developed. We aggregate issuer cities to provinces and replace the single Home–Post interaction with Home interacted with each quarter:

$$\begin{aligned}\text{LGFV bond share}_{apq} = &\sum_{k \neq q_0} \eta_k \, \mathbb{1}\{\text{Home buyer}\}_{ap} \mathbb{1}\{q = k\} \qquad q_0 = 2023\text{Q2}. \\ &+ \mu_{ap} + \mu_{aq} + \mu_{pq} + u_{apq},\end{aligned} \tag{12}$$

Province p is the destination of the investment. LGFV bond share$_{apq}$ measures its bonds relative to the buyer's total bonds, and Home identifies the buyer's own province. Buyer–province, buyer–quarter, and destination-province–quarter effects are the provincial counterparts of the city design. Each plotted η_k is the change in the home–outside allocation gap since 2023Q2, not a holding level (Figure 4).[26]

The event study shows home-bond retreat building as the lower-yield environment persists. Pre-announcement confidence intervals include zero; relative to outside buyers, the home allocation falls by about 2.3 percentage points in 2024Q1, 4.2 by 2024Q4, and 6.1 by 2025Q3. These are province-wide holdings changes, not the per-city average in Table 6. The gradual decline can include nonrenewal of maturing bonds as well as sales. Local banks thus pull

[26]The sample is 2022Q1–2025Q3; outcomes are winsorized at the 1st and 99th percentiles and errors cluster by bank city. Home levels and time-varying bank/province controls are absorbed.

back from the familiar investments whose return advantage had diminished. We turn next to which banks found a replacement.

5 Bank Size, Negotiated Credit, and Competition for Savings

Negotiated credit offered a replacement for lost bond income, but arranging these claims required more than buying a marketable bond. We now ask whether scale distinguished the banks that used this substitute, and how the investment response related to the returns offered to savers and the scale of the WMP business.

5.1 Which local banks shifted into negotiated credit

We first locate the shift across bank types and borrower geography (Table 7). Before implementation, national banks held 82.7% of the identified negotiated-credit value in our sample, mostly outside their headquarters provinces. Local banks' share subsequently rose from 17.3% to 31.3%. Within local banks' portfolios, home negotiated credit increased from 6.1% to 13.9% of included debt holdings, more than the increase outside their provinces. The higher-yielding instrument previously concentrated at national banks thus became a more important way for local banks to finance home borrowers.[27]

Splitting local banks by their pre-program size reveals who made the switch. Among banks observed in both periods, the largest third increased home negotiated credit from 7.45% to 18.26% of included debt holdings. This increase more than offset their decline in home LGFV bonds: together, the two investments rose from 28.1% to 33.2%. At the smallest third, home negotiated credit remained negligible, falling from 0.72% to 0.31%, while the combined

[27]These are descriptive, holding-value-weighted shares over 2022Q1–2023Q3 and 2023Q4–2025Q3. The national–local percentages describe each group's share of all negotiated credit in the sample; the portfolio percentages divide home negotiated credit by each group's included bonds and negotiated credit. They answer different questions and have different denominators.

share fell from 25.8% to 21.1%. The middle group substituted modestly. Larger banks therefore replaced bonds with negotiated contracts, whereas the smallest banks reduced home investments without building a comparable substitute. The size groups use fixed FY2021 assets, so these differences do not arise from reclassifying banks after the program.

To establish what larger banks were substituting into, we match the negotiated borrowers to LGFV bond issuers and financing-platform names. These matches account for 85% of the largest local banks' post-program home negotiated-credit value (Internet Appendix D.1). The shift was thus largely from bonds to less transparent financing within the LGFV sector, not a wholesale move away from familiar local borrowers.[28]

5.2 Bank size and the shift into negotiated credit

The portfolio averages show where the substitute was used. We next ask whether the size difference widened after accounting for observed product, bank, and local economic characteristics. For local-bank product-quarters, we estimate

$$\begin{aligned}\text{Negotiated-credit share}_{iq} = \beta_S \,\text{Log bank size}_b + \delta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_q \\ + \gamma' \,\text{Controls}_{iq} + \alpha_q + u_{iq}.\end{aligned} \tag{13}$$

where Negotiated-credit share$_{iq}$ is product i's total or home negotiated-credit share in quarter q, in percentage points of included holdings. Log bank size$_b$ is its underlying bank's log FY2021 assets in RMB trillion, without standardization. The ten controls Controls$_{iq}$ are those in equation (1); quarter effects α_q remove common movements. The coefficient β_S measures how investment differed with bank size before implementation. The coefficient of interest, δ_S, measures whether larger banks increased investment more afterward. Their sum gives the size difference remaining after implementation (Table D2).[29]

[28] These matches identify sector-level substitution, not the rollover of each bond into a negotiated contract with the same legal entity.

[29] The sample is 2022Q1–2025Q3. Observations are weighted by included debt value, and errors cluster by underlying bank. Quarter effects absorb the Post level. Bank-fixed-effect versions absorb fixed size and retain contemporaneous bank assets as a control. No standard-deviation normalization is used.

The size gradient widens after implementation. The home negotiated-credit slope rises from 4.72 to 12.18 percentage points per log point of assets, a change of 7.47 points. A local bank twice as large therefore increases its home negotiated-credit share by about 5.2 points more. The widening is significant under bank wild-bootstrap inference and remains positive with bank fixed effects. Larger banks do not simply start with more negotiated credit; they expand its portfolio weight disproportionately after bond yield compresses.

A bank with more home investments may have more yield to replace. We distinguish that initial reliance from investment scale by comparing local banks in the same province and quarter, allowing both to affect their response to refinancing:

$$\begin{aligned}\text{Negotiated-credit share}_{iq} = {} & \beta_S \,\text{Log bank size}_b \times \text{Refinancing}^{debt}_{p_b,q} \\ & + \beta_B \,\text{Initial home bias}_a \times \text{Refinancing}^{debt}_{p_b,q} \\ & + \gamma' \,\text{Controls}_{iq} + \alpha_a + \alpha_{p_b,q} + \varepsilon_{iq}.\end{aligned} \tag{14}$$

The outcome Negotiated-credit share$_{iq}$ is product i's home or non-home negotiated credit as a percentage of its included bonds and negotiated credit. Bank b supplies the fixed size measure; issuing institution a supplies the initial reliance measure Initial home bias$_a$, defined in Section 4.4. Refinancing is matched to the underlying bank's headquarters province p_b, not the borrower's province. Institution effects remove persistent portfolio differences, and home-province-by-quarter effects hold regional conditions and the amount refinanced at that date fixed. The two interactions ask which banks respond more to that common environment: those with greater initial reliance, and, separately, those with greater size. A positive β_S means greater substitution by larger banks even after accounting for initial reliance.[30]

Size continues to distinguish the investment response after holding initial reliance and common provincial conditions fixed. As cumulative refinancing rises by ten percentage points

[30]The eleven time-varying controls enter. Observations are weighted by included debt value; errors cluster at the bank-city-by-quarter level. Institution effects absorb the fixed size and reliance levels, while home-province-by-quarter effects absorb the intensity level. Replacing intensity with Post in both interactions gives a before–after comparison conditional on initial reliance and the same regional time effects.

of initial provincial LGFV debt, a bank twice as large increases its home negotiated-credit allocation by about 1.7 points more (Table 8, Panel A). Replacing refinancing intensity with Post gives about 4.1 points more expansion after implementation (Panel B). The first comparison follows the progress of refinancing; the second summarizes the post-program period. Both locate greater substitution at larger local banks.[31]

Does the instrument switch simply offset bond retreat, or leave larger banks with a greater combined allocation to home borrowers? We replace the outcome with the combined share of home LGFV bonds and home negotiated credit, retaining size–Post and reliance–Post interactions, institution and regional time effects, and the controls. The size–Post coefficient is 4.76 (Table A14): a bank twice as large increases its combined home allocation by about 3.3 percentage points more. Larger banks therefore retain a greater portfolio share in the home borrower sector as they change instruments, while smaller banks make less use of the replacement.

These comparisons use holdings-value weights to describe the concentration of investment amounts. Internet Appendix D.9 examines alternative weights, and Appendix E.2 studies pre-existing organizational capacity. These checks distinguish participation in negotiated credit from the size of positions among existing users.

5.3 New products

Existing portfolios may adjust slowly because of their maturity structure or withdrawal terms. New products let banks choose a fresh portfolio. We therefore compare the first eligible holdings of WMPs launched in 2023Q4–2025Q3 across the same fixed size groups, asking

[31] The raw size–intensity coefficient is 24.21 per log point of assets and unit of the debt-based refinancing fraction, giving $24.21 \times 0.10 \times \ln 2 = 1.68$ percentage points. The raw size–Post coefficient is 5.85, giving $5.85 \times \ln 2 = 4.06$ points. These are conditional on initial reliance and regional time effects, unlike the simpler quarter-effect comparison in Table D2.

both whether negotiated credit appears at all and how much of the new portfolio it occupies. This separates limited use of the instrument from an inability to launch products.[32]

The smallest banks did launch new products, but almost none used the substitute. Only two of their 253 new products held home negotiated credit, compared with about 21% at the largest banks (1,118 of 5,248 products; Table 9). Home negotiated credit represented just 0.78% of the smallest banks' new-product debt holdings, versus 27.5% at the largest banks. Home LGFV bonds still accounted for 29.4% at the smallest banks. Thus, their new portfolios continued to use the accessible bond investment rather than taking up negotiated credit. Requiring disclosure in the launch quarter or the next quarter gives the same pattern.

The difference survives a fresh choice of portfolio, consistent with a bank-level cost of sourcing and arranging negotiated credit that launching a new product alone does not remove.

5.4 Relative product returns

Did the banks that depended more on home LGFV bonds lose more return, even when they were similar in size? We fix exposure before the announcement so subsequent portfolio adjustments do not redefine initial reliance. Let Initial home LGFV share$_b$ be bank b's home-province LGFV standard-bond share of included disclosed debt during 2022Q1–2023Q2. This is its actual bond allocation, distinct from the broader benchmark-adjusted Initial home bias$_a$. Let Annualized return$_{bm}$ be its average monthly WMP return in annualized basis points. We estimate

$$\begin{aligned}\text{Annualized return}_{bm} = {} & \beta_L \,\text{Initial home LGFV share}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_m + \gamma' \,\text{Controls}_{b,h(m)} \\ & + \alpha_b + \alpha_{g_b,m} + u_{bm}.\end{aligned} \tag{15}$$

[32] Launch is identified by the income-start date rather than first disclosure. Seven products disclosed before their recorded start date are excluded. Banks retain their FY2021 size groups. Portfolio shares are holding-value weighted; product counts are unweighted.

Here Log bank size$_b$ is fixed FY2021 log bank size and g_b is pre-program subsidiary status. Bank effects remove fixed differences; status-by-month effects allow the two groups separate time paths. Where included, Controls$_{b,h(m)}$ contains eight prior-quarter bank and home-city controls.[33]

Initial bond reliance predicts a larger loss of product return independently of size (Table E7). Among two equally sized banks differing by ten percentage points in initial home-LGFV allocation, the more exposed bank experiences about 13.2 basis points more annualized return decline (wild $p < 0.001$), or 12.6 points with the eight controls ($p = 0.003$). The result also survives province-by-month effects and separate size slopes in every quarter. The return loss is thus greater at banks that had depended more on the affected investment, not only at banks that happened to be small.

We next ask whether larger banks also preserved more of the returns offered to savers. Holding initial reliance on home investments constant lets us compare the return pattern with the earlier size difference in substitution.

We apply the allocation design in equation (14) to monthly returns, adjusting the timing of controls and regional effects accordingly:

$$\begin{aligned} \text{Monthly return}_{im} = {} & \beta_S \,\text{Log bank size}_b \times \text{Refinancing}^{debt}_{p_b,m} \\ & + \beta_B \,\text{Initial home bias}_a \times \text{Refinancing}^{debt}_{p_b,m} \\ & + \gamma' \,\text{Controls}_{i,h(m)} + \alpha_a + \alpha_{p_b,m} + \varepsilon_{im}. \end{aligned} \tag{16}$$

The size measure Log bank size$_b$, prior reliance Initial home bias$_a$, and home-province intensity retain their earlier definitions. The outcome Monthly return$_{im}$ is in basis points, and the eleven controls are measured at the preceding quarter end $h(m)$. The quarterly intensity series is assigned to the months of its quarter. Issuing-institution effects remove persistent

[33] We average NAV series within product-month, then products within bank-month, and annualize monthly log returns. The sample is July2022–September2025, excluding July–September2023; Post begins in October2023. Initial exposure is holdings-value weighted; returns give each bank-month equal weight. Errors and wild-bootstrap draws cluster by bank. Appendix E.4 gives the measurement checks.

performance differences; home-province-by-month effects compare banks facing the same regional conditions. A second version replaces intensity with $\mathbb{1}\{\text{Post}\}_m$ in both interactions. In each version, β_S measures the widening of the size-related return gap, conditional on prior reliance.[34]

The relative return gap widens alongside unequal substitution. After implementation, the annual product-return difference between a bank and one twice as large widens by about 10.3 basis points (Table 8, Panel B). The intensity version gives the same direction: when refinancing rises by ten percentage points of initial provincial LGFV debt, the annual return difference associated with a doubling of bank size widens by about 11.6 basis points (Panel A).[35] For perspective, applying the lost home-LGFV premium to local banks' average initial holdings implies about 19 annual basis points of lost return. The return gap is therefore economically meaningful relative to the bond-income advantage that disappeared.

When did smaller banks begin losing ground in returns? We estimate a size-by-quarter event study for local-bank products:

$$\begin{aligned}\text{Monthly return}_{im} = &\sum_{k \neq q_0} \theta_k \,\text{Log bank size}_b \mathbb{1}\{q(m) = k\} \\ &+ \gamma' \,\text{Controls}_{i,h(m)} + \alpha_a + \alpha_{p_b,m} + \varepsilon_{im},\end{aligned} \qquad q_0 = 2023\text{Q3}. \tag{17}$$

The outcome, size measure, controls, and fixed effects are the same as in equation (16); $q(m)$ identifies the quarter containing month m. The event study replaces the two refinancing interactions with size interacted with quarter indicators, without prior-reliance interactions.

[34]These regressions are unweighted, and errors cluster at the bank-city-by-month level. Issuing-institution effects absorb fixed size and reliance; province-by-month effects absorb intensity or Post levels. A positive size interaction means that larger banks' returns improve relative to smaller banks' returns, not that all banks' returns rise.

[35]The raw size–Post coefficient is 1.2345 monthly basis points per log point of assets, giving $1.2345 \times \ln 2 \times 12 = 10.3$ annual basis points. The size–intensity coefficient is 13.93, giving $13.93 \times 0.10 \times \ln 2 \times 12 = 11.6$ annual basis points. A 0.981-log-point standard deviation instead gives 14.5 and 16.4 annual basis points, calculated from the unstandardized coefficients.

Figure 5 plots θ_k: the change in the size–return relationship from 2023Q3, per log point of assets.[36]

The event study shows the return gap developing gradually rather than jumping at the announcement. There is a short-lived positive estimate in 2022Q2, followed by pre-program fluctuations. The sustained widening begins in 2024Q3 and remains positive through 2025Q3. For a bank twice as large, the plotted changes imply about 2.8 monthly basis points more relative return in 2024Q3 and 3.8 in 2025Q3.[37] This timing fits a reinvestment channel: falling yields can initially raise the prices of existing bonds, while the lower income available on new investments becomes more important as old holdings mature. The later return separation accompanies the unequal shift into negotiated credit as portfolios are reinvested.

5.5 WMP balances

Did the difference in returns also appear in the scale of banks' WMP business? We compare total outstanding WMP balances, combining an underlying bank's own products and its subsidiary's products, including new and continuing products. Unlike the product-flow test in Section 3, this outcome measures the size of the whole WMP business. It asks whether smaller banks lost ground commercially as well as in product performance.

For local bank b at half-year end s, we estimate

$$\begin{aligned}\text{Log WMP balances}_{bs} = \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_s \\ + \gamma' \,\text{Controls}_{bs} + \alpha_b + \alpha_s + \varepsilon_{bs}.\end{aligned} \tag{18}$$

where $\text{Log WMP balances}_{bs}$ is log total WMP balances for local bank b at half-year end s. Size is log FY2021 bank assets in RMB trillion, and Post begins with the December 2023 report. Bank effects α_b remove persistent balance differences, so the test does not simply ask

[36] The sample covers 2022Q2–2025Q3. Observations are unweighted and errors cluster by underlying bank. Fixed size is absorbed by institution effects, and quarter indicators by province–month effects. Unlike the average comparisons in Table 8, this event study traces the broader conditional size–return relationship.

[37] The corresponding raw coefficients are 3.98 and 5.46 monthly basis points per log point of assets; multiplying by $\ln 2$ gives the doubling comparisons.

whether large banks have more WMP assets. Half-year effects α_s remove common growth. The coefficient β_S asks whether the proportional balance change after implementation is greater at initially larger banks. The baseline omits Controls_{bs}; subsequent columns add initial reliance, bank and home-city controls, or province-by-half-year effects.[38]

Smaller banks' weaker relative performance is accompanied by weaker WMP-business growth (Table 10). A bank twice as large experiences about 11% greater proportional WMP-balance growth after implementation, relative to the initial difference. The raw size–Post coefficient is 0.152 log balance points per log point of assets. Comparing banks within the same province and half-year gives 0.144, implying about 10.5% greater growth for a doubling of size. The result also remains with controls for initial reliance or bank and home-city characteristics. Across the fixed size groups, the largest third grows about 40% more than the smallest. The difference concerns growth after implementation, holding persistent bank-level differences fixed, rather than the larger banks' initial scale.[39]

A fixed-bank comparison makes the contraction at the smallest institutions concrete. Among the same 19 banks reporting in June 2023 and December 2024, aggregate WMP balances fall by 14.1%, with declines at 18 banks. Replacing balances with log product counts in the same bank-and-half-year design also shows greater expansion at larger banks (column 6). The business divergence includes differences in the number of products offered, not just the value of an unchanged set of products.

[38]The panel covers 2022H1–2025H2. Fixed size is absorbed by bank effects and Post by time effects. The baseline has no additional controls because the bank-level panel is sparse; errors cluster by bank. For an asset ratio a, the implied relative growth difference is $100[\exp(\beta_S \log a) - 1]\%$. Missing reports are not zeros. Balances include investment returns, product entry and maturity, and investor transactions; they measure WMP business scale rather than individual households switching banks. Internet Appendix E.1 gives coverage and measurement details.

[39]For a 0.981-log-point standard deviation of size, the baseline and province-time estimates imply about 16% and 15% greater relative growth. These are calculated from the raw coefficients; size is not standardized.

To establish when smaller banks lost ground in WMP business, we estimate a second size event study:

$$\text{Log WMP balances}_{bs} = \sum_{\ell \neq s_0} \psi_\ell \, \text{Log bank size}_b \mathbb{1}\{s = \ell\} + \alpha_b + \alpha_s + u_{bs}, \qquad s_0 = 2023\text{H1}. \tag{19}$$

Bank b is observed at half-year reporting date s, and ℓ indexes those dates. Log bank size$_b$ is the same fixed log FY2021 size measure. The event study replaces the size–Post term in equation (18) with size interacted with each reporting date, retaining bank and half-year effects and bank-clustered errors, without additional controls. Figure 6 plots ψ_ℓ, the change in the size–balance relationship relative to June 2023. Zero means an unchanged relationship, not equal balances at large and small banks. These are half-year estimates over 2022H1–2025H2: the quarterly axis places actual reports at Q2 and Q4, without interpolating Q1 or Q3 balances.

The two 2022 estimates are close to zero, and little of the later separation is present by December 2023. The size-related growth gap then increases during 2024 and is larger in 2025. For a bank twice as large, the path implies roughly 10% greater relative growth by December 2024 and 26% by December 2025, compared with June 2023.[40] The late-period difference exceeds the average 11% Post comparison because that average also includes the earlier reports when the gap was much smaller. The event study thus shows a growing business difference, broadly alongside the later separation in product returns.

We check whether this business divergence depends on who reports or on organizational form. Table E3 retains the same 29 banks at every date, separately excludes the final period, and allows banks with different organizational forms separate time paths. The positive size-related growth difference remains, as it does separately below and at or above RMB50 billion of June 2023 WMP balances. The result therefore extends beyond the comparison

[40] The raw log-balance coefficients are 0.141 and 0.336 per log point of assets. The doubling comparisons are $100[\exp(0.141 \ln 2) - 1] \approx 10\%$ and $100[\exp(0.336 \ln 2) - 1] \approx 26\%$. The joint test for the two 2022 coefficients gives $p = 0.41$; Internet Appendix E.1 discusses the longer pre-history.

across this cutoff or between subsidiary and other banks. Appendix E.1 gives the sample definitions and results.

The evidence distinguishes reliance on an investment from the ability to replace it. Initial home-LGFV exposure predicts return losses conditional on size; larger local banks make greater use of negotiated credit and lose less ground in product returns. The WMP-balance results show that smaller banks also fall behind in business growth. We now develop a model that connects the incentive to compete on yield to the scale needed to cover the cost of arranging a substitute.

6 A Model of Unequal Adjustment

The model links two choices: which bank a saver uses and which investments a bank offers. A weaker reputation makes yield more important for winning savers, but a smaller investor base also limits the fee income available to cover the fixed cost of entering negotiated credit. This combination explains why bond-yield compression can weaken smaller banks even when a higher-yielding substitute exists. We take the documented home orientation of local investment as given and derive how investor sorting and entry costs shape the response.[41] Internet Appendix B derives the conditions and computes an illustrative full equilibrium; Internet Appendix C gives proofs.

6.1 Savers and banks

A unit mass of savers each invests one unit with a bank or an outside saving option. Savers differ in their risk aversion $\gamma_h > 0$, which has distribution G across savers. Saver h obtains utility

$$U_{hb} = \gamma_h a_b + \beta\left(R_b - \frac{\gamma_h}{2} V_b\right) + \varepsilon_{hb} \tag{20}$$

[41] The two layers of demand follow Darmouni et al. (2024): savers choose institutions, which choose assets. Heterogeneous saver demand draws on Xiao (2020) and random-coefficient demand models (Berry et al. 1995); fee-based portfolio incentives follow La Spada (2018).

from bank b, and utility ε_{h0} from the outside option. Here R_b and V_b are the expected excess return and the variance of the bank's products, defined below, so $R_b - \frac{\gamma_h}{2} V_b$ is the risk-adjusted return to saver h, and $\beta > 0$ measures how strongly savers' choices respond to it; the flow evidence in Section 3 corresponds to $\beta > 0$. The term a_b is the bank's reputation and distribution network. A more risk-averse saver cares more about the safety of the institution holding the savings and so values reputation more. For simplicity, we set the saver's weight on reputation equal to risk aversion. The term ε_{hb} is saver h's own taste for bank b, such as a nearby branch or an existing account, independent across savers and banks as in the standard logit model of demand.[42] Among savers with risk aversion γ, the share choosing bank b is

$$s_b(\gamma) = \frac{\exp\left(\gamma a_b + \beta(R_b - \gamma V_b/2)\right)}{1 + \sum_j \exp\left(\gamma a_j + \beta(R_j - \gamma V_j/2)\right)}, \tag{21}$$

and the bank's market share is $s_b = \int s_b(\gamma)\, dG(\gamma)$. There are three banks that differ in reputation: a national bank N, a large local bank L, and a small local bank S, with $a_N > a_L > a_S$. Holding portfolios fixed, a stronger reputation attracts more savers. Market share measures scale in the model; the empirical comparisons use the bank's log FY2021 assets, Log bank size$_b$, as predetermined scale.

Each bank invests its products' funds in one portfolio. It places a share x_b in LGFV bonds, a share n_b in negotiated credit linked to the same borrower sector, and the rest in accessible low-yielding assets with riskless return r_f, with $x_b, n_b \geq 0$ and $x_b + n_b \leq 1$. Bonds and negotiated credit earn expected excess returns μ_B and μ_{NS} over r_f, net of expected credit losses. Their variances are σ_B^2 and σ_{NS}^2. We assume a positive covariance $\sigma_{B,NS} > 0$ to capture their common exposure to the LGFV sector. The portfolio's expected excess return and variance are

$$R_b = \mu_B x_b + \mu_{NS} n_b, \qquad V_b = \sigma_B^2 x_b^2 + 2\sigma_{B,NS} x_b n_b + \sigma_{NS}^2 n_b^2. \tag{22}$$

[42] Formally, ε_{hb} is independently and identically distributed type I extreme value, the standard assumption in logit models of demand that yields the shares in equation (21); see Berry (1994) and, for bank deposits, Xiao (2020).

We assume $\sigma^2_{NS} > \sigma^2_B > \sigma_{B,NS}$: negotiated credit has greater return variance than bonds, and the two are imperfectly correlated. All banks face the same asset returns and risks. Their portfolio differences arise from the savers they attract and whether entering negotiated credit is worthwhile.

A bank earns a fee f per yuan of savings. Initiating negotiated credit requires a one-time fixed cost $F > 0$ for sourcing, evaluation, and contracting. The bank pays it when it first arranges negotiated credit, rather than for each product or contract, and the cost is sunk afterward. Taking rivals' portfolios as given, the bank chooses its portfolio to maximize

$$f\, s_b - F\, \mathbb{1}\{n_b > 0\}\, e_b, \tag{23}$$

where $e_b = 1$ if the bank has not arranged negotiated credit before and $e_b = 0$ otherwise. With a common fee rate, portfolio choice maximizes the savings attracted to the bank. Entry must also generate enough additional fee income to pay the fixed cost.

6.2 Portfolio choice

A change in a bank's portfolio moves the savers who are close to indifferent between this bank and their next choice. Differentiating equation (21), when the bank raises the utility its products give savers with risk aversion γ, the share of those savers choosing it rises at rate $s_b(\gamma)\big(1 - s_b(\gamma)\big)$. This rate measures how many such savers are marginal: it is small when the bank serves almost none of them or almost all of them. We denote the mean and variance of risk aversion among a bank's marginal savers by

$$\bar{\gamma}_b = \frac{\int \gamma\, s_b(\gamma)\big(1 - s_b(\gamma)\big)\, dG(\gamma)}{\int s_b(\gamma)\big(1 - s_b(\gamma)\big)\, dG(\gamma)}, \qquad \mathrm{Var}_b(\gamma) = \frac{\int (\gamma - \bar{\gamma}_b)^2\, s_b(\gamma)\big(1 - s_b(\gamma)\big)\, dG(\gamma)}{\int s_b(\gamma)\big(1 - s_b(\gamma)\big)\, dG(\gamma)}. \tag{24}$$

For a positive bond position with $x_b + n_b < 1$, the bond derivative of s_b is zero at an interior optimum. When the bank also holds negotiated credit, its derivative is zero as well. The

first-order conditions are

$$\mu_B = \bar{\gamma}_b\Big(\sigma_B^2 x_b + \sigma_{B,NS} n_b\Big), \qquad \mu_{NS} = \bar{\gamma}_b\Big(\sigma_{B,NS} x_b + \sigma_{NS}^2 n_b\Big). \tag{25}$$

Each condition equates an asset's expected excess return to its marginal risk, valued by the bank's marginal savers. A bank therefore invests as those savers would. To show the choices explicitly, we first hold their composition, and hence $\bar{\gamma}_b$, fixed. Appendix B.6 lets this composition adjust and reproduces the predicted patterns in a numerical equilibrium. The formulas below describe interior choices within each investment regime; the bond-only regime sets $n_b = 0$ and gives

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b\, \sigma_B^2}. \tag{26}$$

If the bank arranges negotiated credit, the bond condition gives $x_b = x_b^B - (\sigma_{B,NS}/\sigma_B^2) n_b$: negotiated credit adds exposure to the same borrower sector, so the bank reduces its bond share by $\sigma_{B,NS}/\sigma_B^2$ for each unit of negotiated credit. Substituting this bond share into the negotiated-credit condition gives

$$\mu_{NS} - \frac{\sigma_{B,NS}}{\sigma_B^2}\,\mu_B = \bar{\gamma}_b\left(\sigma_{NS}^2 - \frac{\sigma_{B,NS}^2}{\sigma_B^2}\right) n_b. \tag{27}$$

The left side is the return that negotiated credit earns beyond the bonds it displaces, and the right side is the risk it adds beyond those bonds. Solving,

$$n_b^{NS} = \frac{\mu_{NS} - (\sigma_{B,NS}/\sigma_B^2)\,\mu_B}{\bar{\gamma}_b\Big(\sigma_{NS}^2 - \sigma_{B,NS}^2/\sigma_B^2\Big)}, \qquad x_b^{NS} = x_b^B - \frac{\sigma_{B,NS}}{\sigma_B^2}\, n_b^{NS}, \tag{28}$$

and negotiated credit raises the risk-adjusted return to the bank's marginal savers by

$$g_b = \frac{\Big(\mu_{NS} - (\sigma_{B,NS}/\sigma_B^2)\,\mu_B\Big)^2}{2\bar{\gamma}_b\Big(\sigma_{NS}^2 - \sigma_{B,NS}^2/\sigma_B^2\Big)}, \tag{29}$$

which is positive when negotiated credit earns more than the bonds it displaces, $\mu_{NS} > (\sigma_{B,NS}/\sigma_B^2)\mu_B$. With marginal savers held fixed, the bank holds (x_b^{NS}, n_b^{NS}) if it arranges negotiated credit and $(x_b^B, 0)$ otherwise.

6.3 Who serves cautious savers

Reputation determines not only how many savers a bank attracts but also which ones. For any two banks, equation (21) gives

$$\log \frac{s_b(\gamma)}{s_{b'}(\gamma)} = \beta\Big(R_b - R_{b'}\Big) + \gamma\Big[\Big(a_b - a_{b'}\Big) - \frac{\beta}{2}\Big(V_b - V_{b'}\Big)\Big]. \tag{30}$$

The bracket combines the reputation gap with the difference in portfolio risk. When it is positive, more risk-averse savers are relatively more likely to choose the more reputable bank. Portfolio choice depends on marginal savers. To obtain transparent comparative statics, we use the small-share approximation: when $s_b(\gamma)$ is small, the marginal-saver weight $s_b(\gamma)(1 - s_b(\gamma))$ is approximately $s_b(\gamma)$. The two saver distributions coincide in this limit. Holding portfolios fixed, reputation then raises marginal risk aversion and market share at the rates

$$\frac{\partial \bar{\gamma}_b}{\partial a_b} = \mathrm{Var}_b(\gamma) > 0, \qquad \frac{\partial \log s_b}{\partial a_b} = \bar{\gamma}_b \tag{31}$$

(Proposition B.1). Larger banks win more of the cautious savers through reputation. Smaller banks compete for savers more willing to accept investment risk, so their optimal bond share in equation (26) is higher. Higher bond returns help compensate for their weaker reputation in attracting savings.

6.4 Why negotiated credit requires scale

A bank that has not yet arranged negotiated credit enters only if the fees from the additional savers it attracts cover the fixed cost. Let s_b^B and s_b^{NS} be its market shares with the bond-only portfolio and with the portfolio it would choose with negotiated credit, holding rivals'

portfolios fixed. The bank enters if and only if

$$f\left(s_b^{NS} - s_b^B\right) \geq F. \tag{32}$$

This exact comparison is the entry rule. For intuition, in the small-share case the share gain is approximately $\beta s_b g_b$, so to first order the bank enters when

$$f\,\beta\, s_b\, g_b \geq F \tag{33}$$

(Proposition B.2). Here βg_b is the approximate proportional increase in demand from offering the substitute. Multiplying it by the investor base s_b gives the additional savings attracted, and multiplying again by f gives the fee income available to cover F. For a given proportional demand response, a larger bank attracts more money and earns more fees, while paying the same fixed cost.

Smaller banks' savers value additional yield more, but that stronger demand per saver need not generate enough fee income to justify entry. Under the scale condition, the larger bank's broader investor base outweighs this difference in preferences.[43] For an intermediate fixed cost, only the large local bank enters. The small bank also wants to offer higher returns, but doing so through negotiated credit would cost more than the additional fees it earns. Launching a new product does not remove this bank-level cost. Figure 7 illustrates the resulting entry threshold.

6.5 Debt resolution and the three predictions

We represent the persistent loss of bond income by a fall in expected excess return from μ_B to $(1-\tau)\mu_B$, with $0 < \tau < 1$. To isolate this yield channel, the benchmark holds negotiated-credit returns, asset risks, and marginal-saver composition fixed. The predictions

[43] Allowing portfolios to adjust, Proposition B.2 gives the sufficient condition $\mathrm{Var}_b(\gamma)(1+\beta R_b) < \bar{\gamma}_b^2$. It ensures that the increase in the investor base with reputation dominates the decline in the value of negotiated credit to more cautious marginal savers.

use the sorting and scale conditions above; Internet Appendix B also allows bond risk to fall and solves a numerical equilibrium with changing saver composition. Throughout, Δ denotes the post-resolution change.

Prediction 1: smaller banks rely more on higher-yielding local bonds. Before entry into negotiated credit, the optimal bond share and its response to reputation are

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b \sigma_B^2}, \qquad \frac{dx_b^B}{da_b} = -\frac{\mu_B}{\bar{\gamma}_b^2 \sigma_B^2} \frac{d\bar{\gamma}_b}{da_b} < 0. \tag{34}$$

Under the sorting condition, higher reputation raises marginal risk aversion, $d\bar{\gamma}_b/da_b > 0$, and lowers the optimal bond share. Smaller banks instead use more bond yield to attract their less cautious savers. Prior entry into negotiated credit further reduces bond holdings through equation (28). Table 1 examines the predicted all-bank size gradient; Table 2 establishes that local banks pursue this strategy mainly through home borrowers.

Prediction 2: scale determines which banks substitute into negotiated credit. For a first-time entrant, the small-share, first-order entry benchmark gives

$$\begin{aligned} n_b^* &= \mathbb{1}\Big\{ s_b/\bar{\gamma}_b \geq C(\tau) \Big\}\, n_b^{NS}, \\ C(\tau) &= \frac{2F\Big(\sigma_{NS}^2 - \sigma_{B,NS}^2/\sigma_B^2\Big)}{f\beta\Big[\mu_{NS} - (\sigma_{B,NS}/\sigma_B^2)(1-\tau)\mu_B\Big]^2}. \end{aligned} \tag{35}$$

Here s_b is the bond-only market share at the comparison date, and n_b^{NS} is the conditional position in equation (28) at the lower bond return. As τ rises, negotiated credit becomes more valuable relative to bonds and the common entry cutoff $C(\tau)$ falls. Under the scale condition, the large local bank has a higher $s_b/\bar{\gamma}_b$: it can earn enough fees to cross the entry threshold while the small bank remains below it. Internet Appendix B gives the fixed-cost range supporting this outcome; earlier entrants do not pay again. Table 8 tests the size difference in negotiated-credit expansion, and Table 9 asks whether it persists in newly launched portfolios (Sections 5.2 and 5.3).

Prediction 3: smaller local banks lose more product return and savings business. Comparing the two initially bond-only local banks when only the large bank enters,

$$\begin{aligned}\Delta R_L - \Delta R_S = \tau(2-\tau)\frac{\mu_B^2}{\sigma_B^2}\Big(\frac{1}{\bar{\gamma}_S} - \frac{1}{\bar{\gamma}_L}\Big) \\ + \Big(\mu_{NS} - \frac{\sigma_{B,NS}}{\sigma_B^2}(1-\tau)\mu_B\Big)n_L^{NS} > 0.\end{aligned} \tag{36}$$

The smaller bank loses more bond income because it relied more on bonds before resolution; this is the first term. Only the larger bank obtains replacement income from negotiated credit; this is the second. The small bank therefore loses part of the return advantage it used to compete against the larger bank's reputation. In the small-share benchmark, holding the national bank's offer fixed, this deterioration in its relative offer also reduces its relative share of savings (Proposition B.5). The empirical tests examine the corresponding outcomes separately: return losses by initial bond exposure in Table E7, relative returns by size in Table 8, and WMP-business growth in Table 10. Figure 5 and Figure 6 show their timing.

7 Conclusion

In China's WMP market, LGFV bonds did more than finance local public projects: they gave smaller banks a source of yield with which to compete against larger rivals. Following the 2023 debt resolution, LGFV bond spreads compressed and the estimated product-return premium associated with home LGFV holdings fell by 69%. Banks initially more exposed to these bonds experienced larger return declines even after controlling for size.

Negotiated credit retained higher contractual yields, but larger local banks made greater use of these opaque claims, largely within the same LGFV borrower sector. Smaller banks substituted little and lost ground in both relative product returns and WMP balances. The model explains why the banks most reliant on bond yield can be least able to replace it. Larger banks' reputations attract cautious savers, leaving smaller banks to compete more heavily on yield. After bond yields fall, the substitute requires a fixed contracting cost. A

larger investor base generates enough additional savings and fee income to justify that cost; a smaller base does not. Smaller banks consequently lose the return margin that helped them compete, while larger banks can offer the replacement investment.

The findings reveal how borrower financing feeds back into competition among the intermediaries providing it. Government support changes not only borrower risk but also the investment opportunities through which banks compete for savings. When replacing lost yield requires scale, stabilization can erode smaller intermediaries' competitive edge and redirect shadow banking toward larger institutions and more opaque credit.

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Table 1
Initial credit specialization by bank size

Dependent variable / specification	Total LGFV bond share (pp)	Total negotiated-credit share (pp)	Home LGFV bond share (pp)	Home negotiated-credit share (pp)
	(1)	(2)	(3)	(4)
Log bank size	−7.130***	6.831***	−5.458***	0.622
	(1.778)	(1.803)	(1.123)	(0.572)
Log product assets	−3.028***	−3.480***	−0.677	−0.820***
	(0.828)	(0.914)	(0.431)	(0.282)
𝟙{Restricted redemption}	2.405	16.943***	−0.264	1.123
	(2.138)	(4.113)	(1.323)	(0.947)
Product risk rating	−3.065**	4.162	0.392	0.467
	(1.316)	(2.956)	(0.867)	(1.117)
Bank nonperforming-loan ratio	−7.968	10.855**	−3.180	−0.675
	(6.171)	(5.471)	(4.021)	(1.338)
Bank cost–income ratio	−0.245*	−0.582***	−0.085	−0.110**
	(0.131)	(0.184)	(0.092)	(0.049)
Bank net interest margin	−5.013	9.229	2.795	3.955
	(7.203)	(8.397)	(4.126)	(2.737)
Home-city GDP (RMB trillion)	0.064	−1.051	0.810	−1.932**
	(3.585)	(1.945)	(2.118)	(0.864)
Home-city GDP growth (%)	−0.381	−0.182	0.527	−0.012
	(1.211)	(1.046)	(0.768)	(0.508)
Home-city LGFV debt/GDP	27.483**	3.752	22.502***	5.983**
	(11.977)	(4.867)	(6.930)	(2.381)
Home-city income (RMB 10,000)	1.325	3.696**	−1.132	1.656**
	(3.213)	(1.595)	(2.049)	(0.767)
Wild p: log bank size	0.002	0.009	< 0.001	0.364
Observations	68,331	68,331	68,331	68,331
Quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column regresses the stated allocation share (pp) on unstandardized log FY2021 parent-bank interest-bearing assets in RMB trillion. A one-unit change is one log point, not one standard deviation. Zero holdings are retained. All ten other controls are displayed; fixed size replaces current bank assets. Quarter effects enter, but no national-bank indicator or bank effects. Weights use included debt value. Parentheses contain parent-bank clustered errors, grouping subsidiaries and aliases; stars indicate 10%, 5%, and 1% significance. Wild tests use 4,999 Rademacher draws. Appendix D.9 defines the comparison. Estimation sample: 2022Q1–2023Q3.

Table 2
Home bias by bank type and asset class

Asset class	All credit	LGFV	NS	Other standard
Dependent variable	Home bias (fraction) (1)	Home bias (fraction) (2)	Home bias (fraction) (3)	Home bias (fraction) (4)
$\mathbb{1}\{\text{Local bank}\}$	0.2713***	0.3205***	0.4885***	0.1611***
	(0.0092)	(0.0144)	(0.0192)	(0.0086)
Log product size	−0.0004	−0.0019	0.0016	−0.0083***
	(0.0018)	(0.0031)	(0.0055)	(0.0021)
$\mathbb{1}\{\text{Restricted redemption}\}$	−0.0182***	−0.0189**	−0.0285	−0.0107*
	(0.0068)	(0.0090)	(0.0260)	(0.0058)
Product risk rating	0.0059*	0.0035	0.0067	−0.0021
	(0.0035)	(0.0070)	(0.0095)	(0.0043)
Log(1 + disclosed positions)	−0.0965***	0.0077	0.0001	−0.0629***
	(0.0145)	(0.0285)	(0.0128)	(0.0182)
Total disclosed weight (%)	0.0007***	0.0016***	0.0008***	0.0003*
	(0.0002)	(0.0003)	(0.0003)	(0.0002)
Log average product maturity	0.0203***	0.0337***	0.0178***	0.0217***
	(0.0027)	(0.0048)	(0.0058)	(0.0027)
Average product risk rating	−0.0513**	−0.1744***	0.1068***	−0.1189***
	(0.0231)	(0.0538)	(0.0369)	(0.0199)
Bank nonperforming-loan ratio (%)	−0.0371***	0.0296	0.0490*	−0.0142
	(0.0121)	(0.0216)	(0.0255)	(0.0116)
Bank cost–income ratio (%)	−0.0008*	−0.0003	0.0019**	−0.0001
	(0.0005)	(0.0007)	(0.0009)	(0.0005)
Bank net interest margin (%)	0.0939***	0.2127***	0.2460***	0.0412**
	(0.0203)	(0.0241)	(0.0330)	(0.0161)
Home-city GDP growth (%)	0.0188***	0.0310***	0.0158*	0.0088**
	(0.0046)	(0.0083)	(0.0086)	(0.0040)
Mean of dependent variable	0.1851	0.2562	0.1944	0.1305
Observations	186,731	108,313	43,803	158,812
Fixed effects	Quarter			
Standard errors	Bank city × quarter			

Notes: NS denotes negotiated credit. This table tests whether local banks overweight home credit beyond what provincial market size predicts. Estimates use equation (2). The unit of observation is a product-quarter. The dependent variable is the product's home-province share within the asset class minus the home province's share of nationwide disclosed holdings in the class, Home bias^{n}_{iq}, as a fraction. Each column uses products with positive holdings in the class. Local bank equals one for products of banks other than state-owned and joint-stock commercial banks. Product size and the number of disclosed positions enter as $\log(1+x)$. Menu term and menu risk are the average contractual term and numerical risk rating of the bank's products in the quarter, weighted by holding records. Standard errors clustered at the bank-city-by-quarter level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Estimation sample: 2022Q1–2025Q3.

Table 3
Home credit exposure and product returns before and after the program

Panel A. Exposure coefficients and implied premiums

Dependent variable / specification	Monthly NAV return (bp) (1)
Home LGFV bond share	8.852^{***} (1.856)
Home LGFV bond share × 𝟙{Post}	-6.133^{***} (1.501)
Home negotiated-credit share	6.765^{***} (1.291)
Home negotiated-credit share × 𝟙{Post}	-2.395 (1.472)
Home other-bond share	0.506 (1.552)
Home other-bond share × 𝟙{Post}	0.077 (1.538)
Non-home LGFV bond share	3.041^{**} (1.348)
Non-home LGFV bond share × 𝟙{Post}	-2.261 (1.398)
Non-home negotiated-credit share	2.776^{***} (0.957)
Non-home negotiated-credit share × 𝟙{Post}	1.382^{*} (0.796)
𝟙{Local bank}	0.193 (0.677)
Home LGFV bond premium after the program	2.719^{*} (1.613)
Home negotiated-credit premium after the program	4.370^{***} (0.996)
Difference in home-premium changes	-3.738^{**} (1.522)

Control coefficients and specification continue on next page

Table 3 (continued)

Panel B. Control coefficients from the same regression

Dependent variable / specification	Monthly NAV return (bp)
	(1)
Log product assets	−0.567***
	(0.142)
𝟙{Restricted redemption}	3.711***
	(0.625)
Product risk rating	−2.326***
	(0.607)
Log bank assets	−0.346
	(0.452)
Bank nonperforming-loan ratio	0.794
	(1.227)
Bank cost–income ratio	0.052*
	(0.027)
Bank net interest margin	0.106
	(1.045)
Home-city GDP (RMB trillion)	0.203
	(0.655)
Home-city GDP growth (%)	0.043
	(0.276)
Home-city LGFV debt/GDP	3.588**
	(1.690)
Home-city income (RMB 10,000)	−0.438
	(0.684)
Observations	554,555
Month fixed effects	Yes
Standard errors	Parent bank

Notes: Both panels report the same regression: Panel A shows exposure coefficients and implied premiums; Panel B shows all eleven control coefficients. The regression relates monthly log adjusted-NAV returns (bp) to lagged exposure shares, Post interactions, controls, and month effects. Non-home other bonds are the reference. Shares are fractions; multiply coefficients by 0.2 for a 20-point allocation difference. Post starts in October 2023. The final premiums and change contrast use this regression's full covariance matrix. Centrally administered issuers are never home. Parentheses contain parent-bank clustered errors; stars indicate 10%, 5%, and 1% significance. Internet Appendix E gives robustness checks. Estimation sample: Apr. 2022–Dec. 2025.

Table 4
Contractual-rate movements: negotiated credit versus LGFV bonds

Panel A. Nationwide annual rate changes (pp)

Dependent variable / specification	NS–bond gap (1)	NS yield (2)	Bond yield (3)
$\mathbb{1}\{\text{Negotiated credit}\} \times \mathbb{1}\{\text{Year} = 2023\}$	-0.146^{***} (0.047)	—	—
$\mathbb{1}\{\text{Negotiated credit}\} \times \mathbb{1}\{\text{Year} = 2024\}$	1.299^{***} (0.072)	—	—
$\mathbb{1}\{\text{Year} = 2023\}$	—	-0.089^{**} (0.037)	0.163^{***} (0.033)
$\mathbb{1}\{\text{Year} = 2024\}$	—	-0.245^{***} (0.053)	-1.436^{***} (0.042)
Observations	102,310	36,251	27,769
Issuer × quarter fixed effects	Yes	No	No
Issuer × debt type fixed effects	Yes	No	No
Maturity category fixed effects	Yes	Yes	Yes
Borrower × creditor fixed effects	No	Yes	No
Issuer × bond type × placement fixed effects	No	No	Yes
Standard errors	Issuer	Issuer	Issuer

Panel B. Additional changes in priority provinces (pp)

Dependent variable / specification	NS–bond gap (1)	NS yield (2)	Bond yield (3)
$\mathbb{1}\{\text{Priority province}\} \times \mathbb{1}\{\text{Year} = 2024\}$	1.042^{***} (0.288)	0.317 (0.240)	-0.725^{***} (0.143)
$\mathbb{1}\{\text{Priority province}\} \times \mathbb{1}\{\text{Year} = 2023\}$	0.251 (0.210)	0.584^{***} (0.182)	0.333^{***} (0.119)
Province wild p: 2024 gap	0.0063	—	—
Observations	2,296	2,296	2,296
Issuer fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Standard errors	Issuer	Issuer	Issuer

Notes: Each column is one regression of annual contractual rates (pp). Panel A reports 2023 and 2024 changes relative to 2022; earlier year terms remain estimated but are not shown. Issuer, channel, and maturity controls follow Appendix D.2; the three samples differ, so their changes are not additive. Panel B uses the same issuer-years in all columns, with issuer and year effects and a separate priority-province 2023 interaction. Its gap is additive. Parentheses contain issuer-clustered errors; stars indicate 10%, 5%, and 1% significance. Estimation samples: Panel A, 2018–2024; Panel B, 2019–2024.

Table 5
Prior reliance, refinancing intensity, and home LGFV allocation

Dependent variable / specification	Home LGFV bond share (pp) (1)	Home LGFV bond share (pp) (2)	Home LGFV bond share (pp) (3)
Initial home bias × refinancing intensity	−46.360*** (7.881)	−47.923*** (7.774)	—
Initial home bias (through 2023Q2) × refinancing intensity	—	—	−47.541*** (7.838)
Log product assets	0.332 (0.644)	0.344 (0.643)	0.345 (0.642)
𝟙{Restricted redemption}	5.012** (2.254)	4.956** (2.259)	4.957** (2.258)
Product risk rating	−0.335 (1.643)	−0.357 (1.688)	−0.351 (1.689)
Log bank assets	4.687 (12.670)	7.792 (13.659)	7.798 (13.618)
Bank nonperforming-loan ratio	6.494 (3.998)	5.223 (3.549)	5.343 (3.585)
Bank cost–income ratio	0.255 (0.336)	0.173 (0.363)	0.174 (0.360)
Bank net interest margin	−4.501 (3.353)	−4.600 (3.203)	−4.508 (3.227)
Home-city GDP (RMB trillion)	−20.580** (9.759)	−24.420* (12.047)	−23.486* (11.857)
Home-city GDP growth (%)	−1.488** (0.537)	−1.541*** (0.540)	−1.493** (0.542)
Home-city LGFV debt/GDP	−22.410 (23.961)	−20.956 (24.057)	−20.922 (24.018)
Home-city income (RMB 10,000)	39.096*** (9.309)	38.962*** (9.548)	38.647*** (9.388)
Initial home-tilt window	Through 2023Q3	Through 2023Q3	Through 2023Q2
Wild p: home tilt × refinancing	0.010	0.043	0.037
Observations	83,051	83,051	83,044
Institution fixed effects	Yes	Yes	Yes
Home province × quarter fixed effects	Yes	Yes	Yes
Standard errors	Home province	Home province	Home province

Notes: Local-bank product-quarters; the outcome is the home-LGFV share of included debt (pp). Initial home bias is the initial benchmark-adjusted home allocation across all included debt, and intensity is cumulative refinancing divided by the bank's home province's 2020 LGFV debt; both are fractions. Columns 2–3 include reliance-by-quarter slopes, not tabulated. Column 3 measures reliance through 2023Q2. Ordinary controls are displayed; institution and home-province-by-quarter effects absorb the baseline terms. Province-clustered errors are in parentheses; wild tests use 27 provinces. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table 6
Home-province LGFV holdings after the program

Specification	Baseline	Own-city split	Net buying	New purchases	City-risk interaction	Pre-period trend
Dependent variable	LGFV holdings / bond book (pp)	LGFV holdings / bond book (pp)	Net holdings change / current book (pp)	Log(1 + new purchases)	Log(1 + new purchases)	LGFV holdings / bond book (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{\text{Home buyer}\} \times \mathbb{1}\{\text{Post}\}$	−0.453**		−0.127**	−0.158**	0.203*	
	(0.193)		(0.053)	(0.079)	(0.117)	
$\mathbb{1}\{\text{Own city}\} \times \mathbb{1}\{\text{Post}\}$		−1.936***				
		(0.626)				
$\mathbb{1}\{\text{Rest of home province}\} \times \mathbb{1}\{\text{Post}\}$		−0.242				
		(0.179)				
$\mathbb{1}\{\text{Home buyer}\} \times \mathbb{1}\{\text{Post}\} \times$ city risk					−0.149***	
					(0.051)	
$\mathbb{1}\{\text{Home buyer}\} \times$ trend						0.003
						(0.051)
Mean of dependent variable	0.784	0.784	−0.052	0.937	0.945	0.816
Observations	73,089	73,089	73,089	73,127	72,196	32,540
Bank and city controls	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Fixed effects	Bank × city; bank × quarter; city × quarter					
Standard errors	Bank city					

Notes: This table tests whether local buyers retreat relative to other buyers facing the same issuer-city bond supply. Estimates use equation (11). The unit of observation is a bank, LGFV issuer city, and quarter over 2022Q1–2025Q3; the panel contains 73,635 cells for 170 banks and 248 issuer cities and covers the cities whose LGFV bonds a bank holds at some point, with zero holdings in quarters without positions in the city. Home equals one when a local bank invests in a city of its own province, and Post equals one from 2023Q4. Holdings are the bank's LGFV holdings in the city divided by its current bond book, in percentage points. Net holdings change includes changes in continuing positions as well as entries and exits, is divided by the current bond book, and is winsorized at the 1st and 99th percentiles. New purchases enter as log(1 + amount in RMB million). Column 2 separates the bank's own city from the rest of its home province. Column 5 interacts Home buyer × Post with the city's pre-program risk, the issue-size-weighted mean CSI spread of its LGFV bonds with maturities of one to five years over 2022Q3–2023Q2, in percentage points, for the 209 cities with at least five bond-quarter observations; the Home buyer × post coefficient in that column is evaluated at a zero spread. Column 6 replaces Post with a linear quarterly trend and uses 2022Q1–2023Q3. Standard errors clustered by bank city are in parentheses; clustering by issuer city or by bank-city pair gives standard errors of 0.151 and 0.146 in column 1. The mean holdings share in home cells is 3.91 before 2023Q4 and 3.43 afterward. Bank and destination-city characteristics are absorbed by the bank-by-quarter and city-by-quarter fixed effects. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Estimation sample: 2022Q1–2025Q3 (columns 1, 2, 3, 4, 5); 2022Q1–2023Q3 (column 6).

Table 7
Negotiated credit holdings by bank type and bank size

Asset category	National banks		Local banks	
	Pre	Post	Pre	Post
Panel A. Composition of included debt holdings, percent				
Home-province LGFV bonds	0.4	0.3	24.7	17.4
Home-province negotiated credit	2.1	1.9	6.1	13.9
Negotiated credit from other provinces	22.0	26.6	3.2	5.7
Product-quarters	34,595	59,691	36,194	57,516
Panel B. Share of the sector's holdings in local banks' products, percent				
Home-province negotiated credit			61.6	83.1
All negotiated credit			17.3	31.3
Home-province LGFV bonds			97.4	97.1
Panel C. Local banks present in both periods, by pre-program size tercile				
Home-province negotiated credit, smallest tercile			0.72	0.31
Home-province negotiated credit, middle tercile			3.45	4.74
Home-province negotiated credit, largest tercile			7.45	18.26
Home-province LGFV bonds, smallest tercile			25.1	20.8
Home-province LGFV bonds, middle tercile			37.7	25.5
Home-province LGFV bonds, largest tercile			20.7	15.0

Notes: NS denotes negotiated credit. This table locates the observed NS expansion by bank type, borrower geography, and local-bank size group. Product-quarter portfolios with positive included debt holdings, 2022Q1–2025Q3. Pre ends in 2023Q3; Post begins in 2023Q4. All panels use holding-value weights. Panels A and B retain all bank groups and classify Guangfa and its subsidiary as national. Panel A reports class shares of each group's included debt holdings; Panel B reports the fraction of sector holdings held by local banks' products. Panel C retains 102 underlying local banks observed in both periods, with 25, 38, 39 banks in the smallest, middle and largest groups. Groups are fixed by FY2021 interest-bearing assets, using the common pre-announcement local-bank universe described in Section 2. Bank financials are assigned to the bank's own products and those of its wealth-management subsidiary; alternative names do not create additional banks. Missing FY2021 size is not filled. Negotiated credit includes trust loans, leasing and other identified negotiated claims.

Table 8
Bank size, negotiated-credit substitution, and relative returns

Panel A. Resolution-intensity interactions

Dependent variable / specification	Home negotiated-credit share (pp)	Non-home negotiated-credit share (pp)	Monthly return (bp)
	(1)	(2)	(3)
Log bank size × refinancing intensity	24.205***	14.531***	13.925***
	(4.877)	(2.733)	(5.265)
Initial home bias × refinancing intensity	5.021	13.152***	−43.242**
	(10.341)	(4.398)	(20.927)
Log product assets	−2.172***	−1.122***	−0.927***
	(0.364)	(0.155)	(0.124)
𝟙{Restricted redemption}	5.538***	4.252***	9.008***
	(1.711)	(0.778)	(0.672)
Product risk rating	3.442***	0.907**	−2.175***
	(1.004)	(0.460)	(0.841)
Log bank assets	−30.179***	−23.281***	14.410*
	(9.590)	(6.046)	(8.658)
Bank nonperforming-loan ratio	−2.621	−4.356***	−8.821**
	(2.474)	(1.400)	(3.885)
Bank cost–income ratio	−0.417*	−0.407**	0.105
	(0.226)	(0.189)	(0.282)
Bank net interest margin	16.147***	1.141	6.847*
	(3.389)	(1.602)	(3.752)
Home-city GDP (RMB trillion)	51.899***	15.888***	9.886
	(9.602)	(4.129)	(12.485)
Home-city GDP growth (%)	−3.070***	−0.469**	0.011
	(0.717)	(0.219)	(0.835)
Home-city LGFV debt/GDP	3.490	13.626**	−9.979
	(13.251)	(5.383)	(32.452)
Home-city income (RMB 10,000)	−43.525***	−17.132***	−17.165
	(10.743)	(4.531)	(14.391)
Observations	83,027	83,027	206,770
Institution fixed effects	Yes	Yes	Yes
Home province × quarter fixed effects	Yes	Yes	No
Home province × month fixed effects	No	No	Yes
Standard errors	Bank city × quarter	Bank city × quarter	Bank city × month

Table 8 (continued)

Panel B. Post-program interactions

Dependent variable / specification	Home negotiated-credit share (pp)	Non-home negotiated-credit share (pp)	Monthly return (bp)
	(1)	(2)	(3)
Log bank size × 𝟙{Post}	5.853***	2.329***	1.234*
	(0.866)	(0.412)	(0.725)
Initial home bias × 𝟙{Post}	7.157**	4.363***	−0.936
	(3.095)	(1.240)	(3.056)
Log product assets	−2.151***	−1.115***	−0.938***
	(0.363)	(0.154)	(0.125)
𝟙{Restricted redemption}	5.456***	4.218***	8.995***
	(1.721)	(0.776)	(0.679)
Product risk rating	3.563***	0.961**	−2.252***
	(1.005)	(0.462)	(0.845)
Log bank assets	−25.758***	−22.166***	14.556*
	(9.830)	(6.004)	(8.713)
Bank nonperforming-loan ratio	−0.336	−3.246**	−7.674**
	(2.356)	(1.418)	(3.850)
Bank cost–income ratio	−0.370*	−0.376*	0.219
	(0.216)	(0.192)	(0.287)
Bank net interest margin	13.616***	0.567	6.640*
	(3.187)	(1.607)	(3.707)
Home-city GDP (RMB trillion)	37.848***	11.336***	12.392
	(9.750)	(4.132)	(12.768)
Home-city GDP growth (%)	−1.711***	0.181	0.043
	(0.563)	(0.247)	(0.855)
Home-city LGFV debt/GDP	−12.207	4.744	−11.188
	(12.462)	(4.921)	(31.374)
Home-city income (RMB 10,000)	−21.873**	−11.227***	−11.239
	(10.434)	(4.209)	(15.128)
Observations	83,027	83,027	206,770
Institution fixed effects	Yes	Yes	Yes
Home province × quarter fixed effects	Yes	Yes	No
Home province × month fixed effects	No	No	Yes
Standard errors	Bank city × quarter	Bank city × quarter	Bank city × month

Notes: Each numbered column is one regression. Size is unstandardized log FY2021 bank assets in RMB trillion; prior reliance is pre-program home tilt. Panel A interacts both with cumulative refinancing divided by the bank's home province's 2020 LGFV debt; Panel B interacts them with Post. Outcomes are allocation shares (pp) or monthly returns (bp). All eleven controls are displayed, with institution and home-province-by-period effects. Allocation columns weight by included debt value; return columns are unweighted. Standard errors cluster by bank-city-by-period; underlying-bank clustered checks retain the size interactions ($p \leq 0.024$). Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3 (columns 1, 2); 2022Q2–2025Q3 (column 3).

Table 9
Negotiated credit in new products, by bank size

First eligible portfolio	Smallest tercile	Middle tercile	Largest tercile
New products	253	1,877	5,248
Parent banks	13	25	33
Products holding home negotiated credit	2	108	1,118
Products holding any negotiated credit	2	134	1,586
Home negotiated-credit share (%)	0.78	6.31	27.46
Non-home negotiated-credit share (%)	0.00	2.73	14.74
Total negotiated-credit share (%)	0.78	9.04	42.20
Home LGFV bond share (%)	29.36	27.94	11.11
Home negotiated-credit share: first disclosure within one quarter (%)	0.59	5.75	27.29
Products in that restriction	234	1,737	4,611

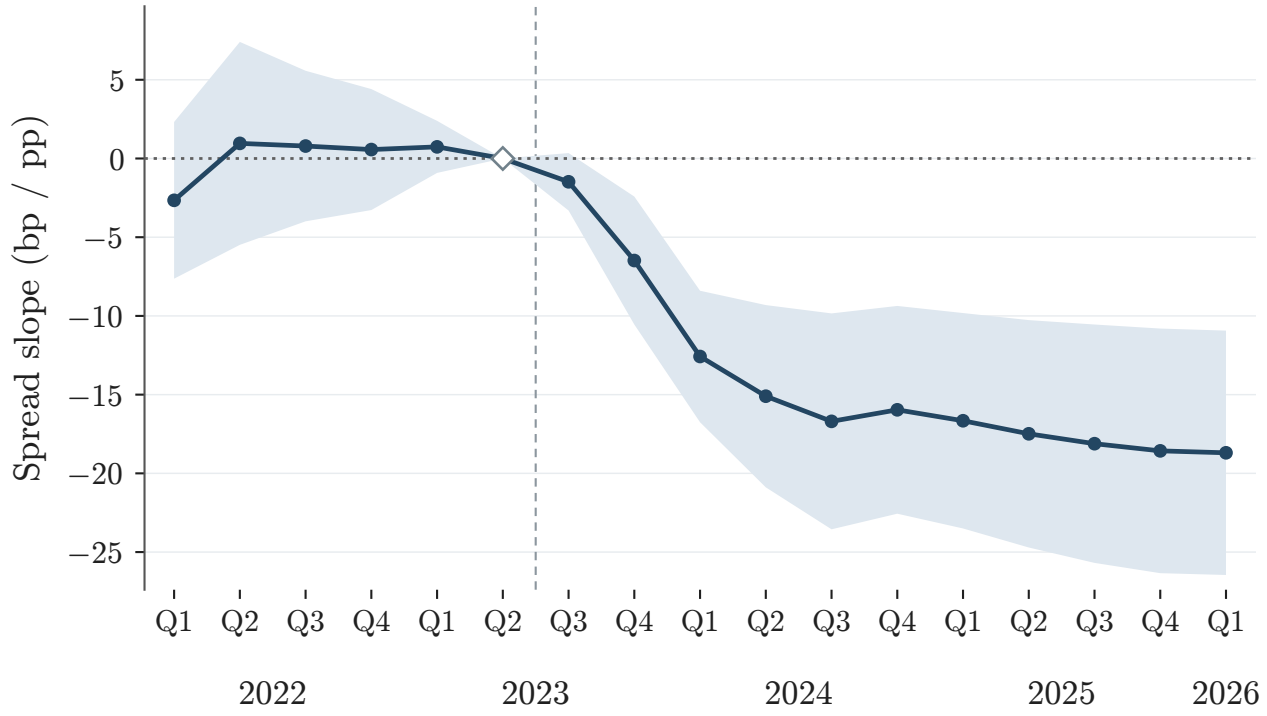
Notes: NS denotes negotiated credit. This table compares the first observed portfolios of products launched after the program by local banks in each size tercile. Launches in 2023Q4–2025Q3 are identified from income-start dates rather than first disclosure, and each product contributes its first eligible portfolio. Shares are holding-value weighted percentages of included debt holdings; counts are unweighted. Size terciles use fixed FY2021 parent-bank interest-bearing assets and the common local-bank cutoffs. Products of a bank's wealth-management subsidiary are assigned to the parent bank. The final two rows restrict to products whose first portfolio is disclosed in the launch quarter or the next quarter. No regressions enter this table.

Table 10
Bank size and WMP business growth after the program

Dependent variable / specification	Log WMP balance	Log WMP balance	Log WMP balance	Log WMP balance	Log WMP balance: terciles	Log product count
	(1)	(2)	(3)	(4)	(5)	(6)
Log bank size × 𝟙{Post}	0.152*** (0.039)	0.170*** (0.044)	0.152*** (0.031)	0.144*** (0.052)	—	0.203*** (0.053)
Initial home bias × 𝟙{Post}	—	0.188 (0.126)	0.298*** (0.098)	—	—	—
𝟙{Middle size tercile} × 𝟙{Post}	—	—	—	—	0.177* (0.093)	—
𝟙{Largest size tercile} × 𝟙{Post}	—	—	—	—	0.333*** (0.088)	—
Log bank assets	—	—	0.048 (0.231)	—	—	—
Bank nonperforming-loan ratio	—	—	0.222*** (0.066)	—	—	—
Bank cost–income ratio	—	—	−0.013 (0.009)	—	—	—
Bank net interest margin	—	—	−0.049 (0.082)	—	—	—
Home-city GDP (RMB trillion)	—	—	1.064*** (0.265)	—	—	—
Home-city GDP growth (%)	—	—	−0.008 (0.011)	—	—	—
Home-city LGFV debt/GDP	—	—	−0.733* (0.404)	—	—	—
Home-city income (RMB 10,000)	—	—	−0.074 (0.184)	—	—	—
Wild-bootstrap p	< 0.001	< 0.001	< 0.001	0.008	< 0.001	—
Observations	571	479	395	494	571	557
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Half-year fixed effects	Yes	Yes	Yes	No	Yes	Yes
Home province × half-year fixed effects	No	No	No	Yes	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

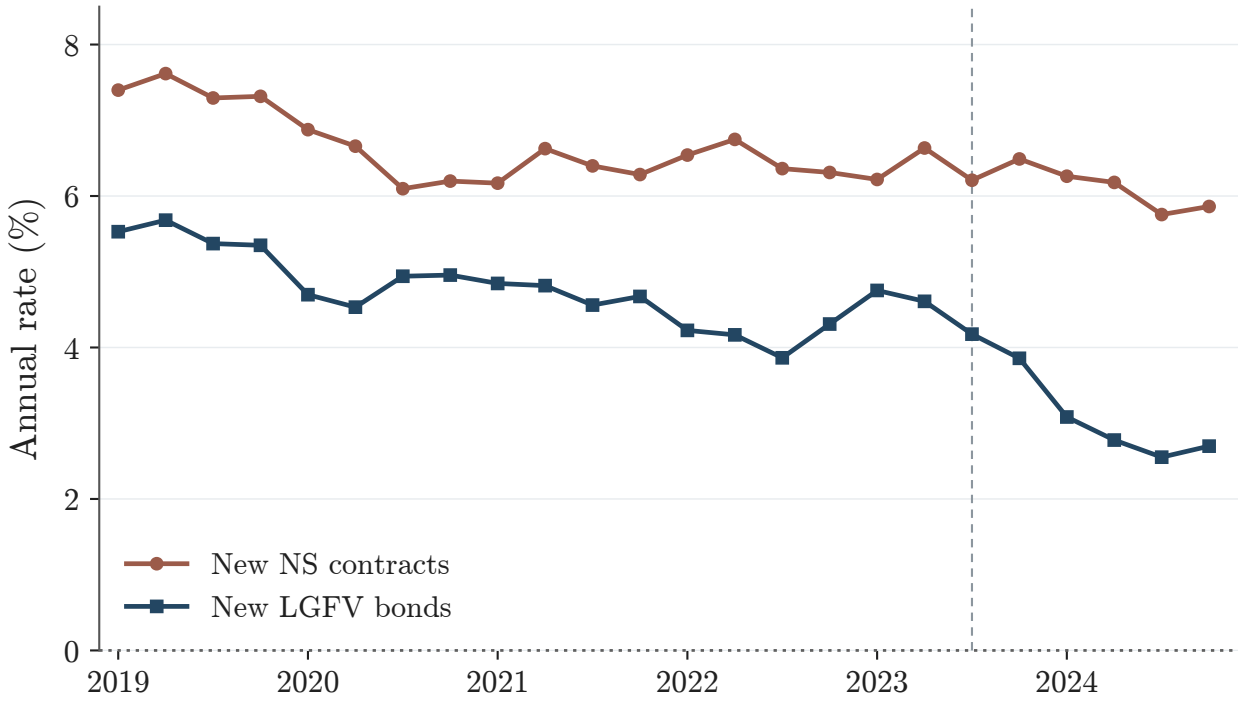
Notes: Each column is one bank-half-year regression. Columns 1–5 use log positive observed WMP balances, combining the bank and subsidiary; column 6 uses log product counts. Size is unstandardized log FY2021 bank assets in RMB trillion. Post begins at year-end 2023. The baseline includes bank and half-year effects, with no additional controls; later columns add reliance, the displayed bank and city controls, province-time effects, or fixed size terciles (smallest omitted). Missing reports are not zeros. Parentheses contain bank-clustered errors; stars indicate 10%, 5%, and 1% significance. Appendix E.1 discusses coverage. Estimation sample: 2022H1–2025H2.

Figure 1
Provincial LGFV spreads by final refinancing intensity



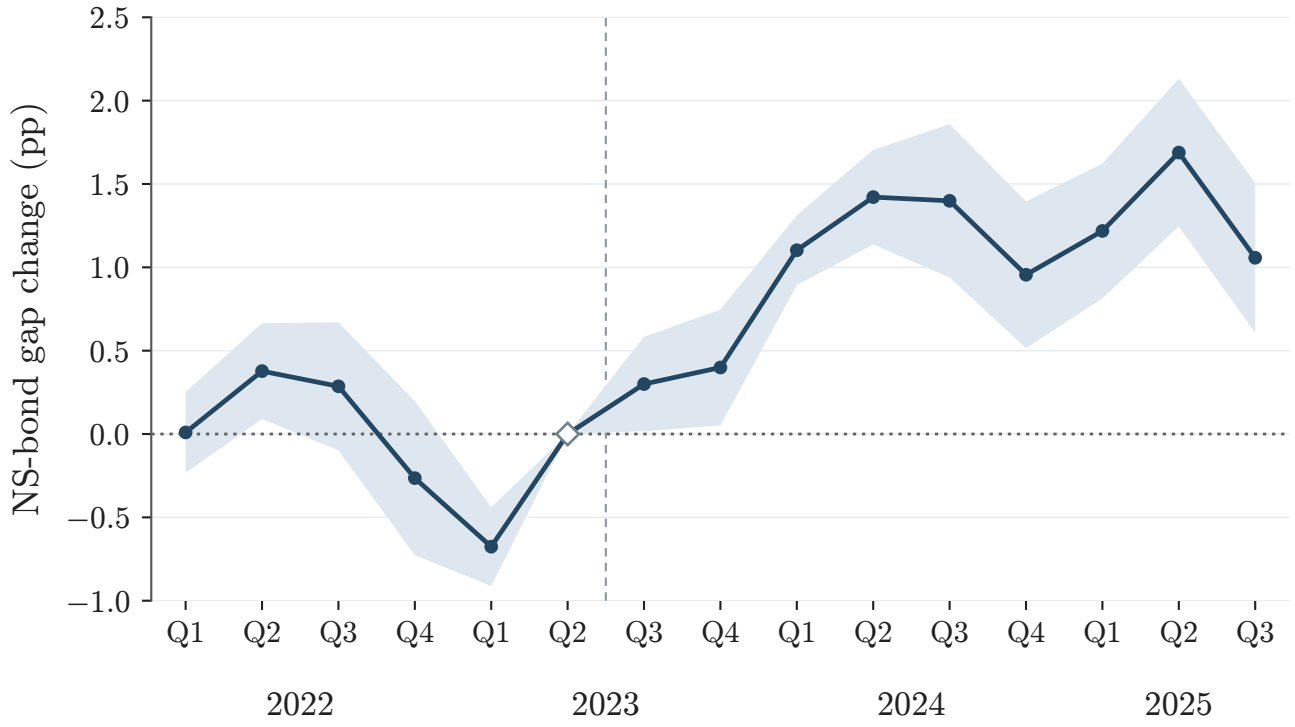
Notes: This event study plots the quarter-specific coefficients β_k in equation (7). Provincial LGFV spreads, in basis points, are regressed on final refinancing intensity interacted with quarter indicators, with province and quarter fixed effects; 2023Q2 is the omitted quarter. Final intensity is cumulative special-refinancing issuance at the end of the intensity series as a percentage of 2020 provincial GDP. Coefficients are basis points per percentage point of final intensity, without standardization. Spreads and intensity are winsorized at the 1st and 99th percentiles, and the sample covers 31 provinces over 2022Q1–2026Q1. The shaded bands are pointwise 95% confidence intervals based on standard errors clustered by province, and the dashed line marks 2023Q3.

Figure 2
Contractual rates on new local-government financing



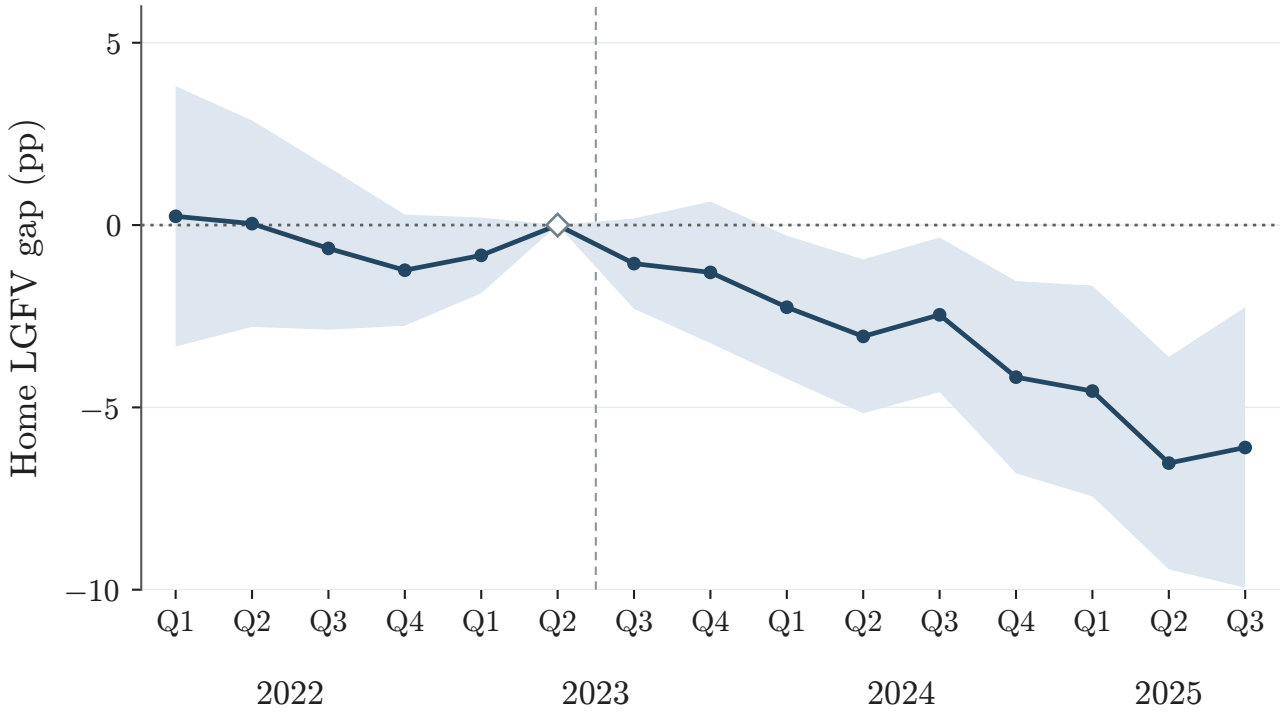
Notes: NS denotes negotiated credit. This figure compares financing-yield opportunities in negotiated credit and LGFV bonds. Quarterly annual contractual yields in percent on new NS contracts and new physical LGFV bond issues, 2019Q1–2024Q4. Each series uses financing reported by LGFV issuers in the deduplicated financing records and weights rate-visible observations by positive known original financing amounts; unknown amounts are not imputed. The series need not contain the same borrowers in a quarter and are not adjusted for changes in composition or tenor. The dashed vertical line marks the 2023Q3 announcement. Table 4 compares rates within reporting issuers and controls for tenor. Incomplete 2025 observations are excluded.

Figure 3
The negotiated-credit–bond rate gap within the same borrower



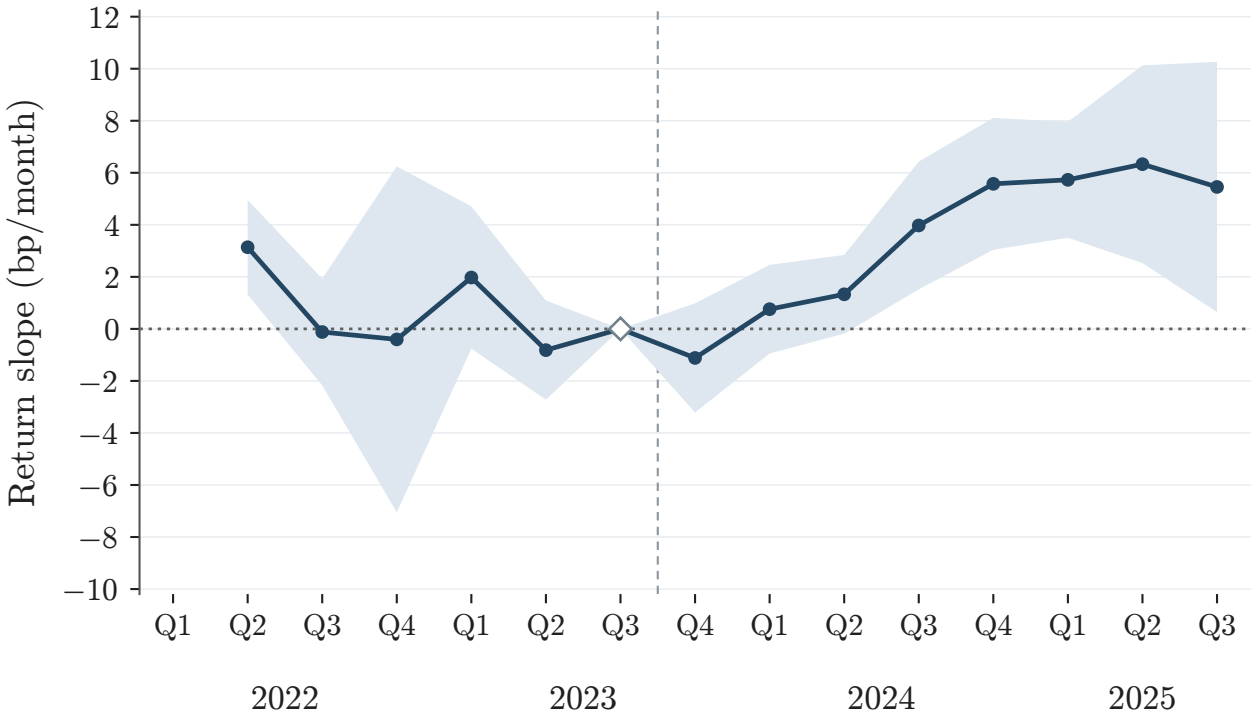
Notes: This within-borrower event study plots $\delta_q = \beta_q - \beta_{2023Q2}$ as defined in equation (9): changes in the annual contractual NS-minus-bond rate gap, in percentage points. Quarterly gap levels are estimated by equation (8), with borrower-by-calendar-month effects and continuous maturity interacted with every quarter; there are no borrower-by-debt-type effects. The plot starts in 2022Q1. The usable estimation sample is 2021Q1–2025Q3 (17,303 financing observations; 589 borrowers), with observed maturity required and fixed-effect singletons removed. Funded-borrower and reporting-issuer DM codes must coincide; affiliates are excluded. Before these restrictions, the pool contains borrower-quarters with both channels and borrowers with at least two matched quarters. Observations are equally weighted. Shading gives conventional pointwise 95% confidence intervals clustered by borrower province, using t_{25} critical values (26 provinces) and the covariance of the gap contrasts. The hollow diamond marks the reference; the vertical dashed line separates 2023Q2 from 2023Q3. The line stays solid through 2025. Coverage falls to 58, 35, and 21 borrowers in 2025Q1–Q3; 19 borrowers have both channels in the same month in 2025Q3. No usable 2025Q4 comparison remains. The NS extract's latest disclosure is April 20, 2026. Earlier changes, 2021Q1–2023Q1, reject joint zero ($p < 0.001$). Tables D4–D5 report specification, inference, and coverage checks.

Figure 4
Home-province gap in local banks' LGFV holdings by quarter



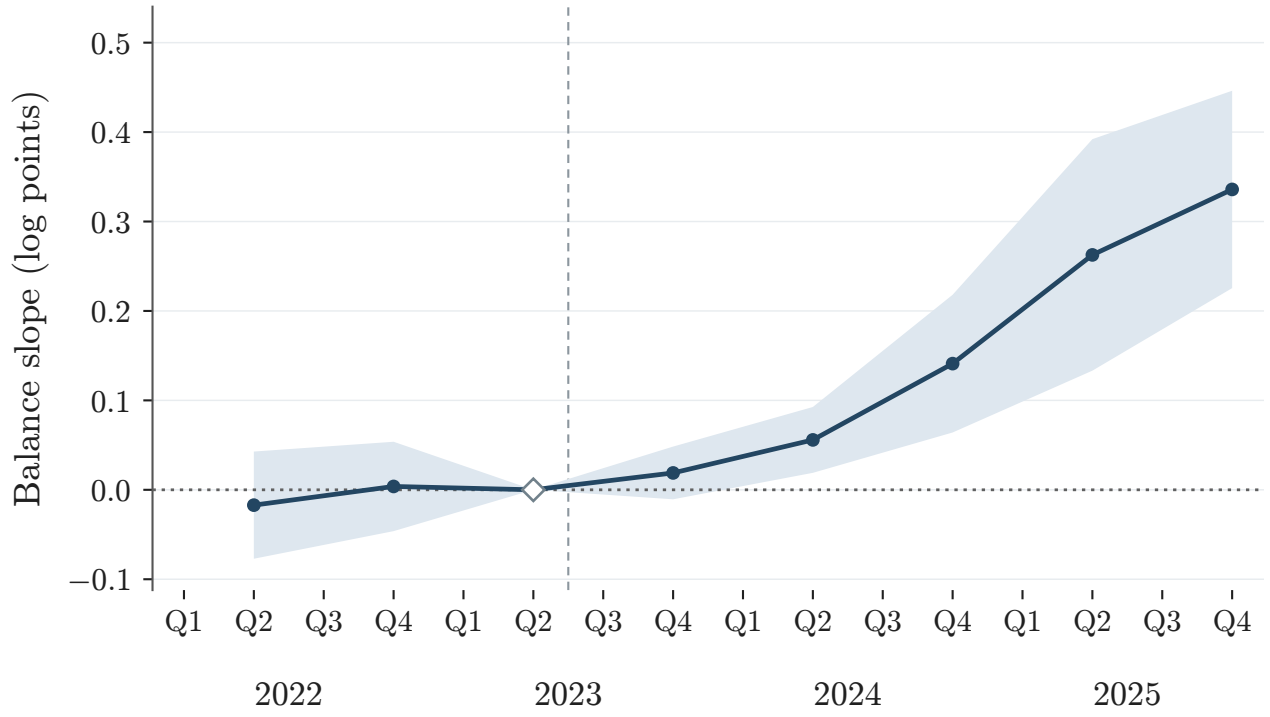
Notes: This event study plots the home-bond allocation coefficients η_k in equation (12). A buying institution's LGFV holdings in a province, as a percentage of its total bond holdings and winsorized at the 1st and 99th percentiles, are regressed on Home interacted with quarter indicators. Buyer-by-province, buyer-by-quarter, and destination-province-by-quarter fixed effects enter; 2023Q2 is omitted. The sample covers 2022Q1–2025Q3. The observation is a buyer, province, and quarter, so the coefficients measure changes in the home gap for an entire province rather than a single city as in Table 6. Home equals one for a local bank's own province. The shaded bands are pointwise 95% confidence intervals based on standard errors clustered by bank city, and the dashed line marks 2023Q3.

Figure 5
Bank size and product returns by quarter



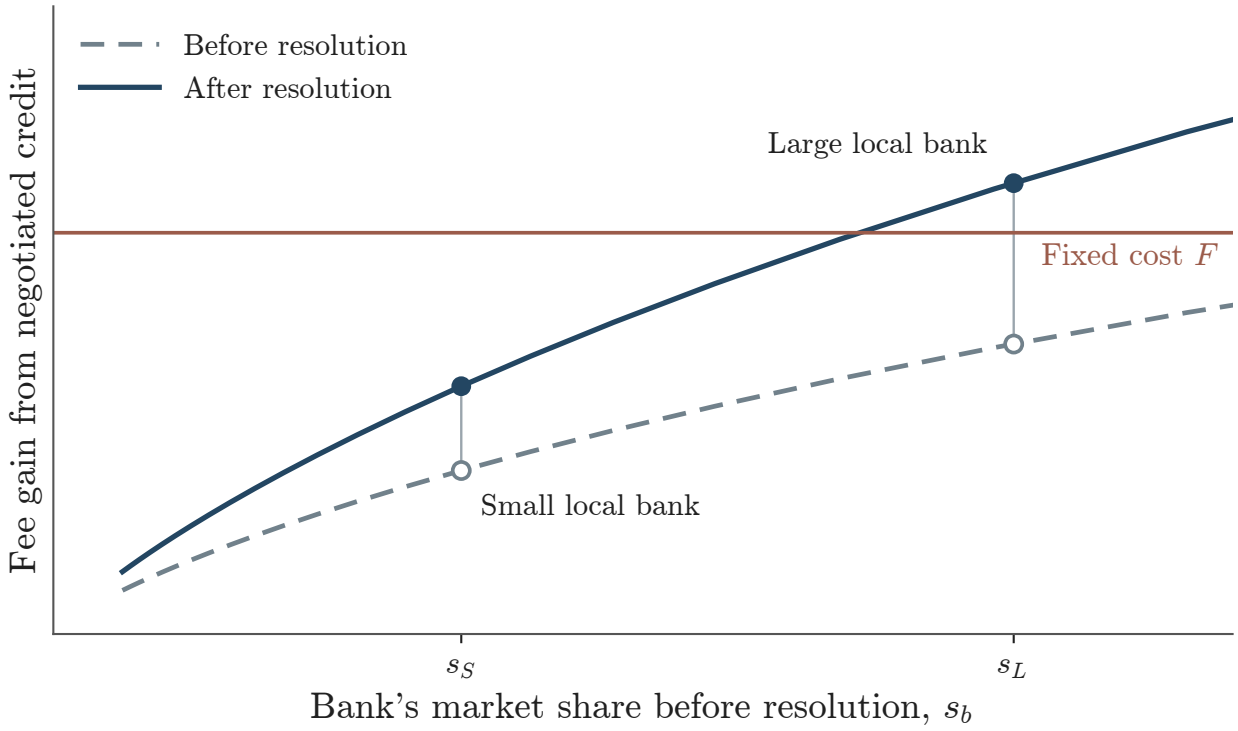
Notes: This size-by-quarter event study plots θ_k in equation (17). Monthly log adjusted-NAV returns (bp) for local-bank products are regressed on bank size interacted with quarter indicators. Each coefficient measures the change in the size–return slope relative to 2023Q3, per log point of FY2021 underlying-bank interest-bearing assets in RMB trillion, without standardization. The regression includes the eleven time-varying controls, institution effects, and bank-home-province-by-month effects, but no prior-reliance interactions. The sample is 2022Q2–2025Q3 (206,770 NAV-series-months; 101 underlying banks). Shading shows pointwise 95% confidence intervals based on underlying-bank clustered errors. The hollow diamond marks the normalized reference quarter, not an estimated zero. The vertical line marks implementation in 2023Q4.

Figure 6
Bank size and total WMP balances



Notes: This size-by-half-year event study plots ψ_ℓ in equation (19). The quarterly axis locates actual midyear and year-end reports at Q2 and Q4; no Q1 or Q3 balances are imputed. Bank size is interacted with reporting-date indicators in a log-balance regression with bank and half-year fixed effects and no additional controls, for 89 local banks over 2022H1–2025H2 (571 bank-half-years). Each coefficient measures the change in the size–balance slope relative to the June 2023 report (2023Q2), per log point of FY2021 interest-bearing assets in RMB trillion, without standardization. Balances combine a bank's own products and those of its wealth-management subsidiary. Shading shows pointwise 95% confidence intervals based on bank-clustered errors. The hollow diamond marks the reference; the vertical line separates 2023Q2 and Q3. The joint test that the two 2022 coefficients are zero has a p-value of 0.41. Source: Wind institution-level WMP statistics, with the bank-disclosure correction described in Appendix E.1.

Figure 7
The scale threshold for negotiated credit



Notes: This figure illustrates the entry condition in equation (32), using the full equilibrium in Internet Appendix B.6. The curves plot the fee gain from arranging negotiated credit for a bank of varying reputation that faces the three banks' equilibrium portfolios, against its market share before resolution, before resolution (dashed) and after resolution (solid). Resolution raises the gain because the bonds that negotiated credit displaces earn less. A bank enters when its gain covers the fixed cost F. Hollow and filled circles mark the two local banks before and after resolution: the large bank's gain crosses F, and the small bank's does not.

Internet Appendix

Local Credit Reliance and Unequal Adjustment:
Evidence from Wealth Management in China

The appendix separates supporting empirical exhibits (Internet Appendix A), the model solution and proofs (Internet Appendix B and Internet Appendix C), detailed specifications and supplementary results (Internet Appendix D), and return-measurement, balance-coverage, and participation-capacity checks (Internet Appendix E).

Internet Appendix A
Supporting Evidence

A.1. Coverage, investment opportunities, and denominators

The supporting evidence first establishes the coverage of the data and the investment opportunities available to WMPs. Table A1 defines the variables, including the distinction between a home share within one asset class and a home allocation across all included bonds and negotiated credit. Table A2 reports disclosure coverage by date and bank category. The universe contains 171 banks; each regression uses the subset with its required observations and controls. Table A3 summarizes holdings, products, and bond issues separately. Its issuer-size panel counts each issuer once and uses available FY2021 assets. The median LGFV issuer is smaller than other nonfinancial and financial issuers. The local-bond strategy thus concerns relatively small borrowers whose securities remain available to all eligible investors.

Table A15 asks whether these bonds also offer more contractual income. The sample consists of 20,172 unique renminbi fixed-rate bonds issued at par in 2022Q1–2023Q3. Average LGFV issuance yields exceed those of other nonfinancial and financial issuers by 122 and 149 basis points. Regressions of the annual issuance yield on issuer-type indicators, issue-month effects, maturity and its square, issue size, and call provisions retain gaps of 91 and 108 basis points. Adding reported ratings gives 88 and 110 basis points, with errors clustered by issuer. This is a comparison of contractual borrowing costs, not realized or risk-adjusted investment returns.

The supporting portfolio exhibits distinguish investment value, reported portfolio weights, and counts of disclosed positions. Over the full 2022Q1–2025Q3 sample, LGFV bonds account for 16.4% of local banks' summed reported portfolio weights, compared with 5.4% at national banks; negotiated credit accounts for 6.2% versus 11.6% (Table A11). These are shares across all disclosed positions, not the pre-program bond-only value shares in the introduction or the included-debt shares in Table D1. Figure A3 follows LGFV allocations using both broad

disclosed holdings and a bond denominator. The exhibits establish the same local-bank specialization while answering different questions about its scale.

Figure A1 places WMPs in the wider asset-management market and shows that bonds, cash and deposits, and certificates of deposit account for most industry assets. Its market-size categories can invest in one another and therefore are not additive. Figure A2 shows WMP balances over 2013–2025, with a break in industry definitions in 2018. Figure A5 then measures the importance of the WMP–LGFV link in both directions. Applying the disclosed bond mix to industry assets gives estimated LGFV bond holdings of 10.2% of WMP assets in 2024, equivalent to 28.1% of the LGFV bond market. Thus an asset class that is a minority of WMP portfolios can still receive a substantial share of its financing from WMPs. Appendix D.8 explains the extrapolation and the change in the external LGFV denominator source. The plot ends in 2024, the last year with a verified market total.

A.2. The local strategy, investor demand, and borrower selection

Local banks concentrate on products whose redemption terms allow less liquid investments. Restricted-redemption products account for 80.2% of their assets, compared with 47.3% at national banks (Table A6). Equation (3) tests whether this product choice accompanies greater home investment. Conditional on the product, bank, and city controls, the local–national home-allocation difference is 20.0 percentage points wider in restricted-redemption products than in daily-open products, including 7.7 points for LGFV bonds and 7.8 for negotiated credit (Table A7). The result connects product design to the local investment strategy: banks can hold less liquid local claims in products that do not promise daily withdrawals.

Table A8 asks what savers observed from this strategy beyond average returns. Using the full-period specification in equation (D.8), a 20-point higher home-LGFV tilt corresponds to roughly eight additional annual basis points and a 0.8-percentage-point lower probability of a negative month, against a 9.9% mean. Reported volatility and downside volatility are also lower. Home-bond exposure therefore combines higher reported returns with fewer observed monthly losses; these NAV measures do not establish lower underlying credit risk. Appendix D.6 defines the outcomes and distinguishes this full-period comparison from the pre-program premium.

Higher returns matter commercially if they attract money. Table A10 compares next-quarter flows across prior-return fifths within the same risk and redemption categories, using category-by-quarter effects and lagged controls. Top-ranked products receive 12.2% more of beginning assets in net flows than bottom-ranked products, and their probability of positive net inflows is 8.1 points higher. The top–bottom flow difference is larger in daily-open products than in restricted products (45.7% versus 4.4%); adding lifecycle controls to the main comparison retains a 12.3% difference. Equal bank-quarter weights and province-specific

local-bank peer groups give smaller, imprecise differences. Performance predicts flows in the full product market, with the strength of the association varying by product design and comparison group.

The pricing and distress tests ask why banks seek yield at home rather than buying similarly risky bonds elsewhere (Tables A12 and A13). The pricing outcome is the public valuation yield minus the issue yield: a larger concession means a higher purchase price and cheaper financing for the issuer. Greater ownership by banks elsewhere in the issuer's province is associated with better purchase terms relative to valuation; ownership by banks in the issuer's own city has the opposite association. Within issuer-city-by-quarter cells, shifting one percentage point of issue ownership toward same-city banks raises the concession by 1.01 basis points. The difference is consistent with local knowledge outside the issuer's city and additional borrower support within it. Holding public valuation risk and bond characteristics fixed, same-city ownership also predicts fewer first issuer credit events over the next eight quarters. This distress contrast is less precisely estimated than the pricing contrast. Together, the tests allow informed selection and support to coexist rather than treating either as the sole explanation for home investment.

Table A4 checks whether the distress result depends on the event horizon or on counting several bonds from one issuer. The same-city-minus-other-city contrast remains negative for four-quarter events, events observed through 2025Q4, and issuer-quarter aggregation, with issuer-clustered p-values of 0.084, 0.054, and 0.115. Its direction is stable, although precision varies across designs. Table A9 then asks how home banks respond when borrowers face repayment pressure. Per percentage point of the maturity wall, the home interaction is -0.00696 for net buying in percentage points and 0.00116 for the guarantee indicator. A ten-point larger wall therefore corresponds to about 0.07 points less home net buying and a 1.16-point higher guarantee probability on new home purchases; the new-purchase-count interaction is not significant. Home banks reduce net investment and seek more protection, showing that local support can coexist with selective lending. Appendix D.4 gives the definitions and calculates the corresponding standard-deviation comparisons.

A.3. Yield compression and substitution

The yield-compression exhibits establish what changed in the opportunity used by local banks. Figure A4 groups provinces by their initial LGFV spread; the highest–lowest gap falls from 564 basis points in 2023Q1 to 64 in 2025Q4. Table A5 relates province-quarter spreads to cumulative refinancing/GDP, with province and quarter effects. The baseline coefficient is -17.67 basis points per percentage point of refinancing/GDP. Controls for convergence from initial spreads and for provincial trends give coefficients of -2.43 and -8.65, respectively. Spreads decline more where refinancing is greater under each comparison, although the estimated magnitude depends on the trend controls.

Table A14 asks whether negotiated-credit expansion allowed larger local banks to maintain their overall investment in home borrowers. The outcome combines home LGFV bond and home negotiated-credit shares; the regression includes size–Post and reliance–Post interactions, institution and home-province-by-quarter effects, and the baseline controls. The size–Post coefficient is 4.76 percentage points per log point of assets, implying about 3.3 points for a doubling of bank size. The substitution is therefore not simply an equal exchange of negotiated credit for bonds: larger banks maintain more financing of the home borrower sector after its marketable bonds lose yield.

Table A1
Variable definitions

Variable	Definition
$\mathbb{1}\{\text{Local bank}\}_b$	Local-bank indicator: equals one for banks other than state-owned and joint-stock commercial banks.
$\text{Home bias}_{iq}^{n}$	Within-class home share minus the province's weight in the nationwide market portfolio. Positive values indicate home overweighting (equation (2)).
$\text{Home allocation bias}_{iq}^{n}$	Home class-n holdings as a fraction of all included debt, less the corresponding nationwide allocation to that province and class (equation (D.10)). Without n, pools all classes.
$\text{Initial home bias}_a$	Product-assets-weighted mean home allocation bias for issuing bank or subsidiary a, 2022Q1–2023Q3. Bank size is matched to underlying bank b.
Log bank size_b	Natural log of FY2021 underlying-bank interest-bearing assets (RMB trillion), without standardization.
$\text{Initial home LGFV share}_b$	Home LGFV bonds / included debt, holdings-value weighted over 2022Q1–2023Q2. Includes only standard bonds, without a market-weight adjustment.
$\mathbb{1}\{\text{Restricted redemption}\}_i$	Restricted-redemption indicator: equals one for products without daily redemption.
Monthly return	Log return computed from month-end adjusted NAVs in consecutive months, in basis points.
Trailing volatility	Annualized standard deviation of monthly returns over the twelve months ending in the outcome month.
Net flow_{iq}	Quarterly net flow: growth in inferred assets net of returns, in percent (equation (D.7)).
$\text{Yield concession}_{jq}$	Yield concession at issuance: 100 times the difference between the CSI valuation spread and the coupon-based issue spread, both stored in percentage points, yielding basis points.
$\text{Same-city ownership}_{jq}$	Share of bond j held by local banks in the issuer's city at the end of its issue quarter q, in percentage points of issue size.
$\text{Other-city provincial ownership}_{jq}$	Share held by local banks elsewhere in the issuer's province at the same date, in percentage points of issue size.
Distress	Equals 100 if the issuer records its first credit event within eight quarters after issuance, and zero otherwise.
$\mathbb{1}\{\text{Home buyer}\}_{ac}$	Equals one when WMP issuer a is a local bank or its subsidiary and invests in issuer city c of the underlying bank's headquarters province.
$\text{Maturity wall}_{cq}$	Scheduled principal on all LGFV bonds maturing in city c and quarter q, as a percentage of 2020 city budget revenue.
$\text{Refinancing}_{pq}^{GDP}$	Cumulative special-refinancing issuance / 2020 provincial GDP (percentage points). Final refinancing fixes this value at the last observed date.
$\text{Refinancing}_{pq}^{debt}$	Cumulative special-refinancing issuance / 2020 provincial LGFV debt (fraction), used for allocations and returns. Neither intensity measure is standardized.
$\mathbb{1}\{\text{Post}\}$	Equals one from 2023Q4 in quarterly data and from October 2023 in monthly data.
$h(m)$	The quarter preceding month m; exposures and controls for month m are measured at the end of that preceding quarter.
$\text{Log WMP balances}_{bs}$	Natural logarithm of the outstanding WMP balance of bank b at the end of half-year s, combining its own products and those of its wealth-management subsidiary.

Notes: Holdings cover 2022Q1–2025Q3. Included debt comprises bonds and identified negotiated claims with known borrower locations, excluding central-government, central-bank, and policy-bank obligations. Other centrally administered issuers are never home in the primary return tests. Issuer credit events include defaults, rating warnings, and guarantor events, not necessarily default on the held bond. bp and pp denote basis points and percentage points.

Table A2
Sample coverage

Panel A. Overall coverage

Measure	Count
Source holding records	2,288,894
Holding records with a usable product identifier	2,279,252
WMP products	60,396
Product-quarter observations	253,484
Reported institution names	193
Underlying banks	171
Bank headquarters provinces	30
Quarters	15

Panel B. Annual coverage

Year	Banks	Source holding records	WMP products
2022	141	421,072	19,514
2023	161	579,703	26,021
2024	149	708,907	30,512
2025 (Q1–Q3)	140	579,212	29,527

Panel C. Bank categories

Bank group	Banks	WMP products	Product-quarters
National banks	18	27,799	131,535
Large state-owned	6	9,571	49,593
Nationwide joint-stock	12	18,228	81,942
Local banks	153	32,597	121,949
Regional commercial	100	23,817	91,937
Rural institutions	53	8,780	30,012
All banks	171	60,396	253,484

Notes: Coverage is 2022Q1–2025Q3. A holding record is a product-asset-quarter observation. Source-record counts include 9,642 records with blank or "NA" product registration codes; usable-record, product, and product-quarter counts exclude these codes. Reported institution names include wealth-management subsidiaries and alternative bank names; the parent-bank crosswalk groups them into 171 banks. Panel B counts banks and products once per year; annual counts therefore need not sum to full-period counts. Panel C uses the source bank-type codes and assigns subsidiary products to the parent bank's category. Regional commercial banks include provincial and city-focused banks, which the source does not code as mutually exclusive geographic subgroups. Rural institutions include rural commercial and cooperative institutions. Guangfa Bank's wealth-management subsidiary is classified with nationwide joint-stock banks. Estimation samples impose the restrictions stated in each regression table.

Table A3
Descriptive statistics

Variable	Observations	Mean	SD	Median	p90
Panel A. Holding-observation characteristics					
Position weight (%)	2,286,025	7.12	12.03	4.04	11.46
Holding value (RMB million)	2,278,335	59.22	345.74	10.39	113.89
Panel B. Standard-bond characteristics					
Issue yield (%)	1,162,002	4.09	1.19	4.00	5.58
Issue maturity (years)	1,162,491	5.09	3.06	5.00	10.00
Panel C. Additional bond-issue characteristics					
$\mathbb{1}\{\text{Guaranteed}\}$	57,429	0.10	0.30	0.00	1.00
$\mathbb{1}\{\text{Callable}\}$	57,429	0.09	0.29	0.00	0.00
Number of WMP holders	57,429	0.23	0.52	0.00	1.00
Issue-quarter WMP ownership (pp)	57,429	1.52	4.59	0.00	5.02
Panel D. Product-quarter characteristics in the NAV sample					
Log product assets	153,563	19.41	1.64	19.38	21.51
Product risk rating	153,563	2.15	0.44	2.00	3.00
Restricted redemption (0/1)	153,563	0.88	0.33	1.00	1.00
Panel E. Pre-implementation issuers' FY2021 assets (RMB billion)					
LGFV issuers	1,939	42.79	48.09	28.41	88.39
Other nonfinancial issuers	2,062	134.48	548.95	35.32	243.34
Financial issuers	570	525.79	2,765.50	52.90	560.29

Notes: Panel A describes holding records and Panel B standard-bond holding records with the characteristic available. Panel C describes all 57,429 unique issues in the 2022Q1–2025Q4 bond database, before estimation restrictions and including issues without WMP ownership; pricing regressions end in 2025Q3. Panel D counts each product-quarter in the return sample once. Log product assets is $\log(1 + \text{inferred assets in RMB})$; product risk is the numerical rating and p90 the 90th percentile. The cross-sectional SD of products' average monthly returns is 10.5 bp. Among local-bank product-quarters before October 2023, mean home and non-home LGFV credit shares are 0.263 (home p90: 0.722) and 0.180. Mean issue rates are 5.62% for home LGFV holdings and 4.93% for other-province holdings; the mean within-product gap is 0.33 pp. Panel E separately describes firms issuing RMB debt in the available Wind issuance file over 2022Q1–2023Q3, not only WMP-held issues. Each issuer receives equal weight and appears once, matched by normalized exact name to reported year-end 2021 total assets. No future balance-sheet values are substituted. The LGFV master identifies the first group; source issuer types identify financial institutions and other enterprises. Central and local governments, policy banks, and unclassified issuers are excluded. Assets are observed for 1,939 of 1,964 LGFV issuers (98.7%), 2,062 of 2,167 other nonfinancial issuers (95.2%), and 570 of 723 financial issuers (78.8%).

Table A4

Issuer distress after issuance: windows, cohorts, and units of observation

Dependent variable / specification	Eight quarters	Four quarters	Through 2025Q4	Issuer-quarter; eight quarters
	(1)	(2)	(3)	(4)
Other-city provincial ownership (pp)	−0.007	0.029	0.027	0.013
	(0.043)	(0.027)	(0.034)	(0.064)
Same-city ownership (pp)	−0.112**	−0.038	−0.098**	−0.145*
	(0.056)	(0.025)	(0.048)	(0.088)
Log issue amount	0.770***	0.253*	0.683***	1.205***
	(0.259)	(0.147)	(0.186)	(0.435)
Issue maturity (years)	−0.351***	−0.195***	−0.329***	−0.450***
	(0.091)	(0.050)	(0.067)	(0.157)
𝟙{Guaranteed}	−0.676	−0.128	−0.346	−0.821
	(0.432)	(0.256)	(0.339)	(0.509)
𝟙{Callable}	−0.268	0.204	0.041	−0.517
	(0.474)	(0.302)	(0.349)	(0.822)
Number of WMP holders	0.607*	0.287	0.331	0.776
	(0.327)	(0.182)	(0.236)	(0.512)
Total WMP ownership (pp)	−0.072**	−0.053***	−0.058**	−0.125**
	(0.034)	(0.018)	(0.026)	(0.060)
𝟙{County-level issuer}	1.041	0.466	1.011	0.336
	(0.823)	(0.437)	(0.617)	(1.187)
𝟙{Prefecture-level issuer}	2.113**	1.420***	2.125***	1.690
	(0.972)	(0.536)	(0.718)	(1.399)
Coupon (%)	−0.544	−0.707*	−0.232	−0.773
	(0.456)	(0.402)	(0.428)	(0.890)
Public valuation spread (pp)	1.391***	1.352***	1.125***	2.090**
	(0.456)	(0.422)	(0.427)	(0.854)
Same-city minus other-city ownership	−0.105*	−0.067**	−0.125**	−0.158*
	(0.064)	(0.031)	(0.054)	(0.090)
Issuers	1,692	1,880	1,880	1,660
Issuers with an event	47	42	51	46
Issuer-clustered p: ownership contrast	0.132	0.084	0.054	0.115
Observations	10,857	16,185	16,185	5,563
Issuer city × issue quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Robust	Robust	Robust	Robust

Notes: Each column is a separate regression of subsequent issuer distress on same-city and other-city provincial ownership and the displayed controls, with issuer-city-by-issue-quarter effects. Ownership is measured in percentage points of issue size; distress is 100 if the stated first credit event occurs and zero otherwise. Column 1 uses a complete eight-quarter window, column 2 a complete four-quarter window, column 3 events observed through 2025Q4, and column 4 aggregates to issuer-quarter. Parentheses and stars use heteroskedasticity-robust errors; the separate row reports issuer-clustered tests of the ownership contrast. Contrasts use the full covariance matrix. The baseline and alternative fixed effects appear in Table A13. Stars denote 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2023Q4 (columns 1, 4); 2022Q1–2024Q4 (columns 2, 3).

Table A5
Refinancing intensity and provincial LGFV spreads

Trend controls	None	Pre-program spread trend	Province trends
Dependent variable	LGFV spread (bp)	LGFV spread (bp)	LGFV spread (bp)
	(1)	(2)	(3)
Refinancing / GDP (pp)	−17.665***	−2.434**	−8.652**
	(3.223)	(1.044)	(3.800)
Pre-program spread × linear trend		−13.550***	
		(1.017)	
Mean of dependent variable	195.694	195.694	195.694
Observations	496	496	496
Fixed effects	Province + quarter	Province + quarter	Province + quarter + province trends
Standard errors		Province	

Notes: This table tests whether provincial spread compression is greater where cumulative refinancing is more intensive. Estimates use equation (D.1). The unit of observation is a province-quarter. The dependent variable is the province's mean secondary-market LGFV spread in basis points. Refinancing intensity is cumulative special-refinancing issuance as a percentage of 2020 provincial GDP, without standardization. Spreads and intensity are winsorized at the 1st and 99th percentiles. Column 2 adds the 2022Q1–2023Q3 mean spread in basis points interacted with a linear quarterly trend, and column 3 includes a separate linear trend for each province. Standard errors clustered by province are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Estimation sample: 2022Q1–2025Q4.

Table A6
Redemption terms by bank group

Institution group	Redemption	Product-quarter observations	Equal-weighted share (%)	Size-weighted share (%)
National banks	Daily	17,294	20.5	52.7
National banks	Non-daily	67,054	79.5	47.3
Local banks	Daily	2,376	2.8	19.8
Local banks	Non-daily	83,273	97.2	80.2

Notes: This table identifies the redemption structure accommodating local banks' credit portfolios. Entries report the number and share of product-quarters by bank group and redemption terms. Daily products allow redemption on every business day; non-daily products are periodic-open or closed-end. Size-weighted shares weight product-quarters by inferred product assets. Shares sum to 100 within each bank group.

Table A7
Home bias and redemption terms

Asset class	All credit	LGFV	NS	Other standard
Dependent variable	Home allocation bias (fraction) (1)	Home allocation bias (fraction) (2)	Home allocation bias (fraction) (3)	Home allocation bias (fraction) (4)
$\mathbb{1}\{\text{Local bank}\}$	−0.063**	0.009	−0.021*	−0.050***
	(0.026)	(0.013)	(0.012)	(0.019)
Log bank assets	−0.016***	−0.004	0.018***	−0.030***
	(0.005)	(0.004)	(0.005)	(0.006)
$\mathbb{1}\{\text{Local bank}\} \times$ restricted redemption	0.200***	0.077***	0.078***	0.045***
	(0.024)	(0.011)	(0.014)	(0.012)
Log product size	−0.001	0.001	0.000	−0.002**
	(0.002)	(0.001)	(0.001)	(0.001)
$\mathbb{1}\{\text{Restricted redemption}\}$	−0.048***	−0.014***	0.010***	−0.045***
	(0.005)	(0.003)	(0.003)	(0.005)
Product risk rating	0.015***	0.000	0.012**	0.003
	(0.005)	(0.003)	(0.005)	(0.003)
Bank nonperforming-loan ratio (%)	−0.074***	−0.005	−0.044***	−0.022**
	(0.018)	(0.014)	(0.009)	(0.009)
Bank cost–income ratio (%)	−0.001***	−0.001**	−0.000	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Bank net interest margin (%)	0.040*	0.026*	0.007	0.004
	(0.022)	(0.016)	(0.010)	(0.011)
Home-city GDP (RMB trillion)	0.003	0.023***	−0.023***	0.003
	(0.008)	(0.006)	(0.003)	(0.004)
Home-city GDP growth (%)	0.013**	0.019***	−0.003	−0.003
	(0.005)	(0.004)	(0.002)	(0.003)
Home-city LGFV debt/GDP	0.189***	0.217***	0.019	−0.049***
	(0.023)	(0.018)	(0.014)	(0.017)
Home-city income (RMB 10,000)	−0.048***	−0.044***	0.007**	−0.010**
	(0.007)	(0.005)	(0.003)	(0.004)
Mean of dependent variable	0.191	0.092	0.031	0.069
Observations	169,997	169,458	168,327	169,997
Fixed effects		Quarter		
Standard errors		Bank city × quarter		

Notes: NS denotes negotiated credit. This table tests whether restricted redemption is associated with a wider local–national home-credit gap. Estimates use equation (3). The unit of observation is a product-quarter. The dependent variable Home allocation bias$_{iq}^{n}$ is the product's home-province holdings of the asset class divided by the value of all its included debt holdings, minus the home province's holdings of the class as a share of nationwide included debt holdings, as a fraction; the first column pools all included debt holdings. Non-daily redemption equals one for products without daily redemption, so the local-bank coefficient measures the difference between local and national banks among daily-open products and the interaction measures the additional difference among other products. All columns use the sample with nonmissing controls. Standard errors clustered at the bank-city-by-quarter level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Estimation sample: 2022Q1–2025Q3.

Table A8
Local LGFV exposure and product returns

Panel A. Monthly returns and trailing volatility

Dependent variable	Monthly NAV return (bp) (1)	Trailing annualized volatility (bp) (2)
Home LGFV allocation bias (fraction)	3.140***	−11.150***
	(0.783)	(2.203)
𝟙{Local bank}	−0.076	5.279***
	(0.627)	(1.568)
Log product size	−0.548***	−3.043***
	(0.103)	(0.190)
𝟙{Restricted redemption}	4.863***	−0.351
	(0.709)	(1.185)
Product risk rating	−2.425**	57.090***
	(1.104)	(1.317)
Log bank assets	−0.489	5.597***
	(0.360)	(0.723)
Bank nonperforming-loan ratio (%)	0.790	−9.203***
	(0.695)	(1.878)
Bank cost–income ratio (%)	0.037**	0.272***
	(0.017)	(0.070)
Bank net interest margin (%)	0.330	11.032***
	(0.889)	(2.738)
Home-city GDP (RMB trillion)	0.233	2.272***
	(0.294)	(0.770)
Home-city GDP growth (%)	0.183	−1.452**
	(0.239)	(0.726)
Home-city LGFV debt/GDP	4.885***	−19.937***
	(1.009)	(2.923)
Home-city income (RMB 10,000)	−0.304	−3.980***
	(0.303)	(0.720)
Mean of dependent variable	23.873	75.758
Observations	554,555	554,555
Fixed effects	Month	
Standard errors	Bank city × month	

Notes: These pooled-period regressions relate home LGFV tilt to product returns and downside risk. Panel A reports equation (D.8) on NAV-series-months over 2022Q2–2025Q4, including all redemption types and products without LGFV holdings. Local LGFV tilt Home allocation bias$^{LGFV}_{i,h(m)}$ is home-province LGFV holdings divided by included debt holdings, minus the home province's LGFV holdings as a share of nationwide included debt holdings. Tilt and controls are measured at the preceding quarter end. Monthly returns are consecutive-month changes in log adjusted NAV. Trailing volatility is the annualized standard deviation of returns over the twelve months ending in month m, requiring at least two available returns. Both outcomes are in basis points. The common sample excludes observations outside the 1st–99th percentile ranges of returns or volatility; it trims observations rather than capping their values. Standard errors clustered by bank-city-by-month cells are in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%. Estimation sample: 2022Q2–2025Q4.

Table A8 (continued)

Panel B. Downside outcomes

Dependent variable	Monthly excess return (bp) (1)	Downside volatility (bp) (2)	Drawdown (log gap ×100) (3)	Negative return (0/1) (4)	Bottom-decile return (0/1) (5)	Drop over 50 bp (0/1) (6)
Home LGFV allocation bias (fraction)	3.817***	−6.942***	−0.061***	−0.039***	−0.039***	−0.008***
	(0.784)	(1.213)	(0.009)	(0.009)	(0.008)	(0.002)
𝟙{Local bank}	0.259	1.929**	0.009	0.012	0.006	−0.001
	(0.618)	(0.893)	(0.007)	(0.008)	(0.007)	(0.002)
Log product size	−0.390***	−1.203***	−0.008***	−0.004***	−0.006***	−0.001***
	(0.102)	(0.137)	(0.001)	(0.001)	(0.001)	(0.000)
𝟙{Restricted redemption}	−1.139**	−3.240***	−0.059***	−0.001	−0.041***	−0.004**
	(0.471)	(0.888)	(0.011)	(0.007)	(0.010)	(0.002)
Product risk rating	−1.943*	27.823***	0.158***	0.110***	0.070***	0.020***
	(1.077)	(1.496)	(0.016)	(0.011)	(0.012)	(0.003)
Log bank assets	−0.356	2.973***	0.013***	0.016***	0.009***	−0.001
	(0.361)	(0.387)	(0.004)	(0.005)	(0.003)	(0.001)
Bank nonperforming-loan ratio (%)	0.975	−5.667***	−0.033***	−0.016**	−0.000	−0.006**
	(0.683)	(1.027)	(0.007)	(0.008)	(0.007)	(0.003)
Bank cost–income ratio (%)	0.046***	0.115**	−0.000	−0.000	0.000	0.000
	(0.016)	(0.052)	(0.000)	(0.000)	(0.000)	(0.000)
Bank net interest margin (%)	0.526	4.497***	0.008	0.015	0.014	0.003
	(0.881)	(1.359)	(0.009)	(0.009)	(0.009)	(0.004)
Home-city GDP (RMB trillion)	0.075	0.885**	0.001	0.002	0.001	0.002*
	(0.296)	(0.427)	(0.003)	(0.003)	(0.003)	(0.001)
Home-city GDP growth (%)	0.218	−1.074***	−0.008***	−0.001	0.002	−0.001
	(0.240)	(0.389)	(0.003)	(0.002)	(0.002)	(0.001)
Home-city LGFV debt/GDP	3.954***	−9.184***	−0.037***	−0.048***	−0.040***	−0.008**
	(1.010)	(1.547)	(0.011)	(0.012)	(0.008)	(0.003)
Home-city income (RMB 10,000)	−0.138	−1.891***	−0.009***	−0.010***	−0.006**	−0.001*
	(0.303)	(0.402)	(0.003)	(0.003)	(0.003)	(0.001)
Mean of dependent variable	0.271	24.489	0.088	0.099	0.092	0.015
Observations	558,455	558,455	558,455	558,455	558,455	558,455
Fixed effects			Month			
Standard errors			Bank city × month			

Notes: The six columns in Panel B share a common NAV-series-month sample, distinct from Panel A's return–volatility sample. Tilt, timing, and controls are as in Panel A. Excess return subtracts the mean monthly return of products with the same redemption terms. Downside volatility is the annualized standard deviation of $\min(\text{Monthly return}_{im}, 0)$ over the twelve months ending in month m, where $\text{Monthly return}_{im}$ is the monthly return. Drawdown is 100 times the gap between the trailing maximum and current log NAV. The final three outcomes are indicators for a negative return, a bottom-decile return within the month, and a return below −50 bp; their coefficients are probability changes, not percentage-point returns. Standard errors clustered by bank-city-by-month cells are in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

Table A9
Refinancing pressure and home-province allocation: all-issuance wall

Panel A. Allocation margins

Dependent variable	Log(1 + new purchases)	New purchases (%)	Net purchases (%)	𝟙{Buy}	Exits (%)	Holdings (%)
	(1)	(2)	(3)	(4)	(5)	(6)
𝟙{Home buyer} × maturity wall	0.00011	−0.00088	−0.0070***	0.00053	0.0045***	−0.0087*
	(0.00218)	(0.00120)	(0.0018)	(0.00046)	(0.0013)	(0.0046)
Mean of dependent variable	0.937	0.120	−0.023	0.261	0.155	0.671
Observations	73,127	73,089	66,013	73,127	66,013	73,089
Bank and city controls	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Fixed effects	Bank × city; bank × quarter; city × quarter					
Standard errors	Bank city					

Notes: Bank-city-quarter regressions. Home denotes a local bank investing in its own province. The wall is scheduled principal on all city LGFV bonds maturing in the quarter as a percentage of 2020 city budget revenue. Bank-city, bank-quarter, and issuer-city-quarter effects enter. The Buy indicator equals one for any new purchase. Net buying removes valuation changes; exits include nonrenewal. Parentheses contain bank-city clustered errors. Appendix D.4 defines each outcome. Stars indicate 10%, 5%, and 1% significance. Estimation samples: Panel A, 2022Q1–2025Q3 (columns 1, 2, 4, 6); 2022Q2–2025Q3 (columns 3, 5); Panel B, 2022Q1–2025Q3.

Panel B. Characteristics of purchased bonds

Dependent variable	Issuer rating (1–4)	𝟙{Guaranteed}	Maturity (years)	Issue spread (pp)	Log issue amount
	(1)	(2)	(3)	(4)	(5)
𝟙{Home buyer}	−0.1972***	0.0389**	−0.0676	0.1507***	−0.0901***
	(0.0316)	(0.0172)	(0.0599)	(0.0454)	(0.0254)
𝟙{Home buyer} × maturity wall	0.00119*	0.00116**	0.0027*	−0.00040	0.000042
	(0.00060)	(0.00046)	(0.0014)	(0.00080)	(0.000512)
Mean of dependent variable	2.8405	0.1540	3.6470	1.6174	2.0001
Observations	33,209	36,303	36,303	36,303	36,303
Bank and city controls	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Fixed effects	Bank × quarter; issuer city × quarter				
Standard errors	Bank city				

Notes: New bond purchases. Outcomes are characteristics of the bonds purchased; issuer rating, guarantees, maturity, spread, and size are not city characteristics. The regressions include Home, its wall interaction, bank-quarter and issuer-city-quarter effects, but not bank-city effects. The wall has the same percentage-point definition as in Panel A; neither panel standardizes it. Spreads are percentage points; guarantees are indicators. Parentheses contain bank-city clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table A10
Relative performance and subsequent investor flows

Panel A. Performance ranks and alternative flow measures

Dependent variable	Net flow (% initial assets) (1)	Net flow (% initial assets) (2)	Symmetric net flow (pp) (3)	Positive-flow probability (pp) (4)
Prior-quarter performance: bottom fifth omitted				
$\mathbb{1}\{\text{Return quintile} = 2\}$	1.915***	1.913***	1.425***	1.715***
	(0.632)	(0.609)	(0.513)	(0.636)
$\mathbb{1}\{\text{Return quintile} = 3\}$	2.942***	2.955***	2.873***	4.471***
	(0.806)	(0.770)	(0.539)	(0.934)
$\mathbb{1}\{\text{Return quintile} = 4\}$	5.218***	5.449***	4.754***	5.800***
	(1.178)	(1.185)	(0.768)	(1.056)
$\mathbb{1}\{\text{Return quintile} = 5\}$	12.177***	12.815***	8.711***	8.146***
	(2.723)	(2.783)	(1.521)	(1.672)
Log beginning product assets	−7.512***	−7.773***	−4.611***	−3.795***
	(1.010)	(1.002)	(0.573)	(0.340)
Product risk rating	Absorbed	Absorbed	Absorbed	Absorbed
Restricted redemption	Absorbed	Absorbed	Absorbed	Absorbed
Log bank assets	4.122***	4.523***	3.307***	3.454**
	(1.136)	(1.303)	(0.825)	(1.462)
Bank nonperforming-loan ratio (%)	−1.112	−0.585	−0.852	0.919
	(2.048)	(3.376)	(1.504)	(2.418)
Bank cost–income ratio (%)	−0.153**	−0.135*	−0.130**	−0.082
	(0.069)	(0.080)	(0.055)	(0.060)
Bank net interest margin (%)	0.948	−0.293	−0.932	2.060
	(2.093)	(1.950)	(1.283)	(2.465)
Home-city GDP (RMB trillion)	−0.764	0.700	−0.699	2.301
	(0.766)	(2.095)	(0.613)	(1.434)
Home-city GDP growth (%)	−0.222	−0.206	−0.104	0.129
	(0.438)	(0.605)	(0.323)	(0.686)
Home-city LGFV debt/GDP	−1.385	−3.140	1.713	4.399
	(2.887)	(3.851)	(1.933)	(5.510)
Home-city income (RMB 10,000)	−0.036	−3.750**	−0.325	−3.055**
	(0.685)	(1.812)	(0.476)	(1.342)
$\mathbb{1}\{\text{Local bank}\}$	4.467*	3.805	5.177***	15.079***
	(2.652)	(3.268)	(1.969)	(3.914)
Underlying banks	129	129	129	129
Mean of dependent variable	1.764	1.760	−4.648	43.102
Top-fifth wild-bootstrap p	< 0.001	—	< 0.001	< 0.001
Observations	123,745	123,738	123,745	123,745
Category × quarter fixed effects	Yes	Yes	Yes	Yes
Home province × quarter fixed effects	No	Yes	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one regression of next-quarter flows on prior-quarter return fifths within risk and redemption categories; the bottom fifth is omitted. Controls are displayed and dated before flows; category-by-outcome-quarter effects enter, and column 2 adds home-province-by-quarter effects. Parentheses contain parent-bank clustered errors. Ordinary flows are capped at 1/99; symmetric flows are bounded and uncapped. Positive flow is in percentage points. Appendix D.5 defines the measures. Stars indicate 10%, 5%, and 1% significance. Estimation samples: Panel A, 2022Q3–2025Q3.

Table A10 (continued)

Panel B. Next-quarter net flow (%)

Specification / outcome	Daily-open (1)	Restricted (2)	Lifecycle (3)	Equal bank-quarter (4)	Local province peers (5)
$\mathbb{1}\{\text{Return quintile} = 2\}$	2.947	1.593***	1.326**	1.110	−0.447
	(2.156)	(0.558)	(0.657)	(1.512)	(1.089)
$\mathbb{1}\{\text{Return quintile} = 3\}$	−0.040	3.450***	2.489***	0.198	2.012*
	(2.838)	(0.628)	(0.797)	(1.207)	(1.189)
$\mathbb{1}\{\text{Return quintile} = 4\}$	10.293***	4.250***	5.152***	2.341	1.517
	(3.240)	(0.899)	(1.148)	(2.050)	(1.561)
$\mathbb{1}\{\text{Return quintile} = 5\}$	45.721***	4.449***	12.331***	2.622	1.059
	(7.666)	(1.072)	(2.630)	(1.904)	(1.738)
Baseline control groups	Yes	Yes	Yes	Yes	Yes
Observations	21,101	102,644	123,745	123,745	29,934
Lifecycle controls	No	No	Yes	No	No
Category × quarter fixed effects	Yes	Yes	Yes	Yes	Yes
Home province × quarter fixed effects	No	No	No	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Panel B reports the key ranking estimates from five separate regressions, retaining the baseline control groups and peer-category-by-quarter effects. Lifecycle adds age and remaining-maturity bins. Equal-bank-quarter weights sum to one within each bank-quarter. The last column ranks local-bank products within provincial peers and absorbs home-province-by-quarter effects. Other control coefficients are omitted in this robustness panel. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table A11
Composition of disclosed WMP holdings by bank group

Asset category	National banks	Local banks	All banks
Panel A. Mean quarterly reported value (RMB trillion)			
Standard LGFV bonds	0.15	0.27	0.43
Other standard bonds	0.91	0.47	1.39
Negotiated credit	0.32	0.11	0.43
Other disclosed assets	5.34	1.41	6.75
Panel B. Portfolio-weight share (%)			
Standard LGFV bonds	5.4	16.4	11.6
Other standard bonds	16.3	23.4	20.3
Negotiated credit	11.6	6.2	8.5
Other disclosed assets	66.7	54.0	59.6
Panel C. Holding-observation share (%)			
Standard LGFV bonds	8.6	25.6	16.6
Other standard bonds	31.4	37.3	34.2
Negotiated credit	4.5	2.2	3.4
Other disclosed assets	55.4	34.9	45.8

Notes: This table places the debt-allocation regressions in the broader composition of disclosed product assets. Entries describe the composition of disclosed WMP holdings for products issued by national banks (state-owned and joint-stock commercial banks), local banks (all other banks), and all banks, pooled over 2022Q1–2025Q3. Each holding record is assigned to one asset category. Panel A reports the average quarterly disclosed value in RMB trillion, Panel B the share of disclosed portfolio weight, and Panel C the share of holding records, all within bank group. Portfolio weights are the position weights reported by products. Other disclosed assets include deposits, funds, trusts, and other investment vehicles.

Table A12
Local WMP ownership and issue pricing

Specification	LGFV: baseline	LGFV: controls	LGFV: province × quarter	LGFV: city × quarter	Non-LGFV
Dependent variable	Yield concession (bp)	Yield concession (bp)	Yield concession (bp)	Yield concession (bp)	Yield concession (bp)
	(1)	(2)	(3)	(4)	(5)
Other-city provincial ownership (pp)	−0.0740	−0.2456***	−0.2821***	−0.4039***	−0.0812
	(0.0883)	(0.0942)	(0.0994)	(0.1006)	(0.1334)
Same-city ownership (pp)	0.8596**	0.5311	0.3606	0.6088*	0.0293
	(0.3508)	(0.3422)	(0.3483)	(0.3261)	(0.0898)
Log issue amount		0.2738	−0.1392	−0.5084	1.2756***
		(0.5531)	(0.5284)	(0.5133)	(0.4907)
Issue maturity (years)		5.2971***	5.3751***	5.8458***	0.2808***
		(0.1970)	(0.2000)	(0.1897)	(0.0907)
𝟙{Guaranteed}		−9.2684***	−7.8799***	−7.9945***	1.9214*
		(0.8286)	(0.8151)	(0.8492)	(1.0386)
𝟙{Callable}		−0.0602	0.5658	2.6258**	−8.5134***
		(1.0159)	(1.0828)	(1.0473)	(0.6729)
Number of WMP holders		−1.9114***	−1.6755***	−1.2945**	−1.4005**
		(0.6499)	(0.6349)	(0.6413)	(0.6700)
Total issue-quarter WMP share (pp)		−0.0286	−0.0395	−0.0312	−0.0303
		(0.0645)	(0.0654)	(0.0656)	(0.0456)
𝟙{County-level issuer}		1.7540	7.7765***	7.0148***	
		(1.6906)	(2.2917)	(2.0447)	
𝟙{Prefecture-level issuer}		1.6979	7.2669***	3.3418	
		(1.7640)	(2.4089)	(2.0974)	
Coupon (%)		11.0794***	7.4932***	3.4766***	4.3664***
		(0.6636)	(0.5914)	(0.6285)	(1.1202)
Same city minus other city	0.9335***	0.7767**	0.6427*	1.0127***	0.1105
	(0.3611)	(0.3444)	(0.3579)	(0.3367)	(0.1525)
Equality test: $p(\beta_C = \beta_P)$	0.0097	0.0242	0.0725	0.0026	0.4685
Mean of dependent variable	19.4802	19.4802	19.4358	19.1572	11.7806
Observations	21,116	21,116	21,090	20,526	30,660
Fixed effects	Issue quarter	Issue quarter	Province × quarter	City × quarter	City × quarter
Standard errors			Robust		

Notes: Equation (6) compares same-city and other-city provincial ownership to assess information and borrower support. Observations are issues in 2022Q1–2025Q3; later issues lack holdings coverage and are excluded, not assigned zero ownership. Yield concession is 100 times the difference between the CSI valuation spread and the coupon-based issue spread, both in percentage points, giving bp. A larger concession means a lower issue yield, hence a higher price relative to the valuation benchmark. Ownership is issue-quarter-end WMP holdings, in pp of issue size, not primary orders. Same-city ownership is positive for 2.7% of LGFV issues; its conditional mean is 9.8 pp. Columns 1–4 use LGFVs and column 5 other issuers. The common LGFV sample has 21,116 issues; removing singletons leaves 20,526 in column 4. All nine controls are displayed; blanks mean absent or absorbed. The contrast is $\beta_C - \beta_P$. Parentheses and stars use heteroskedasticity-robust inference; issuer clustering gives contrast SE 0.355 and $p = 0.0044$ in column 4. Stars denote 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table A13
Local WMP ownership and subsequent issuer distress

Specification	LGFV: baseline	LGFV: controls	LGFV: province × quarter	LGFV: city × quarter	Non-LGFV
Dependent variable	Distress within 8 quarters (%)	Distress within 8 quarters (%)	Distress within 8 quarters (%)	Distress within 8 quarters (%)	Distress within 8 quarters (%)
	(1)	(2)	(3)	(4)	(5)
Other-city provincial ownership (pp)	−0.1018***	−0.1186***	−0.0561	−0.0072	−0.0026**
	(0.0279)	(0.0358)	(0.0392)	(0.0432)	(0.0011)
Same-city ownership (pp)	−0.1221***	−0.2163***	−0.0997**	−0.1117**	−0.0045***
	(0.0179)	(0.0374)	(0.0444)	(0.0560)	(0.0017)
Log issue amount		1.1398***	0.6484**	0.7698***	−0.0310*
		(0.2570)	(0.2565)	(0.2587)	(0.0164)
Issue maturity (years)		−0.5663***	−0.2167**	−0.3512***	−0.0162*
		(0.0900)	(0.0878)	(0.0906)	(0.0085)
𝟙{Guaranteed}		−0.3580	−0.6696*	−0.6761	−0.0705***
		(0.3952)	(0.3998)	(0.4316)	(0.0241)
𝟙{Callable}		−1.3133***	−0.7909	−0.2680	−0.0780
		(0.4775)	(0.5090)	(0.4737)	(0.0629)
Number of WMP holders		0.5738	0.7151*	0.6072*	0.0295
		(0.4056)	(0.3651)	(0.3266)	(0.0258)
Total issue-quarter WMP share (pp)		−0.0400	−0.0789**	−0.0721**	−0.0015
		(0.0330)	(0.0330)	(0.0338)	(0.0013)
𝟙{County-level issuer}		2.6641***	3.7025***	1.0408	
		(0.5277)	(0.7866)	(0.8230)	
𝟙{Prefecture-level issuer}		4.6174***	4.1417***	2.1131**	
		(0.5760)	(0.8690)	(0.9716)	
Coupon (%)		0.0466	−0.6141	−0.5440	0.0772*
		(0.5078)	(0.4694)	(0.4558)	(0.0448)
Public valuation spread (pp)		1.6083***	1.4589***	1.3906***	0.0286
		(0.4807)	(0.4580)	(0.4561)	(0.0322)
Same city minus other city	−0.0203	−0.0978***	−0.0436	−0.1045*	−0.0019*
	(0.0314)	(0.0370)	(0.0498)	(0.0635)	(0.0011)
p-value, issuer clusters	0.575	0.044	0.448	0.132	0.305
Mean of dependent variable	2.6679	2.6679	2.6712	2.7171	0.0338
Observations	11,170	11,170	11,156	10,857	14,776
Fixed effects	Issue quarter	Issue quarter	Province × quarter	City × quarter	City × quarter
Standard errors			Robust		

Notes: Equation (6) tests whether favorable same-city pricing accompanies lower subsequent issuer distress conditional on observed issue risk. Observations are bonds issued in 2022Q1–2023Q4 by previously undistressed issuers. The outcome equals 100 if the issuer records its first credit event within eight quarters, and zero otherwise. Events include defaults, rating warnings, and guarantor events, observed through 2025Q4; every bond has a complete window. Ownership and controls follow Table A12, adding the CSI valuation spread at issuance to control for ex ante risk. Columns 1–4 use LGFV bonds and column 5 other issuers. Same city minus other city reports $\beta_C - \beta_P$; the next row gives its issuer-clustered p-value. Parentheses contain heteroskedasticity-robust standard errors. Stars denote 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2023Q4.

Table A14
Bank size and combined home local credit

Dependent variable / specification	Home LGFV plus NS share (pp)
	(1)
Log bank size × 𝟙{Post}	4.761***
	(0.628)
Initial home bias × 𝟙{Post}	−6.493**
	(2.879)
Log product assets	−2.289***
	(0.292)
𝟙{Restricted redemption}	12.213***
	(1.582)
Product risk rating	5.048***
	(0.909)
Log bank assets	−9.400
	(6.278)
Bank nonperforming-loan ratio	3.607
	(2.695)
Bank cost–income ratio	−0.143
	(0.178)
Bank net interest margin	5.396*
	(2.863)
Home-city GDP (RMB trillion)	10.200
	(9.296)
Home-city GDP growth (%)	−1.645***
	(0.565)
Home-city LGFV debt/GDP	−25.496**
	(12.169)
Home-city income (RMB 10,000)	46.474***
	(12.043)
Observations	83,027
Institution fixed effects	Yes
Home province × quarter fixed effects	Yes
Standard errors	Bank city × quarter

Notes: One regression of the combined home-LGFV and home-NS share of included debt (pp), for local-bank products. Size and prior reliance are interacted with Post, which begins in 2023Q4. Size is unstandardized log FY2021 parent-bank assets in RMB trillion. All eleven controls, institution effects, and home-province-by-quarter effects are included and displayed. Parentheses contain bank-city-by-quarter clustered errors; the size interaction has an underlying-bank clustered p-value of 0.00002. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3. Regressions weight observations by included debt value.

Table A15
LGFV bond issuance yields relative to other issuers

Panel A. Fixed-rate bonds issued at par, 2022Q1–2023Q3

	LGFV	Other nonfinancial	Financial
Mean issuance yield (%)	4.26	3.03	2.77
Median issuance yield (%)	3.92	2.79	2.75
Mean maturity (years)	2.28	1.79	2.12
Bond issues	8,044	10,832	1,296
Issuers	1,581	1,523	205

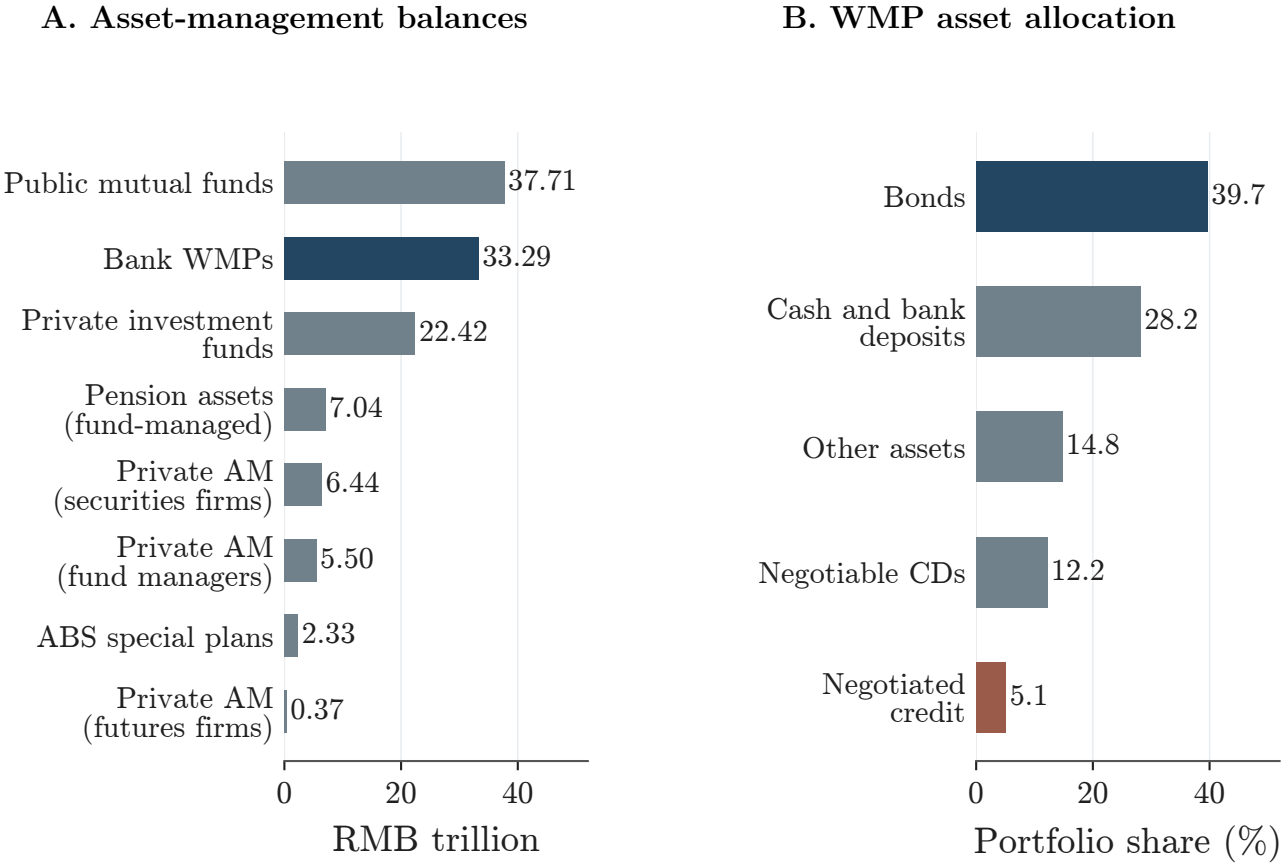
Panel B. Yield differences and bond characteristics

Dependent variable / specification	Issuance yield (pp) (1)	Issuance yield (pp) (2)	Issuance yield (pp) (3)
$\mathbb{1}\{\text{LGFV issuer}\}$	1.224***	0.910***	0.875***
	(0.069)	(0.089)	(0.093)
$\mathbb{1}\{\text{Financial issuer}\}$	−0.265***	−0.173***	−0.226***
	(0.054)	(0.057)	(0.063)
Maturity (years)	—	0.282***	0.277***
		(0.023)	(0.028)
Maturity squared	—	−0.009***	−0.009***
		(0.001)	(0.001)
Log issue amount	—	−0.280***	−0.264***
		(0.038)	(0.036)
$\mathbb{1}\{\text{Callable}\}$	—	0.378***	0.398***
		(0.099)	(0.099)
LGFV minus financial-issuer yield	1.489***	1.084***	1.101***
	(0.061)	(0.099)	(0.117)
Observations	20,172	20,172	20,172
Month fixed effects	No	Yes	Yes
Reported rating fixed effects	No	No	Yes
Standard errors	Issuer	Issuer	Issuer

Notes: Panel A describes unique CNY fixed-rate bonds issued at par in 2022Q1–2023Q3. Panel B regresses issuance yields (pp) on issuer-type indicators, with other nonfinancial issuers omitted. All columns use the same sample; later columns add the displayed issue characteristics, issue-month effects, and reported ratings (including an unreported category). Issuer assets need not be observed. CDs, convertibles, exchangeables, structured, perpetual, subordinated, floating-rate, and reset-rate bonds are excluded. There is no winsorization. Parentheses contain issuer-clustered errors; stars indicate 10%, 5%, and 1% significance. For these fixed-rate, at-par issues, the stated rate equals yield at issuance; it is not a realized return. Estimation sample: 2022Q1–2023Q3.

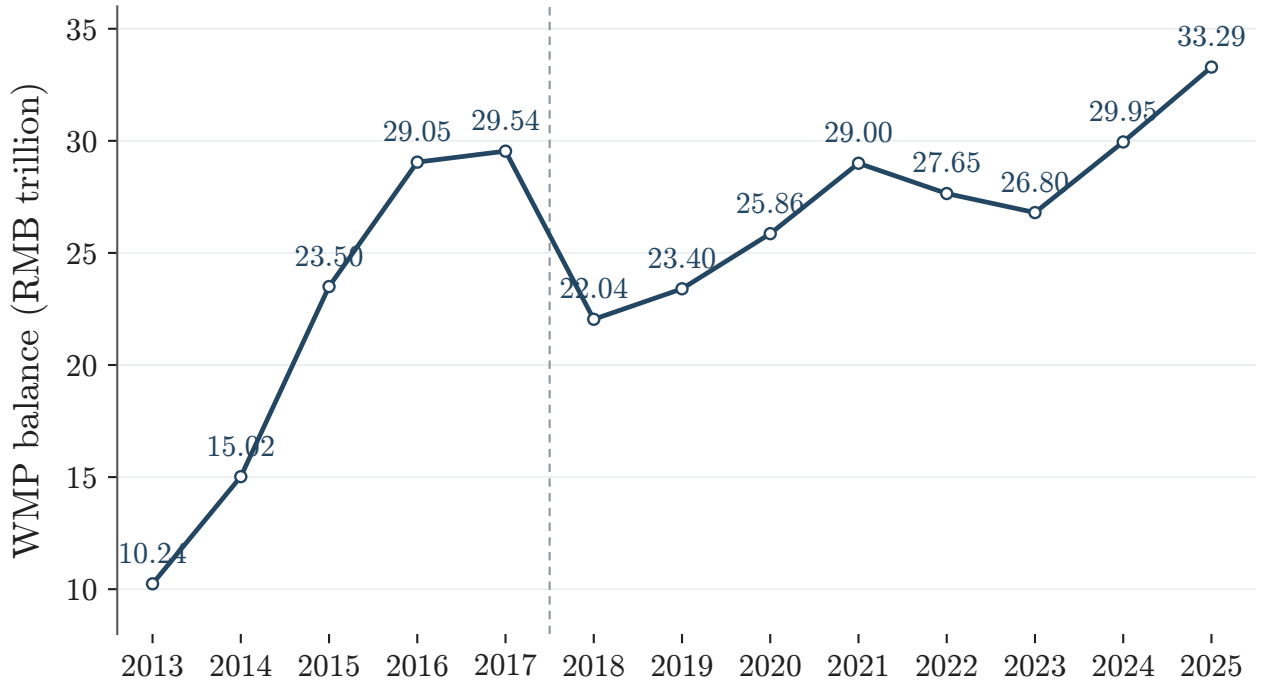
Figure A1

Chinese asset-management products and WMP asset allocation, 2025



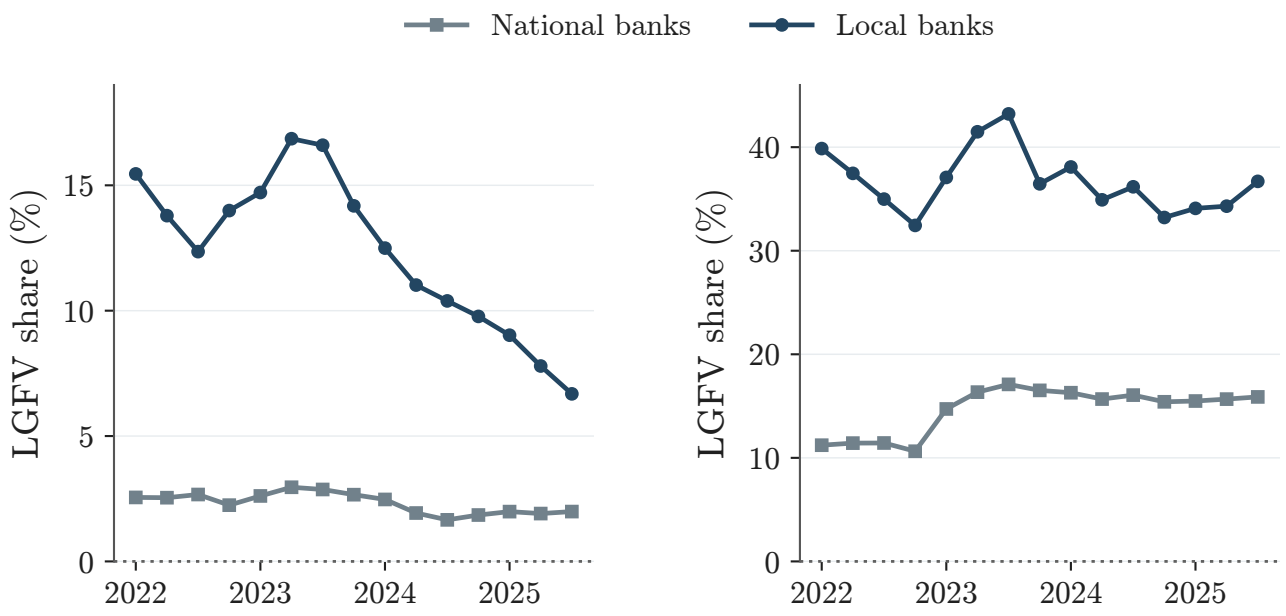
Notes: Panel A reports assets under management of Chinese asset-management products at the end of 2025, in RMB trillion, and Panel B reports the asset allocation of bank WMPs, in percent. Balances overlap through cross-investment. Sources: Asset Management Association of China and annual WMP market reports.

Figure A2
Bank WMP balances, 2013–2025



Notes: The figure plots outstanding bank WMP balances at year end, in RMB trillion. The dashed line marks a change in the statistical definition in 2018, when guaranteed and certain structured products were reclassified outside bank wealth management.

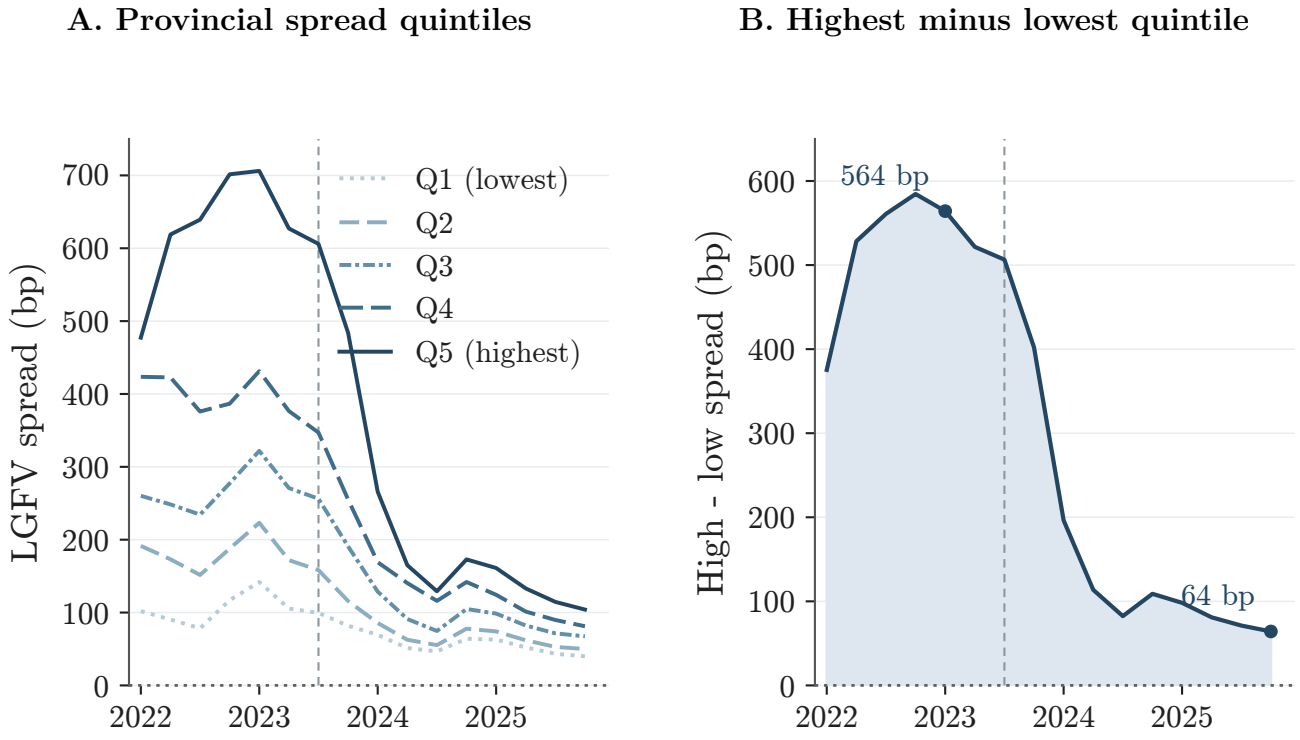
Figure A3
LGFV share of disclosed WMP holdings by bank group



Notes: The figure plots LGFV bonds as a share of the disclosed value of all holdings (Panel A) and of standard bonds excluding certificates of deposit (Panel B), by bank group, at each quarter end.

Figure A4

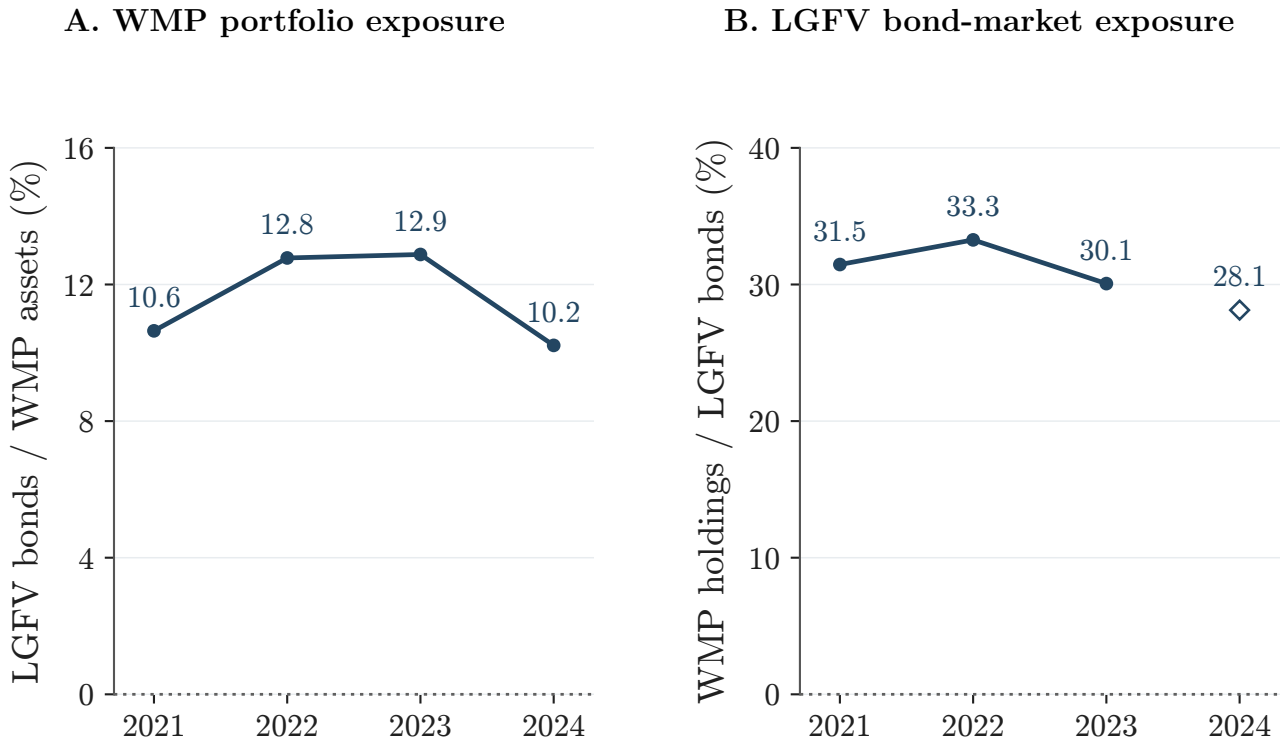
Provincial LGFV spreads by pre-program spread quintile



Notes: Provinces are grouped into quintiles of their mean LGFV spread over 2022Q1–2023Q2. Panel A plots the mean spread of each quintile, and Panel B plots the difference between the highest and lowest quintiles. The highest-minus-lowest gap falls from 564.2 bp in 2023Q1 to 64.2 bp in 2025Q4. The dashed line marks 2023Q3.

Figure A5

The scale of the WMP–LGFV bond link, 2021–2024



Notes: Both panels use the holdings-based extrapolation $\widehat{H}_t = B_t q_t$ in Internet Appendix D.8, not directly observed industry holdings. The left panel divides by official WMP net assets under management; the right divides by the external LGFV bond balance. The open diamond marks the 2024 balance-source change. Sources: Wind disclosures, annual WMP market reports, Lianhe Ratings (2024), and Zhang (2025).

Internet Appendix B
Local Reliance and Unequal Adjustment

This appendix derives the results of the model in Section 6; Internet Appendix C gives proofs. Subsection B.6 solves the full equilibrium numerically, without the first-order approximations used in the text. The final subsection adds a conditional benchmark in which information and a preference for supporting borrowers can coexist in issue pricing.

B.1 Savers and market shares

Savers' risk aversion γ has distribution G on $(0,\infty)$, with utility given by equation (20). Under the logit assumption on savers' tastes, the share of savers with risk aversion γ choosing bank b is equation (21). Holding rivals' portfolios fixed,

$$\frac{\partial s_b(\gamma)}{\partial x_b} = \beta s_b(\gamma)\big(1-s_b(\gamma)\big)\big[\mu_B - \gamma\big(\sigma_B^2 x_b + \sigma_{B,NS} n_b\big)\big],$$
$$\frac{\partial s_b(\gamma)}{\partial n_b} = \beta s_b(\gamma)\big(1-s_b(\gamma)\big)\big[\mu_{NS} - \gamma\big(\sigma_{B,NS} x_b + \sigma_{NS}^2 n_b\big)\big], \qquad \frac{\partial s_b(\gamma)}{\partial a_b} = \gamma\, s_b(\gamma)\big(1-s_b(\gamma)\big). \tag{B.1}$$

The small-share approximation treats $s_b(\gamma)$ as small for every bank and saver type, so $s_b(\gamma)(1-s_b(\gamma)) \approx s_b(\gamma)$. A bank's marginal savers then have approximately the same distribution as its savers; the distributions coincide in the small-share limit used for the analytic comparative statics. Subsection B.6 instead retains the full demand system, including a large national-bank share. For any two banks, and whatever their shares, equation (21) gives equation (30):

$$\log s_b(\gamma) - \log s_{b'}(\gamma) = \beta\big(R_b - R_{b'}\big) + \gamma\Big[\big(a_b - a_{b'}\big) - \frac{\beta}{2}\big(V_b - V_{b'}\big)\Big]. \tag{B.2}$$

B.2 Portfolio choice

Integrating equation (B.1) over G gives

$$\frac{\partial s_b}{\partial x_b} = \beta W_b\big[\mu_B - \bar{\gamma}_b\big(\sigma_B^2 x_b + \sigma_{B,NS} n_b\big)\big], \qquad \frac{\partial s_b}{\partial n_b} = \beta W_b\big[\mu_{NS} - \bar{\gamma}_b\big(\sigma_{B,NS} x_b + \sigma_{NS}^2 n_b\big)\big],$$

where $W_b = \int s_b(\gamma)(1-s_b(\gamma))\, dG(\gamma) > 0$ and $\bar{\gamma}_b$ is defined in equation (24). Conditional on the investment regime, the fixed cost does not vary with the portfolio. We assume positive shares in the assets used and an unbinding budget constraint, so the corresponding derivatives are zero, giving equation (25). These conditions are necessary but not generally sufficient

for a best response because market share need not be globally concave in the portfolio. Subsection B.6 checks the computed portfolios numerically.

Holding marginal savers fixed, equations (26) and (28) give the bond-only and negotiated-credit portfolios. The bond-only risk-adjusted return at $\bar{\gamma}_b$ is $\mu_B^2/(2\bar{\gamma}_b\sigma_B^2) = \mu_B x_b^B/2$; adding negotiated credit raises it by g_b in equation (29). In the full demand system, the portfolio changes which savers are marginal, so an interior candidate best response solves equation (25) as a fixed point. The two regimes generally have different values $\bar{\gamma}_b^B$ and $\bar{\gamma}_b^{NS}$. The cross-regime formula $x_b^{NS} = x_b^B - (\sigma_{B,NS}/\sigma_B^2)n_b^{NS}$ and gain g_b hold exactly when these values coincide. Subsection B.6 instead computes each portfolio with the marginal savers it attracts.

B.3 Sorting by reputation

Proposition B.1 (Sorting)**.** *In the small-share case, holding portfolios fixed,*

$$\frac{\partial \bar{\gamma}_b}{\partial a_b} = \mathrm{Var}_b(\gamma), \qquad \frac{\partial \log s_b}{\partial a_b} = \bar{\gamma}_b. \tag{B.3}$$

For a bank without negotiated credit, allowing its bond share to respond,

$$\frac{d\bar{\gamma}_b}{da_b} = \frac{\mathrm{Var}_b(\gamma)}{1 - \beta\sigma_B^2\,(x_b^B)^2\,\mathrm{Var}_b(\gamma)/\bar{\gamma}_b}, \qquad \frac{dx_b^B}{da_b} = -\frac{x_b^B}{\bar{\gamma}_b}\,\frac{d\bar{\gamma}_b}{da_b}, \tag{B.4}$$

both of the sign shown when the denominator is positive. In the small-share case, if $a_b - a_{b'} > \beta\sigma_{NS}^2/2$, then $\bar{\gamma}_b > \bar{\gamma}_{b'}$ for any portfolios, and if neither bank holds negotiated credit, $x_b^B < x_{b'}^B$.

The bond-only portfolio behind Prediction 1 is

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b\,\sigma_B^2}. \tag{B.5}$$

Using the fixed-portfolio sorting derivative in equation (B.3), its dependence on reputation through marginal-investor risk aversion is

$$\frac{\partial \log x_b^B}{\partial a_b} = -\frac{1}{\bar{\gamma}_b}\frac{\partial \bar{\gamma}_b}{\partial a_b} = -\frac{\mathrm{Var}_b(\gamma)}{\bar{\gamma}_b} < 0. \tag{B.6}$$

Equation (B.4) allows the portfolio itself to adjust and gives the corresponding total derivative under its stated condition. The portfolio response reinforces the sorting: a bank that holds fewer bonds offers a safer portfolio, which attracts still more cautious savers. Outside the small-share case, the ordering still holds for all of a bank's savers but need not hold for its marginal savers, whose weights $s_b(\gamma)(1 - s_b(\gamma))$ fall where a bank serves most of a group. The numerical equilibrium in Subsection B.6, in which the national bank has a large share, confirms the ordering of $\bar{\gamma}_b$ for its parameters. The condition in the last statement of Proposition B.1 ensures that the reputation gap outweighs any difference in portfolio risk, because no feasible

portfolio has variance above σ^2_{NS}. In the data, FY2021 bank assets proxy for the reputation and distribution network that determine a bank's share.

B.4 Scale and entry into negotiated credit

Arranging negotiated credit raises the risk-adjusted return to a saver with risk aversion γ by $\Delta v_b(\gamma) = \Delta R_b - \frac{\gamma}{2}\Delta V_b$, where ΔR_b and ΔV_b are the changes in the bank's expected return and variance. Holding rivals' portfolios fixed, the fee gain is

$$G_b = f \int s_b(\gamma) \, \frac{(1 - s_b(\gamma))\left(e^{\beta \Delta v_b(\gamma)} - 1\right)}{1 - s_b(\gamma) + s_b(\gamma) e^{\beta \Delta v_b(\gamma)}} \, dG(\gamma), \tag{B.7}$$

and a bank that has not arranged negotiated credit before does so if and only if $G_b \geq F$. A bank that already holds it keeps it whenever $G_b > 0$, because its cost is sunk. Equivalently, $G_b = f(s_b^{NS} - s_b^B)$, the exact comparison in equation (32).

Proposition B.2 (Scale and entry)**.** *In the small-share case and to first order in $\beta \Delta v_b$, the fee gain is $G_b = f \beta s_b g_b$, with s_b the bank's market share without negotiated credit and g_b in equation (29). For a bank without negotiated credit, allowing its bond share to respond to its reputation,*

$$\frac{d \log G_b}{da_b} = \bar{\gamma}_b - \frac{1}{\bar{\gamma}_b} \frac{d\bar{\gamma}_b}{da_b}, \tag{B.8}$$

which is positive whenever $\mathrm{Var}_b(\gamma)\,(1 + \beta R_b) < \bar{\gamma}_b^2$, where $R_b = \mu_B x_b^B$ is the bank's expected excess return. If this holds for all reputations between a_S and a_L, then $G_L > G_S$: the large bank arranges negotiated credit whenever the small bank does, and for $G_S < F \leq G_L$ only the large bank does.

The condition says that the dispersion of risk aversion among a bank's marginal savers is small relative to its mean. Holding portfolios fixed, it would be $\mathrm{Var}_b(\gamma) < \bar{\gamma}_b^2$; the factor $1 + \beta R_b$ accounts for the portfolio response in equation (B.4), which reinforces the sorting. Market shares rise exponentially with reputation, whereas the value of negotiated credit to marginal savers falls only as their average risk aversion rises, so the scale effect dominates. Costs on existing contracts are sunk, and the cost is paid by the bank, so launching a new product does not remove it.

B.5 Debt resolution, substitution, and market shares

Debt resolution lowers the bond's expected excess return from $\mu_{B,0}$ to $\mu_{B,1} = (1 - \tau)\mu_{B,0}$, with $0 < \tau < 1$. Subscripts 0 and 1 denote values before and after resolution. The return and risk of negotiated credit are unchanged. To allow resolution to make bonds safer as well, the bond variance may fall from $\sigma^2_{B,0}$ to $\sigma^2_{B,1} \leq \sigma^2_{B,0}$, retaining a positive-definite covariance

matrix and interior conditional choices. The propositions hold the composition of each bank's marginal savers, $\bar{\gamma}_b$, fixed across the two states; Subsection B.6 lets it adjust.

Proposition B.3 (Bond retreat)**.** *A bank without negotiated credit changes its bond share by*

$$\Delta x_b = \left[(1-\tau)\frac{\sigma_{B,0}^2}{\sigma_{B,1}^2} - 1\right] x_{b,0}^B, \tag{B.9}$$

which is proportional to its initial bond share and negative if and only if

$$\frac{\mu_{B,1}}{\sigma_{B,1}^2} < \frac{\mu_{B,0}}{\sigma_{B,0}^2}. \tag{B.10}$$

With unchanged bond risk, $\Delta x_b = -\tau x_{b,0}^B$. A bank that holds negotiated credit before and after resolution, with unchanged risks, changes its bond share by $-\tau\sigma_{NS}^2\mu_{B,0}/\left[\bar{\gamma}_b(\sigma_B^2\sigma_{NS}^2 - \sigma_{B,NS}^2)\right]$, which is also proportional to the bond return it loses.

With unchanged bond risk, the supporting bond-retreat comparative static can also be written directly in primitives:

$$\Delta x_b = -\tau\, x_{b,0}^B = -\tau\, \frac{\mu_{B,0}}{\bar{\gamma}_b\, \sigma_B^2}. \tag{B.11}$$

Equivalently, $\partial^2 \Delta x_b/(\partial\tau\, \partial x_{b,0}^B) = -1$: a larger initial position produces a stronger response to resolution. This exposure interaction is examined in Section 4.4 and Table 5. This benchmark applies to banks that stay bond-only, holding marginal savers fixed. Condition (B.10) states that yield compression must outweigh any reduction in bond risk for banks to hold fewer bonds. The empirical reliance measure includes other home credit and subtracts a national benchmark; it proxies initial exposure to the affected investment.

For the remaining results, hold bond risk fixed. Write $g_{b,0}$ and $g_{b,1}$ for the gain from negotiated credit before and after resolution, and $G_{b,0}$ and $G_{b,1}$ for the corresponding fee gains in equation (B.7), evaluated at equilibrium portfolios.

Proposition B.4 (Entry after debt resolution)**.** *Suppose the pre-resolution fee gains satisfy $G_{S,0} \le G_{L,0}$, as under the scale condition. If $\mu_{NS} > (\sigma_{B,NS}/\sigma_B^2)\mu_{B,0}$, the gain from negotiated credit rises with resolution,*

$$\frac{\partial g_b}{\partial \tau} = \frac{\sigma_{B,NS}}{\sigma_B^2}\, \mu_{B,0}\, n_b^{NS} > 0, \tag{B.12}$$

where n_b^{NS} is the negotiated-credit share the bank would choose, so at a given market share and composition of marginal savers, the first-order fee gain $f\beta s_b g_b$ is larger after resolution. The full change in the fee gain also reflects the change in the bank's share: to first order, $dG_b/d\tau = f\beta\Big(s_b\, dg_b/d\tau + g_b\, ds_b/d\tau\Big)$. The national bank holds negotiated credit before resolution, the large local bank enters only after resolution, and the small local bank does not

enter whenever

$$\max\{G_{L,0},\, G_{S,1}\} < F \leq \min\{G_{L,1},\, G_{N,0}\}. \tag{B.13}$$

For an entrant, the combined share of bonds and negotiated credit changes by

$$\Delta(x_b + n_b) = -\tau x_{b,0}^B + \frac{\sigma_B^2 - \sigma_{B,NS}}{\sigma_B^2}\, n_{b,1}, \tag{B.14}$$

which is positive when the new negotiated position is large relative to the bond retreat.

The interval in condition (B.13) is nonempty when both upper bounds exceed its lower bound. This requires enough post-resolution fee income to make entry worthwhile for the large local bank, while the small bank's gain remains below the cost. Subsection B.6 verifies these comparisons with endogenous market shares rather than inferring them from the yield change alone.

The portfolio expression behind Prediction 2 combines the entry decision with the conditional negotiated-credit share. In the fixed-composition benchmark, after resolution,

$$\begin{aligned} n_{b,1}^* &= \mathbb{1}\{f(s_{b,1}^{NS} - s_{b,1}^B) \geq F \text{ or } e_b = 0\}\; n_{b,1}^{NS}, \\ n_{b,1}^{NS} &= \frac{\mu_{NS} - (\sigma_{B,NS}/\sigma_B^2)(1-\tau)\mu_{B,0}}{\bar{\gamma}_b\left(\sigma_{NS}^2 - \sigma_{B,NS}^2/\sigma_B^2\right)}. \end{aligned} \tag{B.15}$$

Here $s_{b,1}^B$ and $s_{b,1}^{NS}$ are the bank's post-resolution market shares under the two candidate portfolios, holding rivals' portfolios fixed; e_b records whether it has already paid the initiation cost, as in equation (23). Substituting the gain g_b into the small-share, first-order entry condition (33) gives, for a first-time entrant,

$$\frac{s_b}{\bar{\gamma}_b} \;\geq\; \frac{2F\left(\sigma_{NS}^2 - \sigma_{B,NS}^2/\sigma_B^2\right)}{f\beta\left(\mu_{NS} - (\sigma_{B,NS}/\sigma_B^2)(1-\tau)\mu_{B,0}\right)^2}. \tag{B.16}$$

In this approximation, s_b is the bank's bond-only share at the date of the comparison. Holding the portfolio used to evaluate investor composition fixed, equation (B.3) gives $\partial \log(s_b/\bar{\gamma}_b)/\partial a_b = \bar{\gamma}_b - \mathrm{Var}_b(\gamma)/\bar{\gamma}_b$, positive when $\mathrm{Var}_b(\gamma) < \bar{\gamma}_b^2$. Proposition B.2 gives the stronger condition when that portfolio adjusts. Holding market shares and marginal savers fixed as resolution intensifies, the right side of equation (B.16) falls: a lower bond return makes negotiated credit more valuable and lowers the entry cutoff. The full numerical solution lets both market shares and marginal savers change.

Proposition B.5 (Returns and market shares)**.** *Suppose neither local bank holds negotiated credit before resolution and only the large bank enters afterward. Relative to the small bank,*

the entrant's expected return changes by

$$\Delta R_L - \Delta R_S = \tau(2-\tau)\,\mu_{B,0}\Big(x^B_{S,0} - x^B_{L,0}\Big) + \Big(\mu_{NS} - \frac{\sigma_{B,NS}}{\sigma_B^2}\mu_{B,1}\Big)n_{L,1} > 0, \tag{B.17}$$

and, in the small-share case, to first order and holding the national bank's offer fixed, its market share grows faster:

$$\Delta \log s_L - \Delta \log s_S = \beta\Big[\frac{\tau(2-\tau)}{2}\,\mu_{B,0}\Big(x^B_{S,0} - x^B_{L,0}\Big) + g_{L,1}\Big] > 0. \tag{B.18}$$

Both terms are positive. The small bank held more bonds (Proposition B.1), so it loses more bond income, and only the large bank gains from negotiated credit.

The individual-bank decomposition behind Prediction 3 makes these two channels explicit. For either local bank b, which holds no negotiated credit before resolution, holding marginal savers fixed,

$$\begin{aligned}\Delta R_b = \; & -\tau(2-\tau)\,\mu_{B,0}\,x^B_{b,0} \\ & + \mathbb{1}\{b \text{ enters}\}\Big(\mu_{NS} - \frac{\sigma_{B,NS}}{\sigma_B^2}\mu_{B,1}\Big)n^{NS}_{b,1}.\end{aligned} \tag{B.19}$$

The first term is lost bond income; the second is replacement income available only to the entrant. For the two local banks, in the small-share approximation, to first order and holding the national bank's offer fixed,

$$\begin{aligned}\Delta \log s_b - \Delta \log s_{b'} &= \beta\Big(\Delta v_b - \Delta v_{b'}\Big), \\ \Delta v_b &= -\frac{\tau(2-\tau)}{2}\,\mu_{B,0}\,x^B_{b,0} + \mathbb{1}\{b \text{ enters}\}\,g_{b,1}.\end{aligned} \tag{B.20}$$

Here Δv_b is the change in risk-adjusted return evaluated at the bank's fixed marginal-investor risk aversion, and $g_{b,1}$ is the post-resolution gain from negotiated credit. Finally, substituting equation (B.5) into the relative-return comparison gives

$$\begin{aligned}\Delta R_L - \Delta R_S = \; & \tau(2-\tau)\,\frac{\mu^2_{B,0}}{\sigma_B^2}\Big(\frac{1}{\bar\gamma_S} - \frac{1}{\bar\gamma_L}\Big) \\ & + \Big(\mu_{NS} - \frac{\sigma_{B,NS}}{\sigma_B^2}\mu_{B,1}\Big)n^{NS}_{L,1} > 0.\end{aligned} \tag{B.21}$$

The first term is positive under the marginal-investor sorting condition, and the second is positive when only the large local bank enters. At that bank, the conditional share $n^{NS}_{L,1}$ equals its actual negotiated-credit share $n_{L,1}$ in Proposition B.5. The national bank, which already holds negotiated credit before resolution, is not part of this entrant–nonentrant comparison.

B.6 Numerical solution of the full equilibrium

The propositions use the small-share case, first-order approximations, and a fixed composition of marginal savers. We now compute a three-bank equilibrium of the full demand system for an illustrative parameterization. Savers' risk aversion is lognormal with log mean 1.0 and log standard deviation 0.6, truncated to $[0.4, 15]$; $\beta = 60$, $f = 1$, $\mu_B = 0.012$, $\mu_{NS} = 0.022$, $\sigma_B^2 = 0.012$, $\sigma_{NS}^2 = 0.040$, $\sigma_{B,NS} = 0.008$; reputations are $a_N = 0$, $a_L = -0.75$, and $a_S = -1.25$; and resolution lowers the bond return by $\tau = 0.5$. Each bank's portfolio is a numerically computed best response to its rivals' portfolios, with market shares obtained by integrating equation (21) over the distribution of risk aversion. The fixed cost $F = 0.0077$ lies in the interval of condition (B.13).

Table B1
Full equilibrium of the model

	Before resolution			After resolution		
	National	Large local	Small local	National	Large local	Small local
Bond share x_b	0.226	0.510	0.629	0.047	0.079	0.307
Negotiated-credit share n_b	0.125	0	0	0.160	0.266	0
Market share s_b	0.483	0.073	0.031	0.472	0.077	0.028
Marginal savers' risk aversion $\bar{\gamma}_b$	3.24	1.96	1.59	3.24	1.95	1.63
Expected excess return R_b (bp)	54.6	61.1	75.5	38.1	63.2	18.4
Fee gain from negotiated credit	0.0130	0.0058	0.0032	0.0213	0.0090	0.0048

Notes: Numerically computed Nash equilibrium of the model in Section 6 with the illustrative parameters in the text. The fee gain is the increase in a bank's fee income from arranging negotiated credit, holding rivals' portfolios fixed; the fixed cost is 0.0077. Before resolution only the national bank's gain exceeds the fixed cost; after resolution the large local bank's gain does as well, the small local bank's does not, and the national bank, which paid the cost before resolution, keeps its negotiated credit. Before resolution, the bond shares of the three banks without negotiated credit would be 0.308, 0.510, and 0.629.

All three predictions hold in this numerical example (Table B1). Bond shares fall with reputation, and the more reputable banks' marginal savers are more risk averse. The supporting bond-retreat comparison also holds: without negotiated credit, resolution lowers the three bond shares by 0.154, 0.259, and 0.321, close to $\tau x_b^B = 0.154$, 0.255, and 0.315. The fee gain from negotiated credit rises with reputation and rises after resolution, and only the large local bank's gain crosses the fixed cost. The large bank's expected return rises by 59 basis points relative to the small bank's, and the log of its market share relative to the small bank's rises by 0.17. The ratio $\mathrm{Var}_b(\gamma)(1 + \beta R_b)/\bar{\gamma}_b^2$ is 0.31 at the large local bank and 0.29 at the small one, well below one, so the condition of Proposition B.2 holds.

We check the computed portfolios numerically for each bank, before and after resolution, both with and without negotiated credit. Holding rivals' portfolios at their equilibrium values, we search the whole feasible set $\{x_b, n_b \geq 0,\ x_b + n_b \leq 1\}$ in steps of 0.002. Every computed portfolio is feasible, and no portfolio on the grid attains a higher market share. Optimizations started from six different portfolios converge to the same choice within 10^{-7}, providing further

numerical support for the computed best responses. The marginal savers' risk aversion differs only slightly between the two regimes; for the large local bank after resolution, it is 1.995 with the bond-only portfolio and 1.953 with negotiated credit.

B.7 Information and borrower support in issue pricing

The portfolio mechanism does not require information to be the sole source of local value. To interpret Tables A12 and A13, allow both screening and relationship preferences. A borrower has type $t \in \{-1, +1\}$ with equal probabilities and distress probability $p_0 - dt$, where $0 < d < \min\{p_0, 1 - p_0\}$. Loss given distress is $\Lambda > 0$, so $\delta = \Lambda d$ measures how much the two types differ in expected payoff. A buyer at distance $z \in \{\text{city}, \text{province}\}$ observes a signal $\hat{t}$ that equals the type with probability $(1 + e_z)/2$ and buys borrowers with favorable signals. The buyer chooses accuracy $e_z \in [0, 1]$ at screening cost $c_z e_z^2/2$ per accepted unit, with $c_z > 0$. The ordering $c_{\text{city}} < c_{\text{province}}$ captures the distance-related cost of soft information (Stein 2002; Petersen and Rajan 2002; Agarwal and Hauswald 2010).

An issuer captures fraction $\lambda \in (0, 1)$ of informational and relationship surplus through a yield concession. The relationship benefit $\phi_z \geq 0$ can represent the bank's value of supporting the borrower; it affects preferences but, in this benchmark, not repayment. Optimal screening, pricing, and distress are

$$\begin{aligned} e_z^* &= \min\{(1 - \lambda)\delta/c_z, 1\}, \\ \Omega_z &= \lambda(\delta e_z^* + \phi_z), \qquad \pi_z = p_0 - d e_z^*. \end{aligned} \tag{B.22}$$

Here Ω_z is the concession relative to public pricing of an average borrower, and π_z is distress among selected borrowers.

Proposition B.6 (Conditional pricing and distress contrasts)**.** *Suppose screening is unsaturated and both city buyers and buyers elsewhere in the same province invest. If* $c_{\text{city}} < c_{\text{province}}$ *and* $\phi_{\text{city}} \geq \phi_{\text{province}}$*, then* $\Omega_{\text{city}} > \Omega_{\text{province}}$ *and* $\pi_{\text{city}} < \pi_{\text{province}}$*. If screening costs are equal and only the relationship preference is larger in the city, the pricing contrast is positive but the distress contrast is zero.*

Information improves selection and can increase the surplus shared through lower yields. A pure support preference additionally raises the concession without changing distress under the maintained repayment assumption. Pricing and distress together therefore assess coexistence, rather than assign the empirical concession exclusively to information or support.

Internet Appendix C
Proofs

C.1. Proof of Proposition B.1

In the small-share case, $s_b(\gamma) = \exp(u_b(\gamma))/D(\gamma)$, with $u_b(\gamma) = \gamma a_b + \beta(R_b - \gamma V_b/2)$ and $D(\gamma)$ the denominator in equation (21), whose dependence on bank b's own offer is negligible. The weights defining $\bar{\gamma}_b$ are then proportional to $\exp(\gamma t_b)\, h_b(\gamma)$, where $t_b = a_b - \beta V_b/2$ and $h_b(\gamma) = e^{\beta R_b}\, dG(\gamma)/D(\gamma)$ does not depend on a_b. This is an exponential tilting of a fixed distribution, so the derivative of its mean with respect to the tilt t_b is its variance, which gives $\partial\bar{\gamma}_b/\partial a_b = \mathrm{Var}_b(\gamma)$ holding the portfolio fixed. Similarly, $\partial \log s_b/\partial a_b = \int \gamma\, s_b(\gamma)\, dG / \int s_b(\gamma)\, dG = \bar{\gamma}_b$.

For a bank without negotiated credit, $V_b = \sigma_B^2 (x_b^B)^2$ and $x_b^B = \mu_B/(\bar{\gamma}_b \sigma_B^2)$. Then $d\bar{\gamma}_b/da_b = \mathrm{Var}_b(\gamma)\, dt_b/da_b$, with $dt_b/da_b = 1 - \beta\sigma_B^2 x_b^B\, dx_b^B/da_b$ and $dx_b^B/da_b = -(x_b^B/\bar{\gamma}_b)\, d\bar{\gamma}_b/da_b$. Substituting the last expression into the first and solving gives equation (B.4).

For the final statement, equation (B.2) shows that $s_b(\gamma)/s_{b'}(\gamma)$ increases with γ when $(a_b - a_{b'}) - \frac{\beta}{2}(V_b - V_{b'}) > 0$. Every feasible portfolio has $0 \le V \le (x\sigma_B + n\sigma_{NS})^2 \le \sigma_{NS}^2$, so $a_b - a_{b'} > \beta\sigma_{NS}^2/2$ guarantees the inequality whatever the portfolios. The distribution of bank b's savers then dominates that of bank b''s savers in the likelihood-ratio order. In the small-share case, a bank's marginal savers have the same distribution as its savers, so $\bar{\gamma}_b > \bar{\gamma}_{b'}$. If neither bank holds negotiated credit, equation (26) gives $x_b^B < x_{b'}^B$. □

C.2. Proof of Proposition B.2

In the small-share case, the integrand in equation (B.7) is approximately $s_b(\gamma)\left(e^{\beta\Delta v_b(\gamma)} - 1\right)$, and to first order in $\beta\Delta v_b$ the fee gain is $f\beta \int s_b(\gamma)\Delta v_b(\gamma)\, dG = f\beta s_b\left(\Delta R_b - \frac{\bar{\gamma}_b}{2}\Delta V_b\right)$, because the savers' mean risk aversion equals $\bar{\gamma}_b$ in this case. The bank chooses the negotiated position to maximize this expression, which is the risk-adjusted return at risk aversion $\bar{\gamma}_b$; its maximum over (x_b, n_b), net of the bond-only value, is g_b in equation (29). Hence $G_b = f\beta s_b g_b$.

For the derivative, the bank's bond share maximizes s_b, so by the envelope theorem $d\log s_b/da_b = \partial \log s_b/\partial a_b = \bar{\gamma}_b$ (Proposition B.1). Since g_b is proportional to $1/\bar{\gamma}_b$, $d\log g_b/da_b = -(1/\bar{\gamma}_b)\, d\bar{\gamma}_b/da_b$, which gives equation (B.8). By equation (B.4), $d\bar{\gamma}_b/da_b = \mathrm{Var}_b(\gamma)/(1 - k_b)$ with $k_b = \beta\sigma_B^2 (x_b^B)^2 \mathrm{Var}_b(\gamma)/\bar{\gamma}_b = \beta R_b \mathrm{Var}_b(\gamma)/\bar{\gamma}_b^2$, using $\bar{\gamma}_b \sigma_B^2 x_b^B = \mu_B$. The condition $\mathrm{Var}_b(\gamma)(1 + \beta R_b) < \bar{\gamma}_b^2$ implies $k_b < \beta R_b/(1 + \beta R_b) < 1$, and it is equivalent to $\bar{\gamma}_b^2(1 - k_b) > \mathrm{Var}_b(\gamma)$, so the derivative is positive. If the condition holds for all reputations between a_S and a_L, integrating gives $G_L > G_S$. If the small bank enters, $F \le G_S < G_L$ and the large bank enters as well; if $G_S < F \le G_L$, only the large bank enters. □

C.3. Proof of Proposition B.3

By equation (26) with $\bar{\gamma}_b$ fixed, $x^B_{b,1} = (1-\tau)\mu_{B,0}/(\bar{\gamma}_b\sigma^2_{B,1}) = (1-\tau)(\sigma^2_{B,0}/\sigma^2_{B,1})\,x^B_{b,0}$, which gives equation (B.9). The bracket is negative if and only if $(1-\tau)/\sigma^2_{B,1} < 1/\sigma^2_{B,0}$, which is condition (B.10). For a bank that holds negotiated credit in both states, solving equation (25) gives $x_b = (\sigma^2_{NS}\mu_B - \sigma_{B,NS}\mu_{NS})/[\bar{\gamma}_b(\sigma^2_B\sigma^2_{NS} - \sigma^2_{B,NS})]$; with unchanged risks and μ_{NS}, the change is $-\tau\sigma^2_{NS}\mu_{B,0}/[\bar{\gamma}_b(\sigma^2_B\sigma^2_{NS} - \sigma^2_{B,NS})]$. □

C.4. Proof of Proposition B.4

With $\mu_B = (1-\tau)\mu_{B,0}$ in equation (29),

$$\frac{\partial g_b}{\partial \tau} = \frac{\big(\mu_{NS} - (\sigma_{B,NS}/\sigma^2_B)\mu_B\big)(\sigma_{B,NS}/\sigma^2_B)\mu_{B,0}}{\bar{\gamma}_b\big(\sigma^2_{NS} - \sigma^2_{B,NS}/\sigma^2_B\big)} = \frac{\sigma_{B,NS}}{\sigma^2_B}\,\mu_{B,0}\,n^{NS}_b,$$

using equation (28). It is positive because $\mu_{NS} - (\sigma_{B,NS}/\sigma^2_B)\mu_B \geq \mu_{NS} - (\sigma_{B,NS}/\sigma^2_B)\mu_{B,0} > 0$. Equivalently, by the envelope theorem, a change in μ_B changes the best risk-adjusted return by the bonds held, x^{NS}_b with negotiated credit and x^B_b without it, so $\partial g_b/\partial\mu_B = x^{NS}_b - x^B_b = -(\sigma_{B,NS}/\sigma^2_B)n^{NS}_b$. Because the first-order fee gain $f\beta s_b g_b$ increases with g_b, it is larger after resolution at any given share and composition of marginal savers.

For the entry pattern, every F in the interval (B.13) satisfies $G_{N,0} \geq F$, $G_{L,0} < F \leq G_{L,1}$, and $G_{S,1} < F$. The maintained pre-resolution ordering $G_{S,0} \leq G_{L,0}$ also implies $G_{S,0} < F$. Thus only the national bank enters before resolution; afterward, the large local bank enters and the small bank remains out. All gains are evaluated against rivals' portfolios in the corresponding regime. The interval is nonempty precisely when $\max\{G_{L,0}, G_{S,1}\} < \min\{G_{L,1}, G_{N,0}\}$.

For an entrant, equation (28) gives $x_{b,1} = (1-\tau)x^B_{b,0} - (\sigma_{B,NS}/\sigma^2_B)n_{b,1}$, so $\Delta(x_b + n_b) = -\tau x^B_{b,0} + (1 - \sigma_{B,NS}/\sigma^2_B)n_{b,1}$, which is equation (B.14). □

C.5. Proof of Proposition B.5

Write $\chi = \sigma_{B,NS}/\sigma^2_B$ for the bonds displaced per unit of negotiated credit. A bank without negotiated credit earns $R_b = \mu_B x^B_b = \mu^2_B/(\bar{\gamma}_b\sigma^2_B)$, so its expected return changes by $\big((1-\tau)^2 - 1\big)\mu_{B,0}x^B_{b,0} = -\tau(2-\tau)\mu_{B,0}x^B_{b,0}$. After resolution, the entrant holds $x^B_{L,1} - \chi n_{L,1}$ in bonds and $n_{L,1}$ in negotiated credit, so its expected return is $\mu_{B,1}x^B_{L,1} + (\mu_{NS} - \chi\mu_{B,1})n_{L,1}$, and its change is $-\tau(2-\tau)\mu_{B,0}x^B_{L,0} + (\mu_{NS} - \chi\mu_{B,1})n_{L,1}$. Subtracting the small bank's change gives equation (B.17). The second term is positive because the entrant holds negotiated credit only when $\mu_{NS} > \chi\mu_{B,1}$, and the first is positive because $x^B_{S,0} > x^B_{L,0}$ by Proposition B.1.

In the small-share case, $\log s_b(\gamma) = u_b(\gamma) - \log D(\gamma)$. Holding the national bank's offer fixed, $D(\gamma)$ does not change to first order, because the local banks' own shares are small. A

change in the bank's portfolio then changes $\log s_b$ by β times the change in the risk-adjusted return to its savers, evaluated at their average risk aversion $\bar{\gamma}_b$. The bond-only risk-adjusted return at $\bar{\gamma}_b$ is $\mu_B x_b^B/2$, which changes by $-\tau(2-\tau)\mu_{B,0}x_{b,0}^B/2$, and the entrant gains $g_{L,1}$ in addition. Subtracting gives equation (B.18). □

C.6. Proof of Proposition B.6

For a signal $\hat{t} \in \{-1, +1\}$ with $\Pr(\hat{t} = t \mid t) = (1+e)/2$, symmetry implies

$$\Pr(t = +1 \mid \hat{t} = +1) = (1+e)/2, \qquad \mathbb{E}[t \mid \hat{t} = +1] = e, \qquad \Pr(\text{distress} \mid \hat{t} = +1) = p_0 - de.$$

An accepted borrower has informational surplus δe. After conceding the fraction λ of informational and relationship surplus to the issuer, the bank's per-unit objective, apart from terms independent of e, is $(1-\lambda)\delta e - c_z e^2/2$. Strict concavity on $[0,1]$ gives the optimal accuracy in equation (B.22). For unsaturated screening,

$$e^*_{\text{city}} - e^*_{\text{province}} = (1-\lambda)\delta\left(c^{-1}_{\text{city}} - c^{-1}_{\text{province}}\right) > 0.$$

Consequently,

$$\begin{aligned}
\Omega_{\text{city}} - \Omega_{\text{province}} &= \lambda\left[\delta(e^*_{\text{city}} - e^*_{\text{province}}) + \phi_{\text{city}} - \phi_{\text{province}}\right] > 0, \\
\pi_{\text{city}} - \pi_{\text{province}} &= -d(e^*_{\text{city}} - e^*_{\text{province}}) < 0.
\end{aligned}$$

With equal screening costs, accuracies and distress probabilities are equal. A strictly larger city relationship benefit then raises only the concession. The latter conclusion uses the maintained assumption that relationship preferences do not alter repayment probabilities. □

Internet Appendix D
Specifications and Supplementary Evidence

This appendix supplies the measurement and specification details behind the argument. Subsections D.1–D.2 establish how negotiated claims relate to LGFV borrowers and how the two financing channels are compared. Subsections D.3–D.7 explain the supporting investment, demand, and performance tests. Subsection D.8 reconciles market-wide exposure with shares of included debt holdings. Subsection D.9 and Tables D1–D3 separate initial specialization from subsequent replacement. Tables D4–D5 assess the financing-rate comparison; Figure D1 explains its 2023Q1 movement in rate levels. Return and business-balance checks are in Internet Appendix E.

D.1. Who borrows through negotiated credit?

The substitution argument concerns a change in financing instrument, not necessarily a move away from local borrowers. To identify who receives negotiated financing, we extract borrower names from asset descriptions, locate them using company registration records, and match them to LGFV bond issuers over 2022Q1–2025Q3. These matches account for about 60% of local banks' home negotiated-credit value before the program and 59% afterward. Adding a keyword screen for financing-platform names raises the shares to 79% and 86%; among the largest FY2021 size tercile, the corresponding broad shares are 77% and 85%. Most of this substitution therefore remains within the LGFV sector, although sector-level matches do not trace each bond into a contract with the same legal borrower.

D.2. Spread compression and financing-rate comparisons

The spread regressions ask whether the loss of LGFV bond yield is greater in provinces that refinance more debt. For province p in calendar quarter q, the specification in Table A5 is

$$\begin{aligned}\text{Bond spread}_{pq} = \beta\,\text{Refinancing}_{pq}^{GDP} + \alpha_p + \alpha_q \\ + \text{Trend controls}_{pq} + \varepsilon_{pq}.\end{aligned} \tag{D.1}$$

where the outcome is the province-quarter LGFV bond spread in basis points and intensity is cumulative refinancing as a percentage of 2020 provincial GDP. The effects α_p and α_q compare changes within provinces after removing common quarterly movements. The term $\text{Trend controls}_{pq}$ contains the trend controls reported in each column: none, the pre-program spread interacted with a linear trend, or province-specific linear trends. These additions distinguish the refinancing association from convergence related to initial spreads and from existing provincial trends. Both reduce its magnitude but retain a negative association.

Table A5 thus relates compression to refinancing, while Figure A4 shows the decline in the initial high–low spread gap.

To assess the income available from a substitute, the next comparison places negotiated-credit yields alongside bond issuance yields for the same reporting issuers. Both are annual contractual rates on new financing. For bonds, the source records the stated issue rate; these comparisons do not use secondary-market yields or infer changes in the contractual payments on outstanding bonds. Table 4, Panel A, estimates

$$\begin{aligned}\text{Yield}_{jfq} = \alpha_{fq} + \alpha_{f,d(j)} + \alpha_{\text{mat}(j)} \\ + \sum_{t \neq 2022} \delta_t \mathbb{1}\{\text{Negotiated credit}\}_j \mathbb{1}\{\text{Issue year}_j = t\} + e_{jfq}.\end{aligned} \tag{D.2}$$

Here j is a contract or bond issue, f its reporting issuer, and q its origination quarter; Yield_{jfq} is the annual contractual rate in percentage points. The functions $d(j)$ and $\text{mat}(j)$ identify its debt type and maturity bin. The effects α_{fq} compare financing within an issuer and quarter; $\alpha_{f,d(j)}$ absorb persistent differences between that issuer's financing channels. The year indicator uses the record's origination year, with t indexing calendar years. Thus δ_t measures the change in the negotiated-credit-minus-bond gap from 2022. The gap widens by 1.30 percentage points in 2024. This reporting-issuer sample can include financing by an issuer's affiliates, unlike the exact-borrower event study.

The separate-channel columns in Panel A instead regress each record's rate on origination-year indicators, with 2022 omitted. The negotiated-credit regression absorbs borrower-by-creditor and maturity-bin effects; the bond regression absorbs issuer-by-bond-type-by-placement and maturity-bin effects. Their year coefficients therefore compare rates within the relevant financing relationships or issue types. Both cluster errors by reporting issuer. These are distinct regressions with different samples, rather than components that must sum to the gap estimate.

The quarterly comparison in Figure 3 instead holds the legal borrower fixed. It requires the funded borrower's DM code to equal the reporting issuer's code and matches bonds on that same code. Its initial pool contains borrower-quarters with both channels and borrowers with at least two matched quarters. We then require observed continuous maturity, restrict financing to 2021 onward, and remove borrower-month singletons. The resulting sample contains 17,303 financing records at 589 borrowers in 26 provinces. Contracts are dated by origination rather than disclosure; repeated contracts and physical bond issues are deduplicated, without further rate winsorization or outcome-based trimming. Because these restrictions can remove one channel, not every retained month or quarter still contains both. Mixed-channel months identify the direct rate gap, while other multi-observation months also inform the maturity slopes.

Equation (8) compares annual rates within borrower-by-calendar-month cells, controlling for continuous maturity with a separate slope in each quarter. It does not include borrower-

by-debt-type effects. Quarterly coefficients estimate gap levels; the plotted changes subtract the 2023Q2 gap using the full covariance matrix. Table D4 asks whether the widening depends on maturity adjustment, financing frequency, or borrower composition. Column 2 allows separate maturity slopes by debt type and quarter and evaluates the gap at a common three-year maturity. Column 3 gives each borrower-quarter-debt-type cell total weight one, using inverse-frequency weights on the baseline sample. Column 4 retains the 101 borrowers with both channels in 2023Q2 and at least one later quarter after the baseline restrictions, without requiring a balanced panel. All four comparisons show a wider gap in 2024Q4 and 2025Q3. The earlier gaps also move, so the event study documents divergence within legal borrowers rather than a parallel-trends causal effect.

The pre-program dip in 2023Q1 raises a separate question: did negotiated-credit yields fall, or did bond issuance yields rise? Figure D1 plots both rates for the same 22 legal borrowers, each receiving equal weight in every quarter from 2022Q4 through 2023Q3. Each borrower has both channels in at least one common month of every quarter. Bond issuance yields rise from 4.25% in 2022Q4 to 4.78% in 2023Q1, then fall to 4.49% in Q2. Negotiated-credit rates instead move from 6.27% to 6.32% and 6.40%. The Q1 narrowing therefore reflects higher bond issuance yields rather than a collapse in negotiated-credit yields. A broader fixed set of 76 borrowers observed in both Q1 and Q2 gives the same direction: bond issuance yields fall 47 basis points while negotiated-credit rates rise 4. These descriptive comparisons fix borrowers and apply identical weights to the two channels, but contracts and maturities can change; the conditional regression separately adjusts for maturity.

Table D5 asks whether the late-period widening depends on how repeated financing records and regional dependence are treated. The baseline uses one-way province clustering, allowing arbitrary dependence across borrowers and months within a province. Figure intervals are $\hat{\delta}_q \pm t_{25,.975}\, SE(\hat{\delta}_q)$, with 26 province clusters. Additional inference checks use borrower-clustered errors, a delete-one-province jackknife, and a null-imposed province wild bootstrap with 9,999 Webb-weight draws. Bootstrap confidence sets invert bootstrap-t tests. For the jackknife, the variance is $(G-1)G^{-1}\sum_p(\hat{\delta}_{q,(-p)} - \bar{\delta}_{q,(-p)})^2$, where $G = 26$ and the bar is the mean leave-out estimate; intervals use t_{25}. All 26 leave-one-province estimates remain positive for each reported quarter. Panel A reports the main late-period contrasts and their province-bootstrap confidence sets.

The second regression column of Table D5 gives each borrower-month with both channels one observation. For legal borrower f in calendar month m, let Yield gap$_{fm}$ be its mean NS yield minus mean bond issuance yield and Maturity gap$_{fm}$ its mean NS maturity minus mean bond maturity. As in the main event study, $q(m)$ identifies the quarter containing

month m. We estimate

$$\begin{aligned}\text{Yield gap}_{fm} = &\sum_q b_q \mathbb{1}\{q(m) = q\} \\ &+ \sum_q \eta_q \,\text{Maturity gap}_{fm} \mathbb{1}\{q(m) = q\} + u_{fm}.\end{aligned} \tag{D.3}$$

and report $b_q - b_{2023Q2}$. Each channel mean weights contracts or physical bond issues equally. Differencing removes the common borrower-month condition, so no borrower effect is added. This regression uses 2,525 borrower-months at 557 borrowers in 25 provinces. It tests whether the wider gap remains when each paired month contributes one observation rather than several financing records. The weighting and maturity adjustment differ from the contract-level regression. Province-clustered errors and 9,999 Webb-weight province draws support widening in both reported quarters. Panel B reports the underlying financing records, borrowers, and provinces by quarter; no usable 2025Q4 comparison survives the monthly design. Together, these checks address sampling uncertainty in the observed divergence, not nationwide shocks that could affect the two debt markets differently.

The priority-province comparison in Table 4, Panel B, uses the same issuer-years for each channel and the difference between them:

$$\begin{aligned}\text{Rate}^d_{ft} = \alpha^d_f + \alpha^d_t + \beta_d \,\mathbb{1}\{\text{Priority province}\}_{p_f} \mathbb{1}\{t = 2024\} \\ + \psi_d \,\mathbb{1}\{\text{Priority province}\}_{p_f} \mathbb{1}\{t = 2023\} + u^d_{ft}.\end{aligned} \tag{D.4}$$

The index f again denotes a reporting issuer and t a calendar year. Superscript $d \in \{NS, B, NS - B\}$ labels the outcome: negotiated-credit yield, bond issuance yield, or their difference; it is not a date. The province p_f belongs to that issuer, not the WMP bank. The priority group comprises Tianjin, Inner Mongolia, Liaoning, Jilin, Heilongjiang, Guangxi, Chongqing, Guizhou, Yunnan, Gansu, Qinghai, and Ningxia. Issuer and year effects enter separately for each outcome, and the 2023 interaction allows a transition-year contrast. The coefficient β_d is the additional 2024 change in priority provinces relative to other provinces. Amount-weighted component rates are winsorized before subtraction, so their changes add to the gap on this common sample. Relative to other provinces, the 2024 gap widens by 1.04 percentage points: bond issuance yields fall an additional 0.73 points while the negotiated-credit rate change is positive but imprecise. The provincial comparison reinforces the distinction between compression in marketable bonds and the yield remaining on negotiated claims.

D.3. Redemption terms and local holdings

The redemption comparison asks whether local banks place more home investments in products that allow them to hold less liquid assets (Table A7; equation (3)). The dependent

variable is Home allocation bias$^{n}_{iq}$ in equation (D.10): the share of the product's included bonds and negotiated credit invested in home class n, minus the corresponding province's share of nationwide included holdings. The all-credit column sums this tilt across classes, giving Home allocation bias$_{iq}$. Products with no position in a class retain a zero raw allocation before benchmark subtraction. Controls are product size and risk; bank assets, nonperforming loans, cost-to-income ratio, and net interest margin; and bank-home-city GDP, growth, LGFV debt to GDP, and income. Quarter effects remove common movements. The local-bank coefficient compares daily-open products; its interaction with restricted redemption measures how much wider the local–national allocation gap is in that segment. The additional gap is 20.0 percentage points overall, 7.7 for LGFV bonds, and 7.8 for negotiated credit. Together with local banks' 80.2% asset share in restricted-redemption products (Table A6), the estimates link their product design to their home investment strategy.

D.4. Home-bond allocation and refinancing pressure

The destination comparison in equation (11) asks whether banks reduce home bond investment more than other buyers facing the same local bond supply (Table 6).

In this destination panel, a indexes the buying WMP-issuing institution (a bank or subsidiary), c the issuer city, and q the calendar quarter. Home identifies a local institution investing in a city of its underlying bank's home province; Post begins in 2023Q4. The effects μ_{ac}, μ_{aq}, and μ_{cq} hold persistent buyer–city ties, buyer-wide conditions, and issuer-city conditions fixed. The baseline outcome is LGFV bond holdings in the city as a percentage of the buyer's current bond holdings; other columns measure net buying and new purchases. The Home–Post coefficient therefore compares changes by home and outside buyers of the same city's bonds. The estimated 0.45-point decline is about 12% of the initial mean allocation to cities in the bank's home province; it is concentrated in the bank's own city rather than elsewhere in its province. The related event study in Figure 4 aggregates destinations to bank–province cells and follows changes relative to 2023Q2.

The maturity-wall test addresses a different source of adjustment: whether home banks provide additional financing when local borrowers have more principal due. In Table A9, the allocation specification is

$$\begin{aligned}\text{Outcome}_{acq} = \beta\, \mathbb{1}\{\text{Home buyer}\}_{ac} \times \text{Maturity wall}_{cq} \\ + \mu_{ac} + \mu_{aq} + \mu_{cq} + \varepsilon_{acq}.\end{aligned} \tag{D.5}$$

The buyer, city, and quarter indices are the same as in equation (11). The wall varies by issuer city c and quarter q: it is principal on all LGFV bonds maturing there as a percentage of 2020 city budget revenue. Home identifies local banks investing in their own province. Bank-by-city, bank-by-quarter, and issuer-city-by-quarter effects absorb persistent ties, common funding conditions, and destination shocks.

Panel B of Table A9 asks what kinds of bonds home banks add when maturities are high. For bond j newly appearing in buyer a's holdings in quarter q, the specification is

$$\begin{aligned}\text{Bond characteristic}_{jaq} = \beta_H \, \mathbb{1}\{\text{Home buyer}\}_{a,c(j)} \\ + \beta_W \, \mathbb{1}\{\text{Home buyer}\}_{a,c(j)} \times \text{Maturity wall}_{c(j),q} \\ + \mu_{aq} + \mu_{c(j),q} + \varepsilon_{jaq}.\end{aligned} \tag{D.6}$$

The function $c(j)$ gives the bond issuer's city. Each regression uses one purchased-bond characteristic as its outcome: issuer rating, guarantee status, maturity, issue spread, or log issue amount. The Home level is included because buyer–city effects are absent; the Wall level is absorbed by issuer-city-by-quarter effects. The coefficient β_W asks whether the home–outside difference in that characteristic changes with scheduled maturities. Both panels use bank-city clustered errors and the same unstandardized percentage-point wall measure.

The home–wall coefficients are -0.00696 for net buying and 0.00446 for exits, with both outcomes in percentage points. A ten-percentage-point larger wall therefore predicts about 0.070 points less net buying and 0.045 points more exits by home banks relative to other buyers of the same city's bonds. Exits include nonrenewal at maturity. The wall's standard deviation in the allocation frame is 18.72 percentage points; multiplying the raw coefficients by this value gives the corresponding -0.130- and 0.083-point comparisons. The new-purchase-count interaction is not significant.

Among new purchases, the guarantee-indicator interaction is 0.00116 per percentage point of the wall. A ten-point larger wall thus widens the home–other buyer gap in guarantee probability by 1.16 percentage points; the purchase frame's 19.04-point standard deviation implies a 2.21-point difference. Issuer-rating and maturity interactions are also positive. Both panels use the same raw wall definition; the standard deviations are used only for these calculations. Home banks therefore reduce net financing and seek more protection under repayment pressure, distinguishing local investment from an unconditional commitment to refinance every local borrower.

D.5. Product assets and net flows

The demand tests distinguish money entering a product from growth generated by its investments. We infer product assets as 100 times the median ratio of a disclosed position's value to its percentage weight. Net flow then removes investment returns from asset growth:

$$\text{Net flow}_{iq} = 100 \frac{AUM_{iq} - AUM_{i,q-1}(1 + R_{iq})}{AUM_{i,q-1}}, \tag{D.7}$$

where R_{iq} is the simple quarterly return expressed as a fraction. A value of five means net inflows equal 5% of beginning assets. We apply the baseline 1st–99th percentile exclusion

to monthly log returns before compounding, require three consecutive monthly returns per share class, and average class-level quarterly returns within each product. Both assets and returns must be observed at the required consecutive quarter ends. Flow tails are capped at the global 1st and 99th percentiles before selecting complete regression observations. These disclosure-based flows capture net investor transactions among continuing observed products; redemptions from products that disappear from the holdings data are outside this sample.

The ranking comparison in Table A10 uses equation (5) in the main text. Products are ranked within their prior risk-rating and redemption category $g(i, q-1)$, with category-by-outcome-quarter effects. Peer sets include national and local banks, require at least 20 products and three underlying banks, and are formed before selecting on subsequent flow availability. Tied returns receive midranks. Column 2 also absorbs home-province-by-outcome-quarter effects. These are cross-bank comparisons without bank effects.

The eleven control variables are log product assets, risk, redemption, log bank assets, nonperforming-loan ratio, cost-to-income ratio, net interest margin, and home-city GDP, GDP growth, LGFV debt/GDP, and urban income; we also include a local-bank indicator. Product assets are measured at the beginning of the flow quarter, and the remaining controls in the preceding quarter. Risk and redemption levels are absorbed by the category effects. Standard errors are clustered by underlying bank, combining a bank and its wealth-management subsidiary. The outcome sample covers 2022Q3–2025Q3; earlier observations supply lags.

Alternative outcomes test whether a few unusually large flow rates drive the ranking result. Write $\widehat{A}_{iq} = AUM_{i,q-1}(1+R_{iq})$. Symmetric flow is $200(AUM_{iq} - \widehat{A}_{iq})/(AUM_{iq} + \widehat{A}_{iq})$, bounded between -200 and 200 without tail capping. Positive flow is $100\,\mathbb{1}\{AUM_{iq} > \widehat{A}_{iq}\}$, so its coefficient is a percentage-point difference in the probability of net inflows. The main top–bottom flow difference is 12.2% of beginning assets, and the positive-flow probability differs by 8.1 points. Panel B compares redemption segments, adds lifecycle controls, gives each bank-quarter equal weight, and restricts peers to local banks in the same province. The last two comparisons yield smaller, statistically insignificant top–bottom differences. The full-market result therefore describes the performance incentive across disclosed products, with different estimates when the average is taken across banks or within provincial peer groups.

A linear benchmark replaces the quintile indicators with the previous quarter's return in percentage points, retaining the controls and category-by-quarter effects. Its coefficient is 9.381 (SE 2.064; 123,804 observations): ten basis points more quarterly return predicts next-quarter inflows higher by 0.94% of beginning assets. Both designs use the same return construction and lagged controls, although their nonmissing sample requirements differ. The ranking contrast compares a much larger performance difference: average prior quarterly returns are 1.275% in the top fifth and 0.246% in the bottom fifth. The two magnitudes therefore answer different questions. Neither design instruments returns or identifies the

policy's effect on household transactions. Likewise, a bank's headquarters identifies its location, not where its investors live.

D.6. Full-period returns and downside outcomes

The return premium describes average reported performance. To assess what savers observed beyond that average, Table A8 also examines volatility and downside outcomes. Its full-period specification is

$$\begin{aligned}\text{Outcome}_{im} = \beta\,\text{Home allocation bias}^{LGFV}_{i,h(m)} + \gamma'\,\text{Controls}_{i,h(m)} \\ + \lambda\,\mathbb{1}\{\text{Local bank}\}_b + \alpha_m + \varepsilon_{im}.\end{aligned} \tag{D.8}$$

The dependent variable Outcome_{im} is monthly NAV return or one of the reported volatility and downside measures. The exposure Home allocation bias^{LGFV} is the home-LGFV bond share of included holdings minus the corresponding provincial benchmark, measured as a fraction at the preceding quarter end. The eleven baseline controls, a local-bank indicator, and month effects enter; errors cluster by bank-city-by-month cells. Returns are consecutive-month changes in log adjusted NAV, and trailing annualized volatility uses the twelve-month window ending in the current month, with at least two available returns. Panel B measures excess returns, downside volatility, drawdowns, and the frequency of negative or unusually low monthly returns, as defined in the table notes.

The question is whether the strategy delivers higher reported performance only by exposing savers to more frequent visible losses. The full-period monthly return coefficient is 3.14 basis points per unit of tilt (SE 0.78), or about 8 annual basis points for a 20-percentage-point difference. For the same difference, the probability of a negative month is 0.8 percentage points lower, against a 9.9% mean. These are associations in reported NAVs, conditional on bank type and the other controls. They describe the full-period experience of the local-bond strategy; Table 3 separately estimates how its return premium changes after the program.

D.7. Pricing, distress, and sample timing

The ownership comparisons in Tables A12 and A13 use equation (6) in Section 3.5. Bond j is observed in its issue quarter q, and $c(j)$ identifies its issuer city. The outcome is either the public valuation yield minus the issue yield in basis points or the subsequent issuer-distress indicator scaled to 100. Both ownership shares are in percentage points of issue size, measured at issue-quarter end. The main specification includes issuer-city-by-issue-quarter effects; the tables also report comparisons with issue-quarter or issuer-province-by-quarter effects in their place. The contrast $\beta_C - \beta_P$ shifts ownership from other-city provincial banks to same-city banks while holding their combined share fixed. Bond and ownership controls are listed in

the tables; distress adds the public valuation spread to compare issuers with similar observed ex ante risk.

The pricing sample ends in 2025Q3, the last quarter of holdings coverage. We exclude 977 otherwise eligible LGFV issues in 2025Q4 rather than treating unobserved ownership as zero. The common LGFV sample has 21,116 issues, with 20,526 remaining after removing singleton city-quarter cells. With bond and ownership controls, the same-city-minus-other-city contrast is 0.78 basis points with issue-quarter effects and 1.01 with issuer-city-by-quarter effects. For the latter, the robust standard error is 0.337 ($p = 0.0026$) and the issuer-clustered standard error is 0.355 ($p = 0.0044$). Table A12 uses the holdings-covered sample throughout. Its non-LGFV column gives a small, imprecise contrast of 0.11 basis points. The pricing difference between the two local ownership groups is therefore concentrated in LGFV bonds, where local public-financing relationships are most relevant.

D.8. The scale of the WMP–LGFV bond link

The introduction reports the share of identified standard-bond holdings invested in LGFV bonds. We sum reported LGFV bond values and divide by the reported value of all positions identified as standard bonds within each bank group over 2022Q1–2023Q3, retaining government and policy-bank bonds and excluding negotiated credit. Using valid product identifiers and the current underlying-bank classification gives 38.5% for local banks and 13.2% for national banks. This broader sample contains 46,772 local-bank and 45,346 national-bank product-quarters. By contrast, Table D1 uses matched product-quarters with positive included debt holdings: 36,194 and 34,595, respectively. Within that matched local-bank sample, LGFV bonds account for 40.4% of identified standard-bond value, 47.5% after restricting the denominator to eligible standard bonds, and 43.1% after adding eligible negotiated credit. The table's holding-value weights also reproduce 43.1%. The bond-only and debt-allocation shares therefore describe different portfolios, not conflicting estimates of one share.

For market context, we reconstruct the year-end 2021–2024 top-ten holdings cross-sections from the original Wind exports. We retain one representative record per product registration code and deduplicate positions within that record, preserving identical investments made by distinct products. We restrict amounts to yuan-denominated positions, identify bonds by asset type, and match issuers to the January 4, 2026 LGFV roster. Unidentified bond issuers remain in the denominator but not the LGFV numerator. In 2024, the retained disclosures contain RMB 0.3859 trillion of LGFV bonds out of RMB 2.3459 trillion of bonds, including negotiable certificates of deposit: a ratio of sums of 16.45%, not an equally weighted average of product ratios.

For each year t, let q_t be this disclosed LGFV-to-bond ratio, B_t the industry's bond assets including negotiable certificates of deposit, and A_t its net assets under management, from the annual WMP market reports. Estimated LGFV bond holdings are $\widehat{H}_t = B_t q_t$, and their share

of WMP assets is $\widehat{H}_t/A_t$. The annual WMP market report published in 2025 gives year-end 2024 values of $B_t = 18.60$ and $A_t = 29.95$ trillion yuan; using unrounded holdings gives $\widehat{H}_t = 3.06$ trillion yuan, or 10.2% of net assets under management. Equivalently, the rounded inputs give $16.45\% \times (18.60/29.95) = 10.22\%$. The extrapolation applies the disclosed bond mix to the industry's bond assets. It excludes negotiated credit and indirect exposure through funds and other investment vehicles.

This total-asset benchmark is not a weighted average of the 43.1% and 12.6% in Table D1. Those are holding-value-weighted shares for local and national banks in 2022Q1–2023Q3, conditional on positive included debt holdings. On a common date and sample, the accounting identity is LGFV/AUM = (LGFV/Debt) × (Debt/AUM), where Debt is the value of included debt holdings. Total WMP assets also include deposits, funds, central-government and policy-bank securities, and other assets outside these holdings. The industry benchmark uses a later date and the year-end bond-disclosure sample, so the initial holding shares cannot be directly aggregated into Figure A5.

The reverse ratio is $\widehat{H}_t/O_t$, where O_t is the published year-end LGFV bond balance. Lianhe Ratings (2024) reports Wind balances of RMB 9.81, 10.62, and 11.48 trillion for 2021–2023. Zhang (2025) reports RMB 10.88 trillion for 2024, implying a 28.1% benchmark holding share that year. Figure A5 leaves the 2024 reverse-ratio point unconnected to mark the change in source. These ratios describe two different economic exposures. Under the extrapolation, LGFV bonds account for about one-tenth of WMP assets in 2024, but WMPs hold more than a quarter of the LGFV bond market. A portfolio allocation within a large savings industry can therefore be an important financing source for the borrowers, even though local bonds do not dominate the industry's total assets.

D.9. Initial credit specialization and subsequent substitution

The allocation tests distinguish two comparisons. Before the program, do smaller banks invest more in LGFV bonds and larger banks more in negotiated credit across the full banking market? Afterward, do larger local banks increase negotiated credit more than smaller local rivals? Table D1 describes the initial portfolios, Table 1 estimates the all-bank size relationship, and Table D2 tests its subsequent change within local banks. Table D3 checks how these results depend on weighting and fixed effects.

The initial comparison uses equation (1). Outcomes are percentage-point shares of included bonds and negotiated credit, retaining products with zero holdings in a class. The fixed size measure Log bank size$_b$ is log FY2021 assets in RMB trillion and replaces contemporaneous assets, with the other ten covariates grouped as Controls. Quarter effects remove common movements. The regression weights by the value of included holdings and clusters by parent bank. It does not include a national-bank indicator, so the size coefficient captures differences both across bank types and within them. Across this market, larger banks hold less in LGFV

bonds, especially home bonds, and more in negotiated credit overall. The relationship with home negotiated credit is much smaller because national banks hold most of their negotiated claims outside their headquarters provinces.

We use two ratios that answer different questions. The first measures geographic concentration within an asset class. Product i is observed in quarter q, j indexes its disclosed positions, and n labels the asset class. Let w_{ijq} be the position's holding weight, p_b the underlying bank's home province, p_j the borrower's province, and $\mathcal{C}_{iq}^n$ the product's positions in class n. Then

$$\text{Home share}_{iq}^n = \frac{\sum_{j \in \mathcal{C}_{iq}^n} w_{ijq} \mathbb{1}\{p_j = p_b\}}{\sum_{j \in \mathcal{C}_{iq}^n} w_{ijq}}, \tag{D.9}$$

is the class's home share. Home bias is its deviation from the market-portfolio weight:

$$\text{Home bias}_{iq}^n = \text{Home share}_{iq}^n - \text{Province share}_{p_b,q}^n.$$

Here $\text{Province share}_{p_b,q}^n$ is the home province's share of nationwide disclosed holdings in that class. Zero home bias means investing at market weights. The measure uses products with positive holdings in the class.

The second ratio measures the share of all included debt holdings invested in each home asset class:

$$\text{Home allocation}_{iq}^n = \frac{\sum_{j \in \mathcal{C}_{iq}^n} w_{ijq} \mathbb{1}\{p_j = p_b\}}{\sum_{j \in \mathcal{C}_{iq}} w_{ijq}}, \tag{D.10}$$

$$\text{Home allocation bias}_{iq}^n = \text{Home allocation}_{iq}^n - \text{Province allocation}_{p_b,q}^n.$$

Here $\mathcal{C}_{iq}$ pools the included asset classes, and $\text{Province allocation}_{pq}^n$ is province p's class-n holdings as a share of nationwide included debt holdings. An exposure of $\text{Home allocation}^n = 0.20$ means that 20% of included debt holdings are in that home class, with zeros retained for products that do not hold it. Both Home share^n and Home allocation^n are raw shares: their denominators differ, but neither subtracts a benchmark. Subtracting the appropriate benchmark gives the within-class tilt Home bias^n or the total-debt-denominator tilt $\text{Home allocation bias}^n$. The total Home allocation bias sums across classes. Combined home exposure, used later, includes home LGFV bonds and home NS but not home other bonds.

The geographic-specialization comparison in Table 2 uses equation (2) in the main text. The outcome is home bias: the product's within-class home allocation in excess of its home province's weight in the nationwide market portfolio. Positive class holdings are required. The coefficient compares local with national banks, conditional on the controls listed in the table.

For subsequent adjustment within local banking, Table D2 uses equation (13) in the main text, with total or home NS as a percentage of included debt holdings as the outcome. Post

begins in 2023Q4. The baseline uses the initial-size comparison's controls, holding-value weights, and parent-bank clusters, but only local-bank product-quarters. Bank-effect versions absorb fixed size and retain contemporaneous bank assets as a control.

Table D2 reports the pre-program size coefficient, its Post interaction, and their sum, which gives the post-program difference per log point of assets. The home negotiated-credit difference widens by 7.47 percentage points, from 4.72 to 12.18 points per log point. Multiplying the widening by 0.981 gives a 7.33-point difference per standard deviation of log assets. Bank-effect specifications also give a positive widening. In Table D3, the initial home-LGFV bond relationship remains negative across the all-bank comparisons; the total negotiated-credit relationship is less precise with home-province-by-quarter effects or a national-bank indicator. The subsequent widening in negotiated-credit allocation remains under the listed holding-value-weighted specifications. Equal-product and equal-bank-quarter weights do not give a significant positive change. Holding-value weights show where invested money moved; equal weights describe the average product or bank-quarter. The central allocation result is therefore a concentration of negotiated-credit investment at larger banks, not an equal increase across all of their products.

Table D1
Initial credit portfolios by bank type and bank size

Share (%)	All provinces		Home province	
	LGFV bonds	NS	LGFV bonds	NS
Panel A. Bank type: all available banks				
National banks	12.6	24.1	0.4	2.1
Local banks	43.1	9.3	24.7	6.1
Panel B. FY2021 size terciles: local banks present in both periods				
Smallest tercile	38.6	1.1	25.1	0.7
Middle tercile	53.3	5.3	37.7	3.5
Largest tercile	40.6	11.4	20.7	7.4

Notes: NS denotes negotiated credit. This table documents the initial allocation levels underlying the all-bank and local-bank size comparisons. Entries are percent of included debt holdings, not all standard bonds or total WMP assets, value weighted, over 2022Q1–2023Q3. All-province shares include home and non-home positions; home means the parent bank's headquarters province. The denominator excludes deposits, funds, other investment vehicles, and central-government, central-bank, and policy-bank obligations. Panel A contains 34,595 national-bank product-quarters from 18 banks and 36,194 local-bank product-quarters from 117 banks. Panel B uses the same 102 local banks and fixed FY2021 interest-bearing-asset terciles as Panel C of Table 7: 25, 38, and 39 banks, contributing 1,965, 9,617, and 22,911 pre-period product-quarters. No positive-holding condition is imposed. Table 1 reports controlled comparisons on complete-case samples. Before implementation, local banks hold 65.0% of the sample's LGFV bonds and national banks hold 82.7% of its NS. After implementation, the corresponding fractions are 62.6% and 68.7%. Ending the initial comparison before the July announcement gives the same descriptive sorting: local and national LGFV shares are 42.4% and 12.2%, while NS shares are 9.5% and 24.0%.

Table D2
Larger local banks and the expansion into negotiated credit

Dependent variable / specification	Total NS: quarter FE (1)	Home NS: quarter FE (2)	Total NS: bank + quarter FE (3)	Home NS: bank + quarter FE (4)
Log bank size	7.382*** (2.542)	4.716*** (1.757)	—	—
Log bank size × 𝟙{Post}	10.153*** (2.673)	7.469*** (2.203)	8.288*** (2.110)	6.033*** (1.702)
Log product assets	−1.780** (0.889)	−1.176* (0.640)	−2.896*** (1.080)	−1.887** (0.856)
𝟙{Restricted redemption}	9.036** (3.509)	4.724* (2.808)	10.623*** (3.722)	5.819* (2.955)
Product risk rating	7.813** (3.427)	6.627** (2.946)	4.784 (3.759)	3.968 (2.853)
Bank nonperforming-loan ratio	−5.396** (2.691)	−6.814*** (2.091)	−3.132 (5.828)	−0.880 (5.325)
Bank cost–income ratio	−0.009 (0.250)	0.006 (0.153)	−1.197* (0.643)	−0.867 (0.526)
Bank net interest margin	−0.963 (3.662)	0.361 (2.957)	8.924 (6.043)	8.055 (5.514)
Home-city GDP (RMB trillion)	−5.989*** (1.627)	−6.074*** (1.116)	7.876 (19.243)	−5.811 (15.541)
Home-city GDP growth (%)	−0.727 (0.655)	−0.821 (0.557)	−0.889 (0.584)	−0.788 (0.526)
Home-city LGFV debt/GDP	3.766 (3.764)	4.860 (3.512)	23.120 (20.629)	20.716 (20.528)
Home-city income (RMB 10,000)	2.880*** (0.936)	2.119*** (0.733)	14.448 (11.020)	10.758 (9.791)
Log bank assets	—	—	−18.928 (16.534)	−5.388 (13.793)
Post-program size gradient	17.535*** (3.300)	12.185*** (2.509)	—	—
Wild p: log bank size × Post	< 0.001	< 0.001	< 0.001	0.017
Observations	85,182	85,182	85,181	85,181
Quarter fixed effects	Yes	Yes	Yes	Yes
Parent bank fixed effects	No	No	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one local-bank allocation regression (pp), with quarter effects, the displayed controls, and holdings weights. Size is unstandardized log FY2021 parent-bank assets in RMB trillion. Post begins in 2023Q4. Columns 1–2 estimate the initial size slope and its change; the post slope sums them using their covariance. Columns 3–4 add bank effects, absorb fixed size, and retain current bank assets. Parentheses contain parent-bank clustered errors; wild tests use 4,999 Rademacher draws. Stars indicate 10%, 5%, and 1% significance. Appendix D.9 provides details. Estimation sample: 2022Q1–2025Q3.

Table D3
Size gradients in credit allocation: alternative specifications

Panel A. Initial home LGFV share (pp)

Specification / outcome	Baseline (1)	Equal products (2)	Equal bank-quarter (3)	Province × quarter FE (4)
Log bank size	−5.458***	−4.496***	−5.586***	−1.991***
	(1.123)	(1.199)	(1.245)	(0.678)
Observations	68,331	68,331	68,331	68,328
Product, bank, and city controls	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	No
Home province × quarter fixed effects	No	No	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Panel B. Initial total NS share (pp)

Specification / outcome	Baseline (1)	Equal products (2)	Equal bank-quarter (3)	Province × quarter FE (4)
Log bank size	6.831***	4.113**	6.941***	4.945*
	(1.803)	(1.604)	(1.397)	(2.617)
Observations	68,331	68,331	68,331	68,328
Product, bank, and city controls	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	No
Home province × quarter fixed effects	No	No	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Table D3 (continued)

Panel C. Change in home NS share (pp)

Specification / outcome	Baseline (1)	Province × quarter FE (2)	Equal products (3)	Equal bank-quarter (4)
Log bank size	4.716***	4.721***	4.516**	3.693***
	(1.757)	(1.018)	(2.094)	(1.260)
Log bank size × 𝟙{Post}	7.469***	6.544***	−0.178	−1.732
	(2.203)	(1.845)	(2.186)	(3.428)
Observations	85,182	85,178	85,182	85,182
Product, bank, and city controls	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	No	Yes	Yes
Home province × quarter fixed effects	No	Yes	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Panel D. Change in total NS share (pp)

Specification / outcome	Baseline (1)	Province × quarter FE (2)	Equal products (3)	Equal bank-quarter (4)
Log bank size	7.382***	6.309***	6.903**	5.847***
	(2.542)	(1.263)	(2.798)	(1.701)
Log bank size × 𝟙{Post}	10.153***	9.426***	1.657	−0.536
	(2.673)	(2.748)	(2.692)	(3.732)
Observations	85,182	85,178	85,182	85,182
Product, bank, and city controls	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	No	Yes	Yes
Home province × quarter fixed effects	No	Yes	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is a separate regression with the ten control variables in Table 1, omitted here for compactness. Panels A–B compare all banks before implementation; Panels C–D compare local banks over 2022Q1–2025Q3. Log bank size is unstandardized log FY2021 interest-bearing assets in RMB trillion; log bank assets, where included, varies over time. Log bank size × Post is the change in the size slope, not its post-program level. Parentheses contain parent-bank clustered errors. Holdings weights use included debt value; equal-bank-quarter weights sum to one within each bank-quarter. Stars indicate 10%, 5%, and 1% significance. Appendix D.9 discusses interpretation and additional specifications.

Table D4
The exact-borrower NS–bond rate gap: alternative comparisons

Dependent variable / specification	Baseline	Debt-specific maturity	Equal cells	Reference cohort
	(1)	(2)	(3)	(4)
Negotiated-credit–bond gap at baseline	1.904***	1.749***	1.789***	1.848***
	(0.101)	(0.102)	(0.114)	(0.107)
$\mathbb{1}\{\text{Negotiated credit}\} \times \mathbb{1}\{\text{Quarter} = 2023Q1\}$	−0.676***	−0.700***	−0.665***	−0.528***
	(0.115)	(0.104)	(0.132)	(0.131)
$\mathbb{1}\{\text{Negotiated credit}\} \times \mathbb{1}\{\text{Quarter} = 2024Q4\}$	0.955***	1.060***	1.337***	0.681**
	(0.214)	(0.248)	(0.204)	(0.303)
$\mathbb{1}\{\text{Negotiated credit}\} \times \mathbb{1}\{\text{Quarter} = 2025Q3\}$	1.057***	1.197***	1.067***	1.107**
	(0.218)	(0.217)	(0.239)	(0.502)
Observations	17,303	17,303	17,303	8,088
Quarter-specific maturity controls	Yes	Debt-specific	Yes	Yes
Borrower × calendar month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Borrower province	Borrower province	Borrower province	Borrower province

Notes: Each column is one regression of annual contractual rates in percentage points, with borrower-by-calendar-month effects and quarter-specific maturity slopes. The reference gap is NS minus bonds in 2023Q2; other rows are changes from that gap. Column 2 allows debt-specific slopes and evaluates the gap at three years; column 3 gives borrower-quarter-debt-type cells equal weight; column 4 retains the 101 reference-quarter borrowers with later matched financing. Affiliates are excluded. Only key contrasts are tabulated; all quarterly interactions and maturity slopes remain estimated. Parentheses contain province-clustered standard errors; stars use t_{G-1} tests at 10%, 5%, and 1%. The full baseline path is in Figure 3. Matching, singleton removal, and coverage are detailed in Appendix D.2. Estimation sample: 2021Q1–2025Q3.

Table D5
Uncertainty in the exact-borrower financing-rate gap

Panel A. Selected regression contrasts and inference

Specification / outcome	Contract-level rates (1)	Paired borrower-month gap (2)
Gap change: 2024Q4	0.955***	1.182***
	(0.214)	(0.245)
Gap change: 2025Q3	1.057***	1.009***
	(0.218)	(0.217)
2024Q4: 95% province wild-bootstrap set	[0.472, 1.676]	[0.699, 1.899]
2025Q3: 95% province wild-bootstrap set	[0.440, 1.883]	[0.414, 1.761]
2024Q4: Province wild-bootstrap p	0.0039	0.0017
2025Q3: Province wild-bootstrap p	0.0056	0.0039
Observations	17,303	2,525
Quarter-specific maturity controls	Contract maturity	Maturity difference
Borrower × calendar month fixed effects	Yes	No
Standard errors	Borrower province	Borrower province

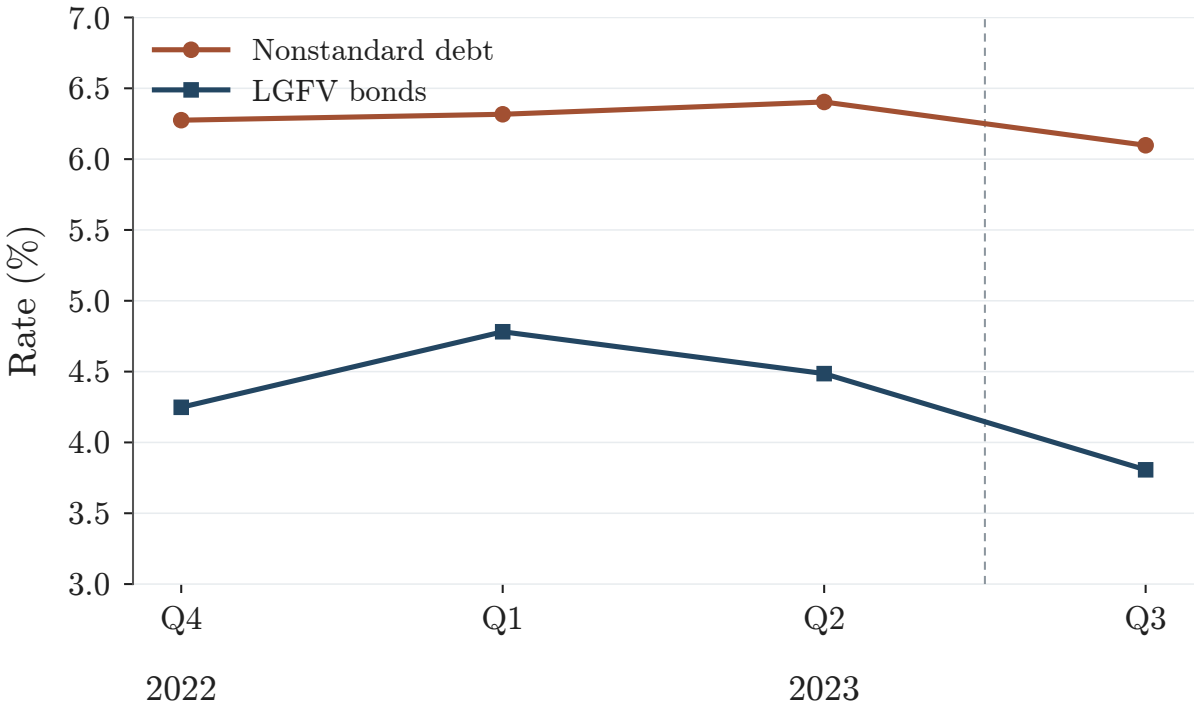
Panel B. Financing coverage (descriptive)

Coverage	2024Q4	2025Q2	2025Q3
NS contracts	360	138	81
Bond issues	140	66	40
Borrowers	75	35	21
Borrowers with both channels in a month	58	33	19
Provinces	13	11	9
Paired borrower-months	68	39	24

Notes: Column 1 uses Table D4' contract-level model; column 2 regresses one borrower-month NS-minus-bond gap on quarter indicators and quarter-specific maturity differences, with no borrower effect. Entries are changes from 2023Q2 (annual pp). Parentheses contain province-clustered errors. Confidence sets invert null-imposed bootstrap-t tests with 9,999 Webb-weight province draws. All quarterly terms remain estimated. Panel B is coverage, not a regression. Appendix D.2 gives the other inference checks. Stars indicate 10%, 5%, and 1% significance. The two regression samples contain 26 and 25 borrower-province clusters, respectively.

Figure D1

Contractual rates for a fixed set of borrowers



Notes: Annual contractual yields on NS and bonds for the same 22 legal borrowers in every quarter from 2022Q4 through 2023Q3, using 799 financing records in seven provinces. Each borrower has both channels in at least one common calendar month of every quarter. Within each borrower-month, contracts or physical bond issues receive equal weight within their channel. Matched monthly means are averaged within borrower-quarter using $h = n_{NS}n_B/(n_{NS} + n_B)$, where n_{NS} and n_B count records in the two channels; the same weights apply to both series. Each borrower then receives weight 1/22 in each quarter. These are observed uncapped contractual rates, without regression or seasonal adjustment. Borrowers and their quarterly weights are fixed, but contracts and maturities can change. The shorter diagnostic window is required by common-borrower coverage; it is not the full sample in Figure 3. Internet Appendix D.2 discusses the Q1–Q2 comparison.

Internet Appendix E
Return, Balance, and Mechanism Checks

The main findings connect a lost source of bond income to unequal substitution and weaker relative performance at smaller banks. This appendix examines return measurement, WMP-balance reporting, existing organizational capacity, and holdings disclosure. The return checks come first; Section E.1 examines balance coverage, Section E.2 distinguishes participation in negotiated credit from investment by existing users, Section E.3 checks holdings coverage, and Section E.4 relates strictly pre-program home-LGFV exposure to subsequent bank-average WMP returns.

Table E1 first asks whether the bonds' contractual income helps account for the decline in their contribution to WMP returns. On the same sample with observed rates, adding the portfolio's average bond issue rate and its Post interaction changes the home-LGFV premium decline from −6.33 to −2.57 monthly basis points per unit share. For a 20-percentage-point exposure difference, the annualized loss falls from about 15 to 6 basis points. Bond income thus accounts statistically for part of the decline, while the post-program negotiated-credit premium remains positive. The fixed sample isolates the change in the estimated premium from differences in rate coverage.

The remaining checks ask whether product age, measured volatility, multiple NAV series, or return tails drive this comparison. Requiring more coverage of bond issue yields, restricting to mature products, controlling for volatility, or retaining one series preserves a negative LGFV premium change and a positive negotiated-credit premium. Across the tail treatments, the LGFV change ranges from −6.03 to −6.61, and the post-program negotiated-credit premium from 3.96 to 4.64 monthly basis points per unit share. The loss of the bond-return premium and the remaining return on negotiated credit therefore survive these changes in measurement and sample construction.

Slow recognition of returns on illiquid investments could make a monthly comparison misleading. Table E2 therefore measures performance over complete six- and twelve-month windows and winsorizes returns rather than dropping their tails. For a 20-point exposure difference, the annualized LGFV-premium loss is about 16 and 13 basis points, respectively, while the remaining negotiated-credit premium is about 13 and 12 basis points. Requiring every monthly observation makes these samples more selective, but the longer windows preserve the same economic distinction: lower returns associated with home bonds and a positive premium on home negotiated credit.

We also check whether the geographic classification of centrally administered borrowers affects the return result. The primary regression builds home shares from individual positions, excludes these issuers from home exposure, classifies Guangfa Bank's subsidiary as national, and groups subsidiaries with their parent for inference. Counting centrally administered issuers in the bank's province as home changes other-bond shares in 27,244 product-quarters

and negotiated-credit shares in 328; home LGFV bond shares are unchanged. With the reclassified exposures, the pre-program home LGFV coefficient is 8.31 and its Post change is −5.32, compared with 8.85 and −6.13 in the primary regression, all in monthly basis points per unit share. The corresponding pre-program non-home LGFV coefficient is 2.29, with equality of the home and non-home coefficients rejected at $p < 0.001$. The larger pre-program premium on home bonds and its subsequent decline remain.

The complete-window specification is the return model in equation (4), with performance measured over $H = 6$ or 12 months rather than one month:

$$\begin{aligned}\text{Window return}_{iw}^{(H)} = {} & \beta_H' \,\text{Asset shares}_{i,h(w)} \\ & + \delta_H' \,\text{Asset shares}_{i,h(w)} \times \mathbb{1}\{\text{Post}\}_w \\ & + \gamma_H' \,\text{Controls}_{i,h(w)} + \lambda_H \,\mathbb{1}\{\text{Local bank}\}_b + \alpha_w + \varepsilon_{iw}.\end{aligned} \tag{E.1}$$

Here w is the first month of the return window and $h(w)$ the preceding quarter, using the same lag convention as the monthly model. For each NAV series, we sum all H monthly log returns in basis points and divide by H; we then average complete series within product to obtain Window return$_{iw}^{(H)}$. Exposure shares and the eleven controls are fixed at the quarter end before the window, and α_w absorbs common window-date conditions. Coefficients are estimated separately for each horizon. Thus δ_H measures the change in an exposure's average-monthly-return premium, not the total return over H months. Table E2 gives the transition exclusions and April–March comparison dates; errors cluster by underlying bank.

Table E1
Home credit exposure and returns: robustness

Panel A. Bond-income controls

Specification / outcome	Rate sample No carry (1)	Rate sample Add carry (2)	50% coverage No carry (3)	50% coverage Add carry (4)
Home LGFV bond share	9.137***	4.186***	9.411***	4.320**
	(1.909)	(1.578)	(2.015)	(1.723)
Home LGFV bond share × 𝟙{Post}	−6.331***	−2.572*	−6.885***	−3.051*
	(1.511)	(1.438)	(1.559)	(1.547)
Home negotiated-credit share	7.813***	5.967***	8.179***	6.524**
	(1.436)	(1.390)	(2.983)	(2.828)
Home negotiated-credit share × 𝟙{Post}	−2.185	−0.778	2.039	2.672
	(1.984)	(1.949)	(3.674)	(3.529)
Home LGFV bond premium after the program	2.806*	1.614	2.527	1.269
	(1.656)	(1.439)	(1.745)	(1.515)
Home negotiated-credit premium after the program	5.628***	5.189***	10.217***	9.196***
	(1.535)	(1.388)	(3.409)	(3.059)
Observations	512,509	512,509	438,024	438,024
Product, bank, and city controls	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Panel B. NAV and sample checks

Specification / outcome	Mature products (1)	Add volatility (2)	One NAV series (3)	No trimming (4)
Home LGFV bond share	9.587***	9.314***	9.245***	8.512***
	(1.735)	(2.160)	(2.005)	(1.938)
Home LGFV bond share × 𝟙{Post}	−6.460***	−5.743***	−6.692***	−6.609***
	(1.684)	(1.705)	(1.523)	(1.718)
Home negotiated-credit share	6.776***	6.263***	7.274***	7.598***
	(1.271)	(1.213)	(1.385)	(1.467)
Home negotiated-credit share × 𝟙{Post}	−2.089	−1.369	−3.328*	−3.637**
	(1.854)	(1.692)	(1.840)	(1.559)
Home LGFV bond premium after the program	3.128**	3.571**	2.553	1.904
	(1.468)	(1.486)	(1.886)	(1.944)
Home negotiated-credit premium after the program	4.687***	4.894***	3.945***	3.961***
	(1.484)	(1.247)	(1.348)	(1.089)
Observations	373,465	377,835	438,077	567,855
Product, bank, and city controls	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column estimates the monthly return model in Table 3, with all five exposure shares, their Post interactions, the local-bank indicator, eleven controls, and month effects. Only focal estimates are displayed; parentheses contain parent-bank clustered errors. Panel A fixes the sample within each no-carry/add-carry pair; carry is the portfolio-weighted mean bond issue rate and its Post interaction. Panel B retains products at least four quarters old with lifecycle controls, adds trailing volatility and its Post interaction, retains one NAV series, or restores untrimmed observations of baseline products. Stars indicate 10%, 5%, and 1% significance. Appendix E describes further tail checks.

Table E2
Return premiums over complete measurement windows

Dependent variable: average monthly return (bp).

Specification / outcome	6 months Raw (1)	6 months Winsorized (2)	12 months Raw (3)	12 months Winsorized (4)	6 months Incumbents (5)	12 months Incumbents (6)
Home LGFV bond share	9.538**	7.267***	5.267**	5.927***	6.607**	5.596**
	(4.081)	(2.070)	(2.069)	(1.949)	(2.647)	(2.435)
Home LGFV bond share × 𝟙{Post}	−9.465**	−6.840***	−5.470**	−5.544**	−6.604**	−5.169*
	(3.942)	(2.242)	(2.140)	(2.201)	(2.935)	(2.693)
Home negotiated-credit share	5.972***	5.050***	5.857***	6.246***	2.782	5.904***
	(2.080)	(1.806)	(1.905)	(1.774)	(2.014)	(1.662)
Home negotiated-credit share × 𝟙{Post}	−0.023	0.534	−1.084	−1.152	5.068	−2.141
	(2.553)	(2.413)	(2.533)	(2.435)	(4.572)	(2.128)
Home LGFV bond premium after the program	0.073	0.427	−0.203	0.383	0.003	0.427
	(2.113)	(1.346)	(2.514)	(2.296)	(1.360)	(1.404)
Home negotiated-credit premium after the program	5.949***	5.585***	4.773**	5.095**	7.851**	3.763**
	(1.561)	(1.305)	(2.215)	(1.981)	(3.322)	(1.606)
Observations	53,469	53,469	11,210	11,210	24,097	3,900
Product, bank, and city controls	Yes	Yes	Yes	Yes	Yes	Yes
Other shares and Post interactions	Yes	Yes	Yes	Yes	Yes	Yes
Return-window date fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one regression with the full controls and asset shares of Table 3; only the home-LGFV and home-NS estimates are displayed. Parentheses contain parent-bank clustered errors. All monthly returns must be observed; exposure and controls precede the window, and complete share-class windows are averaged within product. Winsorization is at 1/99. Incumbents were observed before the program. Windows spanning July–September 2023 are excluded; annual April–March windows compare 2022–2023 with 2024–2025. Post-program premiums sum the level and Post interaction from the same column, using their covariance. Stars indicate 10%, 5%, and 1% significance.

E.1. Coverage and interpretation of bank-level WMP balances

The balance checks ask whether smaller banks' weaker WMP growth reflects changing reporting coverage or the different organizational forms of large and small banks. The specification is equation (18) in the main text.

Yuhang Rural Commercial Bank's June 2025 balance is corrected to RMB 16.909 billion.[44] Omitting it instead gives a baseline coefficient of 0.152 (SE 0.039).

Let $\mathbb{1}\{\text{Pre-subsidiary}\}_b$ indicate that a subsidiary-registered product is observed by 2023Q2. Group-time versions replace half-year effects α_s with $\alpha_{\mathbb{1}\{\text{Pre-subsidiary}\}_b,s}$, allowing separate organizational-group paths. The separate-slope version allows a different coefficient on $\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_s$ in each group, rather than imposing the same size response. Both retain bank effects and the baseline's outcome, predetermined size, Post date, and bank-clustered inference.

Coverage rises from 56 size-matched local banks in 2022H1 to 86 in 2024H2, then falls to 53 in 2025H2. The baseline drops singletons. Table E3 checks this changing composition by excluding the final period, retaining the same 29 banks at every date, and excluding the nine banks with pre-existing subsidiary products. The size-related growth difference remains positive in each sample. Allowing subsidiary banks either a separate Post shift or their own half-year effects gives a coefficient of 0.104 log balance points per log point of assets (SE 0.040–0.041). For a one-standard-deviation size difference, $100[\exp(0.1042 \times 0.981) - 1]$ is about 10.8% greater relative growth; the flexible-time wild p-value is 0.003. With those time effects, the 2022 event coefficients are 0.0173 and 0.0308 (joint $p = 0.48$), followed by 0.0184 in 2023H2, 0.2237 in 2025H1, and 0.2699 in 2025H2. The difference therefore develops after announcement within organizational groups as well as in the pooled comparison.

Estimating separate size–Post slopes gives 0.098 for banks without pre-existing subsidiary products (SE 0.042; wild $p = 0.006$) and 0.237 for the nine banks with them (SE 0.084; wild $p = 0.186$), in log balance points per log point of assets. Status is based on pre-announcement product registrations, not licensing dates. Requiring an observed pre-announcement portfolio leaves 494 bank-half-years at 78 banks. With group-specific time effects, the size coefficient is 0.095 (SE 0.034; wild $p = 0.014$); the separate slope among banks without subsidiary products is about 0.084 (SE 0.035; wild $p = 0.032$). These comparisons hold subsidiary status fixed.

Splitting banks at RMB50 billion of June 2023 WMP balances preserves the size gradient in both groups (Table E3, Panel B). With bank and common half-year effects, the size–Post coefficients are 0.094 below the cutoff and 0.256 at or above it (wild $p = 0.030$ and 0.049), implying 6.8% and 19.4% greater relative growth for a doubling of parent-bank size.

[44] The bank's first-half 2025 report, Table 3, gives 169.09 hundred-million yuan and 191 products; its year-end report confirms the June amount. The Wind entry was 0.0169.

Table E3
WMP balance growth: reporting coverage and sample checks

Panel A. Reporting coverage

Specification / outcome	Baseline (1)	Exclude 2025H2 (2)	Same banks all dates (3)	Exclude subsidiaries (4)	Subsidiary × Post (5)
Log bank size × 𝟙{Post}	0.152***	0.118***	0.134***	0.098**	0.104**
	(0.039)	(0.038)	(0.041)	(0.042)	(0.040)
𝟙{Pre-subsidiary} × 𝟙{Post}	—	—	—	—	0.308***
					(0.085)
Observations	571	521	232	506	571
Bank fixed effects	Yes	Yes	Yes	Yes	Yes
Half-year fixed effects	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Panel B. Organizational composition and the WMP-balance cutoff

Specification / outcome	Subsidiary × time (1)	Separate size slopes (2)	Known status group-time (3)	Cutoff matched (4)	Below RMB50bn (5)	At/above RMB50bn (6)
Log bank size × 𝟙{Post}	0.104**	—	0.095***	0.151***	0.094*	0.256***
	(0.041)		(0.034)	(0.040)	(0.049)	(0.075)
Log bank size × 𝟙{Post}: subsidiary parents	—	0.237***	—	—	—	—
		(0.084)				
Log bank size × 𝟙{Post}: other parents	—	0.098**	—	—	—	—
		(0.042)				
Wild p: log bank size × Post	0.003	—	0.014	< 0.001	0.030	0.049
Observations	571	571	494	496	349	147
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Half-year fixed effects	No	No	No	Yes	Yes	Yes
Subsidiary × time fixed effects	Yes	Yes	Yes	No	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: The outcome is log observed WMP balances. Each column is one bank-half-year regression with the displayed regressors and stated fixed effects; no additional bank or city controls enter. Errors cluster by bank. Size is unstandardized log FY2021 bank interest-bearing assets in RMB trillion. Group-time columns allow pre-program subsidiary status separate half-year paths; column B2 estimates both group-specific size slopes jointly. The same-bank sample contains 29 banks at every date. Columns B4–B6 fix groups using June 2023 WMP balances: 70 matched banks, 50 below RMB50bn, and 20 at or above. All 50 below-cutoff banks lack observed pre-program subsidiary products. The cutoff is not in parent-bank assets and does not identify bank-specific regulatory caps. Missing balances are neither zeros nor assigned to a group. Wild tests use 4,999 bank-level Rademacher draws. Extending the panel to 2021H2 gives a size–Post coefficient of 0.171 but rejects the longer pre-trend test ($p = 0.009$). Appendix E.1 explains definitions and the verified correction. Stars use conventional 10%, 5%, and 1% tests.

E.2. Participation capacity and the interpretation of scale

The fixed-cost mechanism links scale to participation in negotiated credit. We examine this participation margin using pre-existing organizational capacity, then distinguish it from the amount invested by existing participants. In Table E4, $\mathbb{1}\{\text{Pre-subsidiary}\}_b = 1$ if a subsidiary-registered product is observed by 2023Q2; this status applies to all products of underlying bank b. For product i issued by institution a, let $o(i)$ identify whether it is the bank's own product or a subsidiary product. We estimate

$$\begin{aligned}\text{Home allocation}_{iq} = {} & \beta_C \, \mathbb{1}\{\text{Pre-subsidiary}\}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_S \, \text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_q + \beta_B \, \text{Initial home bias}_a \times \mathbb{1}\{\text{Post}\}_q \\ & + \gamma' \, \text{Controls}_{iq} + \alpha_{b,o(i)} + \alpha_{p_b,q} + u_{iq}.\end{aligned} \tag{E.2}$$

Here q is a quarter, p_b is the underlying bank's headquarters province, and $\alpha_{b,o(i)}$ is a separate effect for each bank and product form. The outcome is the home negotiated-credit share, home LGFV bond share, or their sum, in percentage points of included holdings. The coefficient β_C compares post-program adjustment by banks with and without prior subsidiary products, conditional on size and initial reliance. All eleven controls enter, and 2023Q3 is excluded. Bank-by-product-form and home-province-by-quarter effects absorb persistent differences and regional conditions. Errors cluster by parent bank, and observations are weighted by included debt holdings. Nine of 100 banks have this capacity proxy; all are in the largest size tercile, with only one overlapping the other banks' asset range. Size and organizational capacity are therefore closely bundled in this comparison.

Banks with prior subsidiary products increase home negotiated credit by 12.64 percentage points more (wild $p = 0.008$). Adding this proxy changes the size–Post coefficient from 5.08 to -0.38 percentage points per log point of assets: the larger-bank expansion is concentrated among banks with pre-existing organizational capacity.

Table E5 examines size differences among existing participants. Columns 1–4 estimate equation (E.2) within the nine equipped banks, using quarter or province–quarter effects, which absorb capacity–Post. Columns 5–6 use all banks and add the three-way interaction of size, pre-subsidiary status, and Post. The size slope is β_S without prior capacity and $\beta_S + \beta_{SC}$ with it.

These intensive-margin estimates are positive with quarter effects and negative with province–quarter effects, both imprecise under wild-bootstrap inference; the latter variation is concentrated in Jiangsu and Zhejiang. The joint pre-period test gives $p = 0.036$. The organizational-group contrast in Table E4 and the within-group size slopes in Table E5 answer different questions. The fixed-cost mechanism concerns participation; it does not require investment by existing users to rise uniformly with size.

Table E4
Predetermined parent capacity and portfolio adjustment

Specification / outcome	Home NS	Home bonds	Combined	Equal bank home NS	Incumbent home NS	Product FE home NS
	(1)	(2)	(3)	(4)	(5)	(6)
Log bank size × 𝟙{Post}	−0.375	0.257	−0.118	−0.981	0.250	−0.0095
	(1.830)	(1.528)	(1.905)	(1.357)	(1.568)	(1.5536)
𝟙{Pre-subsidiary} × 𝟙{Post}	12.645***	−1.704	10.941***	16.962***	3.286*	3.697**
	(3.117)	(2.491)	(3.314)	(3.606)	(1.838)	(1.843)
Initial home bias × 𝟙{Post}	6.765	−14.457***	−7.692*	2.662	4.710	4.157
	(5.330)	(5.328)	(4.245)	(5.953)	(3.618)	(2.797)
Wild p: pre-subsidiary	0.008	0.586	0.044	—	—	—
Observations	76,713	76,713	76,713	76,713	36,539	36,539
Product, bank, and city controls	Yes	Yes	Yes	Yes	Yes	Yes
Parent-form fixed effects	Yes	Yes	Yes	Yes	Yes	No
Province × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	No	No	No	No	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one regression of home allocation shares (pp). Log bank size is unstandardized log FY2021 interest-bearing assets in RMB trillion; log bank assets, where included, varies over time. Pre-subsidiary equals one if a subsidiary-registered product is observed by 2023Q2, applied to all products of the parent. Nine of 100 parents have this status, all in the largest size tercile. The eleven controls enter throughout; columns 5–6 add lifecycle controls and column 6 product effects. They are omitted from this diagnostic table. All comparisons omit 2023Q3 and include home-province-by-quarter effects. Parentheses contain bank-clustered errors. Wild tests use 999 bank-level Rademacher draws. Size and organizational capacity are bundled here; this is not an independent license experiment. Stars indicate 10%, 5%, and 1% significance. Columns 1–3 weight by included debt value; column 4 gives equal total weight to each parent-quarter; columns 5–6 use fixed pre-program product assets.

Table E5
Does size predict substitution within subsidiary-equipped parents?

Specification / outcome	Home NS Quarter FE (1)	Total NS Quarter FE (2)	Home NS Province × quarter (3)	Total NS Province × quarter (4)	Home NS Pooled (5)	Total NS Pooled (6)
Log bank size × 𝟙{Post}	2.849	4.545	−28.808***	−27.541***	3.229	0.359
	(2.248)	(2.702)	(2.568)	(4.220)	(1.963)	(2.794)
Initial home bias × 𝟙{Post}	82.508***	100.245***	−49.360***	24.342	10.733*	14.824*
	(17.862)	(21.833)	(13.253)	(18.031)	(6.172)	(8.135)
Log bank size × 𝟙{Pre-subsidiary} × 𝟙{Post}	—	—	—	—	−9.737**	−5.550
					(4.053)	(5.551)
𝟙{Pre-subsidiary} × 𝟙{Post}	—	—	—	—	10.730***	19.913***
					(2.827)	(4.362)
Log bank size × 𝟙{Post}: subsidiary parents	—	—	—	—	−6.508*	−5.191
					(3.445)	(5.081)
Wild p: subsidiary size slope	0.352	0.291	0.230	0.225	0.187	0.510
Observations	28,493	28,493	28,493	28,493	76,713	76,713
Product, bank, and city controls	Yes	Yes	Yes	Yes	Yes	Yes
Parent-form fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	No	No	No	No
Province × quarter fixed effects	No	No	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Columns 1–4 use the nine previously equipped parents; columns 5–6 use all size-matched local parents and estimate separate size slopes by capacity status. Outcomes are NS shares (pp). Size is unstandardized log FY2021 assets in RMB trillion, so capacity × Post in columns 5–6 is evaluated at RMB 1 trillion of assets. All regressions retain eleven controls, prior reliance, and parent-form effects; 2023Q3 is omitted. Only focal interactions are displayed. Columns 3–4 use limited within-province size variation, concentrated in Jiangsu and Zhejiang. Parentheses contain bank-clustered errors; wild tests use 999 bank-level Rademacher draws. The implied subsidiary slope in columns 5–6 uses both interaction coefficients and their covariance. This table tests an intensive-margin size gradient, not the entry threshold. Stars indicate 10%, 5%, and 1% significance. Regressions weight observations by included debt value.

E.3. Top-ten disclosure coverage

Top-ten holdings can cover different fractions of product assets. A home-LGFV share calculated from those holdings could therefore vary with disclosure as well as investment choice. Table E6 addresses this concern in two ways: controlling for observed disclosure differences and removing the disclosed-debt denominator. It revisits three headline comparisons, without restricting the sample to high-coverage products or imposing a common position cutoff.

The three comparisons retain their original specifications: initial home-LGFV allocation on log bank size, ten controls, and quarter effects across all banks (equation (1)); home negotiated-credit allocation on log bank size, its Post interaction, and the same controls within local banks (equation (13)); and monthly returns on five lagged allocation shares, their Post interactions, eleven controls, a local-bank indicator, and month effects (equation (4)). The coefficients of interest are log size, log size–Post, and home-LGFV share–Post, respectively. Allocations retain holding-value weights; returns are unweighted. All errors cluster by bank, and each baseline–adjusted pair uses identical observations.

For each product, Coverage is the sum of all reported position weights, expressed as a fraction: disclosed positions totaling 40% give Coverage = 0.40. Log product assets is the natural log of product assets, and Positions counts disclosed positions. Adjusted regressions add

$$\begin{aligned}\text{Disclosure controls}_{it} = &\sum_{k=1}^{3}(\eta_k \,\text{Coverage}_{it}^{k} + \zeta_k \,\text{Log product assets}_{it}^{k}) + \eta_N \,\text{Positions}_{it} \\ &+ \sum_{k=1}^{3}(\tilde{\eta}_k \,\text{Coverage}_{it}^{k} + \tilde{\zeta}_k \,\text{Log product assets}_{it}^{k}) \times \mathbb{1}\{\text{Post}\}_t \\ &+ \tilde{\eta}_N \,\text{Positions}_{it} \times \mathbb{1}\{\text{Post}\}_t.\end{aligned} \tag{E.3}$$

Here t is the outcome quarter or month; in return regressions, Coverage, Log product assets, and Positions come from the preceding quarter end. Terms already in the baseline enter only once. Cubic polynomials allow nonlinear size and coverage relationships; Post interactions allow those relationships to change after implementation.

Panel A retains the original disclosed-debt shares. The initial size coefficient changes from −5.458 to −5.706: a bank twice as large holds 3.8 versus 4.0 percentage points less home LGFV bonds. The negotiated-credit size–Post coefficient changes from 7.469 to 6.766, implying 5.2 versus 4.7 points more expansion at a local bank twice as large. The home-LGFV return interaction changes from −6.133 to −5.863 monthly basis points per unit share. For a 20-point allocation difference, the annual premium loss is therefore 14.7 versus 14.1 basis points. All three adjusted estimates remain significant at 1%.

Panel B instead sums the reported home holdings as weights in product assets, removing the disclosed-debt denominator. Its return regression includes all six disclosed debt categories and their Post interactions, with other assets as the reference. The three disclosure-adjusted coefficients are −2.683, 3.175, and −8.603, all significant at 1%. Both adjustments preserve the headline patterns without reconstructing undisclosed holdings.

Table E6
Disclosure coverage and exposure denominators

Coefficient of interest	Pre-program home LGFV Log bank size		Home negotiated credit Log bank size ×𝟙{Post}		Monthly product return Home LGFV bond share ×𝟙{Post}	
	Baseline (1)	Adjusted (2)	Baseline (3)	Adjusted (4)	Baseline (5)	Adjusted (6)
Panel A. Shares of disclosed eligible debt						
Estimate	−5.458***	−5.706***	7.469***	6.766***	−6.133***	−5.863***
	(1.123)	(1.065)	(2.203)	(1.419)	(1.501)	(1.296)
Observations	68,331	68,331	85,182	85,182	554,555	554,555
R^2	0.424	0.433	0.261	0.341	0.305	0.307
Panel B. Reported product-asset weights						
Estimate	−2.863***	−2.683***	3.451**	3.175***	−8.097***	−8.603***
	(0.703)	(0.656)	(1.428)	(0.983)	(2.366)	(2.594)
Observations	68,331	68,331	85,182	85,182	554,555	554,555
R^2	0.382	0.401	0.199	0.306	0.309	0.311
Baseline controls	10	10	10	10	11	11
Flexible disclosure controls	No	Yes	No	Yes	No	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	No	No
Month fixed effects	No	No	No	No	Yes	Yes
Standard errors	Bank	Bank	Bank	Bank	Bank	Bank

Notes: Each column in each panel is one regression. Columns 1–2 use all banks before implementation; 3–4 use local banks before and after; 5–6 use monthly returns. Designs follow equations (1), (13), and (4); adjusted columns add equation (E.3). Allocations are in pp; returns in monthly bp. Log bank size is log FY2021 underlying-bank interest-bearing assets; log bank assets is the time-varying control. Coverage is the sum of all disclosed position weights, not availability of issue yields. Panel B uses reported product-asset weights directly (fractions in return regressions), not independently observed full portfolios. Its return regression includes all six disclosed debt categories and their Post interactions. All other controls, allocation weights, and underlying-bank clustering are unchanged. Products with no disclosed holdings in a category remain in the regressions. Stars denote 10%, 5%, and 1% significance.

E.4. Initial home-LGFV reliance and return erosion

The exposure comparison asks whether banks initially more dependent on home LGFV bonds lost more return independently of size, rather than inferring that loss from their post-program portfolios. Equation (15) relates strictly pre-program home-LGFV exposure to subsequent WMP-return changes, conditional on size. Returns use the full corrected CNY NAV data, averaging series within product-month and products within bank-month, without requiring a holdings match. The existing global 1/99 monthly-return exclusion is unchanged. In Table E7, columns 3–4 hold the 93-bank sample fixed while adding the eight prior-quarter controls.

Adding pre-program non-home LGFV share–Post checks whether the decline is specific to home exposure rather than initial LGFV investment generally; the home-exposure coefficient is −128.25 (wild $p < 0.001$). Allowing initial size a separate slope in every quarter checks whether a changing size–return relationship explains the result and gives −132.79 ($p < 0.001$). Relative to 2023Q2, the joint pre-period exposure test gives $p = 0.257$; size and combined tests reject ($p = 0.021$ and 0.002), motivating the flexible-size check. The outcome here is product performance; WMP-business growth is examined separately in Appendix E.1.

Table E7
Initial home-LGFV exposure and WMP-return erosion

	Broad sample (1)	Province comparison (2)	Matched sample (3)	Add controls (4)
Initial home LGFV bond share × 𝟙{Post}	−131.555***	−151.670***	−123.616***	−126.340***
	(31.887)	(48.946)	(40.263)	(40.923)
Log bank size × 𝟙{Post}	3.832	6.397	0.158	4.879
	(9.131)	(11.726)	(9.099)	(10.658)
Prior-quarter log bank assets	—	—	—	22.883
				(57.437)
Bank nonperforming-loan ratio	—	—	—	−7.916
				(20.482)
Bank cost–income ratio	—	—	—	−4.043
				(3.205)
Bank net interest margin	—	—	—	19.709
				(36.457)
Home-city GDP (RMB trillion)	—	—	—	100.631
				(74.537)
Home-city GDP growth (%)	—	—	—	−5.459
				(4.105)
Home-city LGFV debt/GDP	—	—	—	102.736
				(132.922)
Home-city income (RMB 10,000)	—	—	—	58.608
				(45.709)
Wild p: initial home LGFV share × Post	< 0.001	0.002	0.002	0.003
Wild p: log bank size × Post	0.693	0.557	0.985	0.671
Observations	2,821	2,821	2,006	2,006
Bank fixed effects	Yes	Yes	Yes	Yes
Subsidiary status × month effects	Yes	Yes	Yes	Yes
Home province × month effects	No	Yes	No	No
Standard errors	Bank	Bank	Bank	Bank

Notes: Each column is one regression of bank-average annualized monthly log WMP returns (bp), July2022–September2025, excluding July–September2023. Initial exposure is the 2022Q1–2023Q2 home-LGFV standard-bond share of included disclosed debt, not total WMP assets. A ten-point share difference implies 0.10 times its coefficient. Log bank size is unstandardized log FY2021 bank interest-bearing assets;

the log bank assets control varies over time. Post starts in October2023. All eight controls are lagged one quarter. Products receive equal weight within bank-months; bank-months receive equal estimation weight. Parentheses show bank-clustered errors; wild tests use 4,999 bank draws. Stars denote conventional 10%, 5%, and 1% significance.